

Systematic Review of Quantum Entanglement Applications to Network Synchronization

John Campbell

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1 Part 1: Conceptualizing Quantum Networks

The rapid evolution of telecommunication networks has historically been driven by classical graph-theoretic principles, with architectures often characterized by hierarchical, cyclic, and scale-free topologies. Classical networks, exemplified by the Internet, frequently exhibit properties such as robustness against random failures and the presence of hub nodes, which emerge naturally through preferential attachment mechanisms described by the Barabási-Albert (BA) model [14]. These scale-free networks are typically cyclic and directed, allowing feedback loops and complex growth dynamics that facilitate efficient routing and redundancy [30].

With the advent of quantum information science, there is an increasing focus on integrating quantum communication capabilities into existing telecommunication infrastructures. Quantum networks, however, fundamentally differ from their classical counterparts due to quantum mechanical constraints such as the no-cloning theorem, coherence length limitations, and the necessity for maintaining entanglement. These constraints inherently restrict quantum networks to acyclic, directed topologies—specifically, directed acyclic graphs (DAGs)—to prevent issues like quantum state duplication and decoherence-induced loss of quantum correlations [31,32]. This shift from cyclic, scale-free architectures to hierarchical, DAG-based structures highlights a paradigm change in network design, with profound implications for routing, scalability, and security.

Understanding these differences is crucial for designing hybrid classical-quantum networks capable of leveraging the strengths of both paradigms. Classical networks grow efficiently through preferential attachment, forming hubs that facilitate high-throughput data aggregation [14]. In contrast, quantum networks rely on localized entanglement distribution and quantum repeaters, which impose topological constraints that favor hierarchical, layered architectures more resistant to decoherence [33]. The transition from classical

to quantum architectures involves not only technological challenges—such as quantum memory development, low-loss transmission, and standardization—but also conceptual shifts in how networks are modeled, managed, and visualized.

Furthermore, the unique properties of quantum channels—such as degradability, superactivation, and entanglement-binding behavior—add additional layers of complexity to network design and analysis [19]. Visual and analytical techniques, including the use of transformation matrices and phase-space representations, are instrumental in understanding quantum state evolution within these networks [26]. Certification protocols for quantum memories and entanglement preservation are vital for ensuring the integrity of quantum information as it propagates through the network, with recent advances enabling measurement-device-independent verification [20].

In summary, the shift from classical, cyclic, scale-free topologies to directed, acyclic, hierarchical quantum network architectures necessitates rethinking fundamental design principles. This transition promises enhanced security, scalability, and functionality—yet also poses significant technical and conceptual challenges. As quantum technologies mature, an integrated approach that combines classical backbone infrastructure with localized quantum links will be essential for realizing the full potential of a global quantum communication network. Furthermore one is forced to come to the conclusion that these networks, though artificial in construction, will inherently display emergent characteristics wrt synchronization and memory. It is hoped that this research will convey that these emergent features of these specialised networks are in fact a resource to be utilized rather than ignored.

2 Telecom Network Architecture with Classical and Quantum Topologies

This section provides a conceptual overview of a three-layer telecom network architecture, emphasizing the differences between classical scale-free networks and quantum networks. Classical architectures typically involve cyclic, directed graphs with complex growth patterns similar to the Barabási-Albert model, whereas quantum networks are inherently acyclic due to their directionality constraints.

2.1 Network Layers and Signalling

The network consists of three primary layers:

- **Long-Haul Interexchange Network** (Layer 1): Connects distant cities or regions via high-capacity fiber links.
- **Metropolitan Interoffice Network** (Layer 2a): Connects central offices within a metropolitan area.
- **Metropolitan Access Network** (Layer 3): Connects end-users or local data centers.

Data exchange across these layers involves sophisticated signalling methods encoding information through phase, amplitude, and possibly time-binned bits (e.g., via interferometry). Classical networks often exhibit a **directionally cyclic** topology, enabling feedback loops and complex growth dynamics that can follow a **scale-free** (power-law) distribution, as described by the Barabási-Albert model [14]. This growth mechanism relies on preferential attachment, leading to the emergence of hub nodes with high connectivity.

In contrast, **quantum networks** are constrained by coherence length limits and are fundamentally **directionally acyclic**. These constraints significantly influence their graph topologies, which are inherently DAGs (Directed Acyclic Graphs). This distinction has profound implications for routing, scalability, and network management.

2.2 Implications for Classical and Quantum Networks

Classical scale-free networks are capable of dynamic growth, exhibiting emergent properties such as robustness and efficient routing through hub nodes [14]. Quantum networks, however, rely on local, point-to-point entanglement and cannot support cycles due to the no-cloning theorem and coherence limitations [?]. Consequently, quantum networks are better represented by DAGs.

This fundamental difference influences various aspects of network design:

- **Routing:** Bottlenecks in quantum networks are more critical because of limited entanglement distribution pathways.
- **Data encoding:** Quantum data utilize degrees of freedom such as polarization, angular momentum, phase, amplitude, and wavelength division multiplexing (WDM). These multiplexing techniques help mitigate bandwidth bottlenecks.
- **Scalability:** Classical networks can grow via preferential attachment, while quantum networks often require localized, hierarchical architectures.

Modern telecommunication networks are often conceptualized as multi-layered systems, facilitating efficient data exchange over vast geographical areas. A typical three-layer architecture includes:

- **Long-Haul Interexchange Network:** Connects national and international backbone points, primarily utilizing optical fiber links for high-capacity transmission.
- **Metropolitan Interoffice Network:** Connects central offices within metropolitan regions, often employing dense wavelength division multiplexing (DWDM) and metro-access technologies.
- **Metropolitan Access Network:** Links end-users or local data centers to the metropolitan backbone, including fiber-to-the-home (FTTH), fiber-to-the-curb (FTTC), and wireless access points.

3 Cyclic and Graph Network Topologies in Quantum Networks

In the study of quantum communication networks, understanding the underlying topology is crucial for optimizing quantum information transfer, entanglement distribution, and network robustness. Unlike classical networks that often rely on hierarchical or tree-like structures, modern quantum networks can adopt **directionally cyclic** topologies, which incorporate cycles and loops. These cyclic structures are represented mathematically by generalized graph models:

$$G = (V, E), \tag{1}$$

where

$$V$$

is the set of vertices (nodes or quantum repeaters), and

$$E \subseteq V \times V$$

is the set of edges (quantum links). The edges can be directed, i.e.,

$$E \subseteq V \times V, \tag{2}$$

signifying unidirectional quantum channels, which are common in quantum networks due to the directional nature of certain quantum processes such as entanglement swapping or photon transmission.

3.1 Advantages of Cyclic Topologies in Quantum Networks

Cyclic topologies introduce redundancy and multiple pathways for quantum information transfer, enhancing the network’s robustness against node or link failures. This redundancy is particularly vital in quantum networks where entanglement distribution must be resilient over long distances, and quantum error correction protocols are employed. Cycles facilitate load balancing and enable multiple routes for entanglement swapping, thereby increasing network throughput and reliability.

3.2 Quantum Coupling and Entanglement in Cyclic Structures

The fundamental resource enabling quantum communication across such networks is **entanglement**. Quantum links (edges) are characterized by their capacity to establish and maintain entanglement between nodes. The strength and quality of these links depend on physical parameters such as signal attenuation, noise, and the quantum coupling mechanisms involved.

In this context, the **quantum coupling**—the interaction strength between quantum nodes—is often modeled as a function of the physical distance

$$d$$

between nodes, following a power-law decay:

$$J(d) \propto d^{-\alpha}, \tag{3}$$

where

$$\alpha > 0$$

is the decay exponent. This power-law behavior arises from fundamental quantum interactions such as dipole-dipole coupling, photon-mediated interactions, or other physical processes governing entanglement distribution.

3.3 Relation to Power Laws and Network Topology

The power-law decay of coupling strength has direct implications for the topology and scalability of quantum networks. When coupling strengths decay as in Equation (3), the network’s effective connectivity is heavily influenced by the distribution of inter-node distances.

- **Short-range interactions** (

$$\alpha$$

large): Favor local entanglement, leading to networks with many small, densely connected clusters.

- **Long-range interactions** (

$$\alpha$$

small): Enable the formation of hubs and global entanglement distribution, similar to classical scale-free networks.

This physical decay behavior influences the emergent **degree distribution**

$$P(k)$$

of the network, where

$$k$$

is the degree (number of links per node).

3.4 Scale-Free Growth in Quantum Network Topologies

Analogous to classical scale-free networks modeled by the **Barabási–Albert (BA)** model [14], quantum networks can exhibit *preferential attachment* mechanisms driven by entanglement capacity and quantum coupling strengths.

- **Preferential attachment:** Nodes with higher entanglement capacity or more robust links tend to attract additional connections, reinforcing the formation of hubs.
- **Degree distribution:** The probability

$$P(k)$$

that a node has degree

$$k$$

follows a power-law:

$$P(k) \sim k^{-\gamma}, \quad (4)$$

where

$$\gamma \approx 3$$

. This indicates a few highly connected nodes (hubs), while most nodes have low degree, facilitating efficient quantum communication over large-scale networks.

with regard to scale free networks of nodes i.e. those forming networks resembling or within the structure that can be compared to Barabasi-Albert type network models, we also begin to see the emergence of condensate like behaviour.

The mapping of a network to a degree of energy levels resembling a Bose gas can be represented by the following diagram modeled after the Barabási-Albert (BA) process, and the condensation phenomena observed in a Bose gas. On the left, a growing scale-free network exhibits preferential attachment, leading to a power-law degree distribution. On the right, a plot demonstrates how this distribution follows a power-law form, characteristic of such networks:

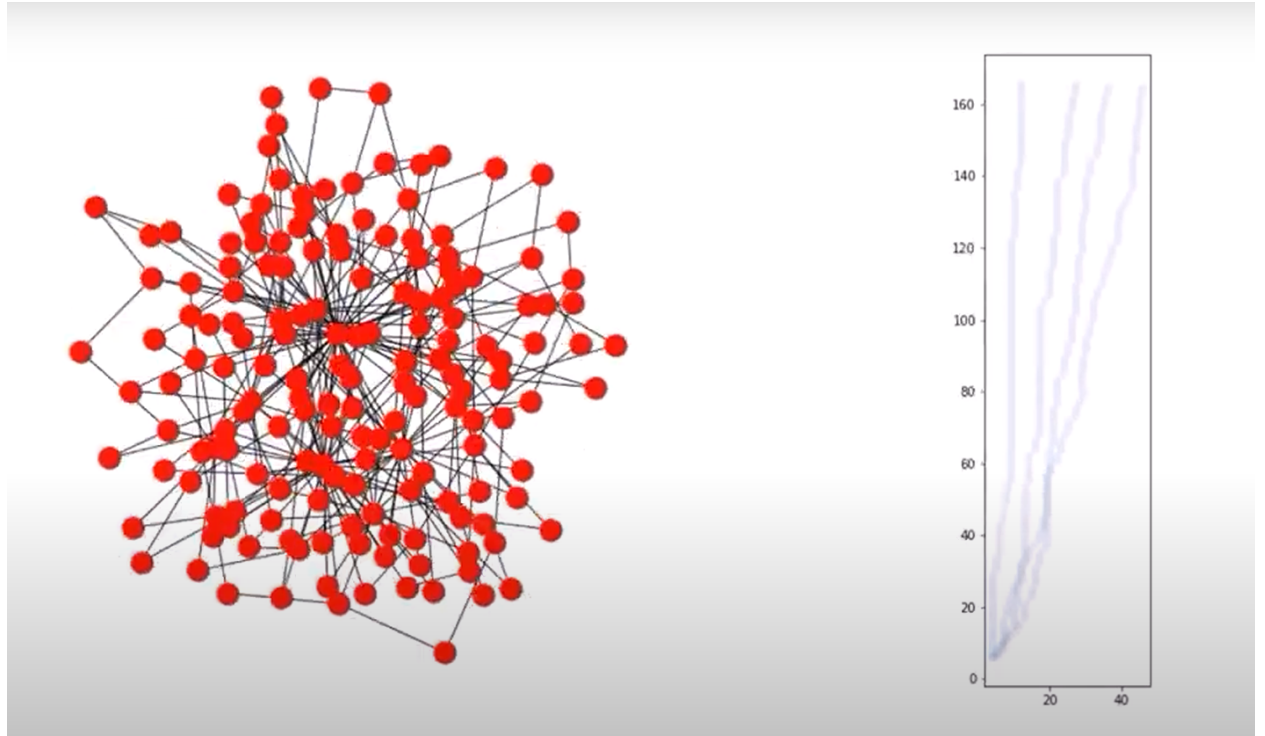


Figure 1: Left: A visual representation of a growing scale-free network generated via preferential attachment, resembling a Bose-Einstein condensate in the context of network growth, where a few nodes (hubs) accumulate a large number of connections. Right: The resulting degree distribution plotted on a log-log scale, illustrating the emergent power-law behavior.

This analogy highlights how the network's evolution mirrors the statistical mechanics of a Bose gas undergoing condensation, with the degree distribution akin to the occupancy distribution of particles in a Bose-Einstein condensate.

Relation between Bose Gas Growth and Network Growth in the Barabási-Albert Model

The growth of a Bose gas and the evolution of a scale-free network share underlying statistical features, particularly in their degree or occupancy distributions.

In a Bose gas at thermal equilibrium, the average occupation number

$$\langle n_i \rangle$$

of a single-particle energy level

$$\varepsilon_i$$

is given by the Bose-Einstein distribution:

$$\langle n_i \rangle = \frac{1}{e^{(\varepsilon_i - \mu)/k_B T} - 1}, \quad (5)$$

where: -

$$\mu$$

is the chemical potential, -

$$k_B$$

is Boltzmann's constant, -

$$T$$

is the temperature.

The total number of particles

$$N$$

is:

$$N = \sum_i \langle n_i \rangle. \quad (6)$$

3.4.1 Mapping Between Network and Bose Gas

The core idea of the mapping is to associate each node's degree

$$k$$

with an energy level

$$\varepsilon$$

. Specifically, higher degrees correspond to lower energy states, reflecting the tendency of some nodes to acquire many links (hubs), analogous to bosons condensing into the ground state.

Define a function linking degree

$$k$$

to energy

$$\varepsilon$$

:

$$\varepsilon(k) = \varepsilon_0 - \alpha \ln(k), \quad (7)$$

where

$$\varepsilon_0$$

is a reference energy and

$$\alpha$$

is a positive constant controlling the energy scaling.

The probability that a node has degree

$$k$$

then corresponds to the occupation number

$$\langle n(\varepsilon) \rangle$$

at energy

$$\varepsilon(k)$$

:

$$P(k) \propto \langle n(\varepsilon(k)) \rangle. \quad (8)$$

Using the Bose-Einstein distribution (5), we have:

$$P(k) \propto \frac{1}{e^{(\varepsilon(k)-\mu)/k_B T} - 1}. \quad (9)$$

Substituting (7):

$$P(k) \propto \frac{1}{e^{(\varepsilon_0 - \alpha \ln k - \mu)/k_B T} - 1}. \quad (10)$$

Rearranged:

$$P(k) \propto \frac{1}{e^{(\varepsilon_0 - \mu)/k_B T} \cdot k^{-\alpha/k_B T} - 1}. \quad (11)$$

For large

$$k$$

, the dominant behavior simplifies to a power-law:

$$P(k) \sim k^{-\gamma}, \quad (12)$$

where

$$\gamma = \frac{\alpha}{k_B T} + 1. \quad (13)$$

This demonstrates how the power-law degree distribution emerges naturally from the Bose-Einstein occupation statistics when the energy levels are logarithmically related to the node degrees.

Mapping this to network growth, the probability

$$\Pi(k)$$

that a new node connects to an existing node with degree

$$k$$

(preferential attachment) follows:

$$\Pi(k) \propto k + \delta \tag{14}$$

where

$$\delta$$

accounts for initial attractiveness. Over time, this leads to a degree distribution:

$$P(k) \sim k^{-\gamma} \tag{15}$$

with the power-law exponent

$$\gamma$$

related to the model parameters. The accumulation of connections at high-degree nodes (hubs) parallels the Bose-Einstein condensation, where particles tend to occupy the lowest energy state, leading to macroscopic occupation and a condensate. Similarly, in the network, a few nodes dominate the connectivity, forming a "hub" analogous to the condensate in the Bose gas.

This analogy underscores how the statistical mechanics of Bose-Einstein condensation can provide insights into the emergence of hubs in scale-free networks, revealing a deep connection between quantum many-body physics and complex network growth phenomena.

3.4.2 Condensation Phenomenon and Hubs

At sufficiently low "temperatures" (or high scaling energy coupling α), a Bose-Einstein condensation occurs where a macroscopic number of particles occupy the lowest energy state, corresponding to the emergence of a highly connected hub in the network. This phenomenon models the formation of super-connected nodes within the network structure.

Through this mapping, the network's degree distribution can be viewed as a Bose-Einstein distribution over energy levels, with the links between nodes in a network can be ranked as an entanglement between nodes, themselves represented as density matrices of quantum harmonic oscillators. These nodes could be practically realised as sources, memories, quantum computing clients in a network etc however they will all be required to be quantum harmonic oscillators that can effectively measure and "carry" entanglement onboard as eigenvalues in their matrix elements. The central criterion for a quantum network is that it must be, by monogamy of entanglement conditions [5], be directionally acyclic. This refers to the fact that the entangled states can only travel in a one-way entropic arrow of time between source and measurement, with the memories and other resources, i.e. repeaters, carrying the paths as a sum-over histories throughout the network provided they are synchronized using the path integral formalism discussed.

We must now examine how entanglement itself is represented in these network structures.

3.5 Implications for Quantum Network Design

Understanding the interplay between the power-law decay of quantum coupling and resulting network topology informs strategies for:

- Optimal placement of quantum repeaters,
- Maximizing entanglement distribution efficiency,
- Ensuring robustness against failures,
- Designing scalable quantum internet architectures.

By leveraging the inherent power-law behaviors of physical quantum interactions, one can engineer network topologies that balance local entanglement fidelity with global connectivity, ultimately supporting resilient and high-capacity quantum communication systems.

3.6 Implications for Signaling and Network Robustness

The scale-free nature means that:

- **Emergence of Hubs:** Few nodes have extremely high connectivity, facilitating efficient data aggregation.
- **Vulnerability and Resilience:** While robust against random failures, targeted attacks on hubs can fragment the network.

Below are TikZ diagrams contrasting classical scale-free cyclic networks with quantum acyclic networks.

3.6.1 Classical Cyclic Network

The architecture of telecommunications networks can be conceptualized as a multi-layered, cyclic graph, comprising long-haul interexchange, metropolitan interoffice, and metropolitan access networks. These layers facilitate data exchange through signaling mechanisms based on phase, amplitude, and potentially time-binned encoded bits [26]. Understanding the topology and growth dynamics of such networks provides insights into their scalability, robustness, and future evolution, especially in the context of classical versus quantum communication paradigms.

4 Classical Telecom Network Architecture

4.1 Layered Network Model

Classical telecom networks are typically structured into three primary layers:

1. **Long-Haul Interexchange Network** (Layer 1): Facilitates large-distance data transmission between major cities or regions, often employing wavelength-division multiplexing (WDM) and optical amplifiers [27].
2. **Metropolitan Interoffice Network** (Layer 2a): Connects regional hubs within metropolitan areas, using dense wavelength division multiplexing (DWDM) over metropolitan fibers.
3. **Metropolitan Access Network** (Layer 3): Links end-users to local exchanges, often employing passive optical networks (PONs) or fiber-to-the-home (FTTH) solutions.

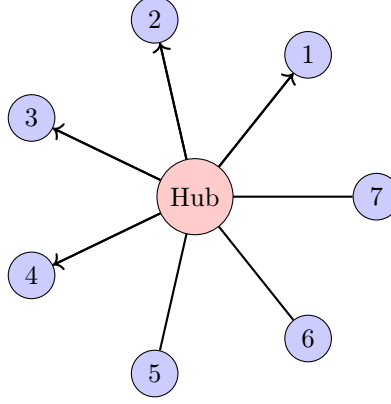


Figure 2: Classical cyclic, scale-free network with feedback loops.

Data exchange across these layers involves complex signaling schemes, including phase, amplitude, and time-binned encoding. Such methods enhance spectral efficiency and enable advanced modulation formats [28]. The signaling graph topology can be highly cyclic and directionally complex, often forming scale-free networks that follow power-law degree distributions akin to the Barabási–Albert model [29].

4.2 Scale-Free Growth and Topology

The growth dynamics of classical telecom networks exhibit preferential attachment, leading to a scale-free topology. This results in robustness against random failures but vulnerability to targeted attacks at highly connected nodes [30]. The network's evolution is driven by the "pull" for connectivity, enabling the emergence of hubs that dominate traffic flow.

4.2.1 Quantum Acyclic Network

Quantum networks differ fundamentally in their growth dynamics. Due to coherence length limitations, they are not expected to follow scale-free growth in the same manner. Instead, Quantum data transmission involves:

- **Quantum Phase, Amplitude, and Wavelength Encoding:** Using entanglement and superposition.
- **Additional Degrees of Freedom:** Polarization, angular momentum, and spin states expand encoding possibilities, alleviating bottlenecks.

4.3 Planting Quantum Networks in Existing Network Infrastructure

- Classical networks with scale-free, cyclic topologies are typically embedded within core and metropolitan layers, facilitating robust data routing and resilience.
- Quantum networks are more likely to be integrated at the metropolitan access or local data center levels, utilizing DAG topologies to manage entanglement distribution. They can scale via hierarchical, localized growth strategies [?].

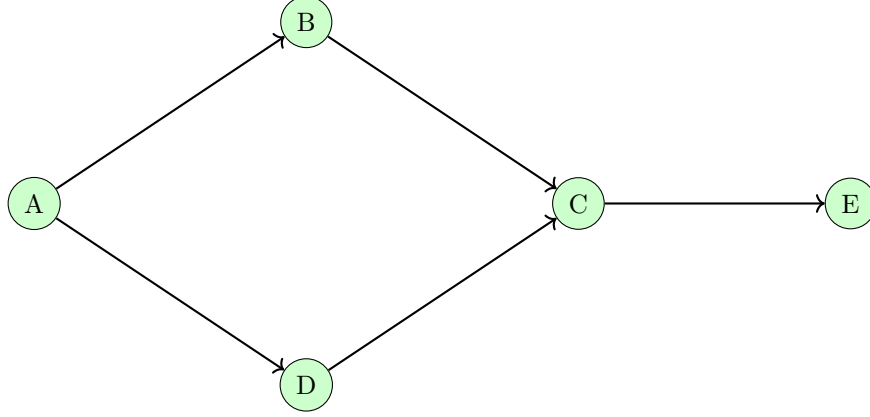


Figure 3: Quantum network represented as a directed acyclic graph (DAG).

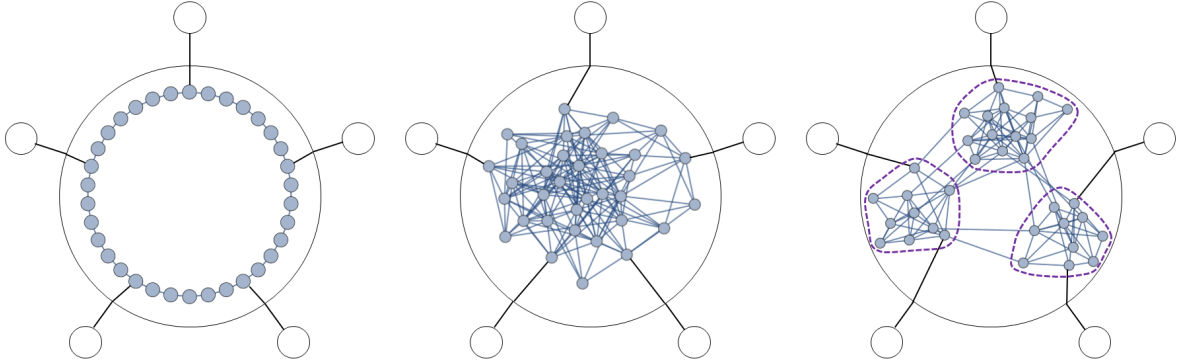


Figure 4: Left to right: A visual representation of a growing scale-free quantum network embedded in a central hub node. generated at first within the classical hub node (i.e. a data center) via random preferential attachment, resembling first a ring network then a Reyni toward BA network and finally, with quantum-directed acyclic nature emergent from scale limiting factors (limits on coherence, memory, repeaters, measurements) etc leading to a Watts–Strogatz type model with small world properties

As quantum communication technologies mature, integrating quantum networks into existing classical infrastructure presents both opportunities and challenges. This section discusses key considerations for deployment, scalability, and the evolution of hybrid classical-quantum networks.

4.4 Hybrid Classical-Quantum Network Architectures

Given the constraints on quantum network topology—namely, their directed and acyclic nature—and their limited scalability due to coherence length restrictions, a promising approach involves hybrid architectures:

- **Classical Backbone with Quantum Endpoints:** Utilizing classical infrastructure for data routing

and control, while deploying quantum links selectively for secure key distribution or quantum sensing applications [34].

- **Entanglement Swapping and Repeaters:** Implementing quantum repeaters at strategic nodes to extend entanglement distribution, effectively creating a "quantum extension" of classical networks [35].

4.5 Routing and Network Management

The acyclic nature of quantum networks necessitates novel routing protocols that optimize entanglement distribution paths while accounting for the limited coherence length:

- **Localized Routing:** Emphasizing local entanglement swapping and storage to mitigate decoherence over long distances.
- **Resource Allocation:** Dynamic management of quantum memory and repeater placement to maximize network throughput and reliability.

4.6 Scalability and Technological Challenges

Achieving scalable quantum networks requires overcoming several technical hurdles:

- **Quantum Memory Development:** Improving coherence times and storage capacities.
- **Low-Loss Fiber and Wavelength Conversion:** Enhancing transmission fidelity and intercompatibility with existing infrastructure.
- **Standardization and Protocols:** Developing universal standards for quantum network components and operations [32].

5 Quantum Channels and their Degradability

Many of the intriguing quantum channels exhibit **degradability**. These channels are characterized by the fact that their *coherent information* can be optimized via the maximization over input states $\varphi_{A \rightarrow A}$, such that:

$$Q(\mathcal{N}) = \max_{\varphi_{A \rightarrow A}} I(A)B)_{\mathcal{N}(\varphi)},$$

where $I(A)B)$ denotes the coherent information of the channel $\mathcal{N}$.

5.1 Degradable Quantum Channels

Channels such as *dephasing* and *entanglement-breaking* channels are known to be degradable. Notably, these channels often exhibit *non-additivity* of their quantum capacity, which is a significant departure from classical intuition.

Light signals for example can travel through an optical fiber via different paths. The impossibility of knowing the exact path a light ray has taken leads to an effective dephasing as well as depolarization noise

To overcome the effects of noise, one can incorporate redundancy in the message to ensure that the quantum information can still be retrieved at the receiving end. This is similar to saying "Alpha, Beta, Charlie" instead of "A, B, C" when speaking on the phone. Although the transmitted message is longer, the redundancy ensures that the information is understood correctly. Moreover the entangled phase and polarization qubits

can have noises self-corrected via squeezing using a type of encoding of qubits known as The Gottesman-Kitaev-Preskill (GKP) code, a quantum error-correcting code that encodes quantum information into an infinite-dimensional space using continuous-variable systems [8].

GKP codes belong to the family of bosonic codes, which leverage the properties encoding information in the continuous variables (quadratures) of a bosonic mode, like a harmonic oscillator. They are designed to be robust against displacement errors, which are common in many quantum systems

The GKP encoding of quantum channels represents the current state-of-the art in transmitting flying continuous-variable (CV) qubits across optical fiber.

Stabilizer measurements, like S_z and S_x , are used to detect and correct these displacements, effectively "pinning" the qubit to the correct position on its lattice are showing promise in quantum network stabilization and synchronization due to their inherent error correction capabilities. These qubits, encoded in bosonic modes, can be stabilized against displacement errors, including those caused by loss, via stabilizer measurements. This leads to improved coherence times and the potential for fault-tolerant quantum computation and networking. The inherent error correction capabilities of GKP qubits make them promising for scaling up quantum networks.

6 Graph Network Routing as Random vs Quantum Walks

In a certain context Graph Network Routing can be framed as a problem of random walk distributions in a network's topology.

One can then draw on this analogy to connect graph network routing to the contrast between random walks and quantum walks. The fundamental difference lies in how they explore the graph's topology and, consequently, how they arrive at a probability distribution over the nodes.

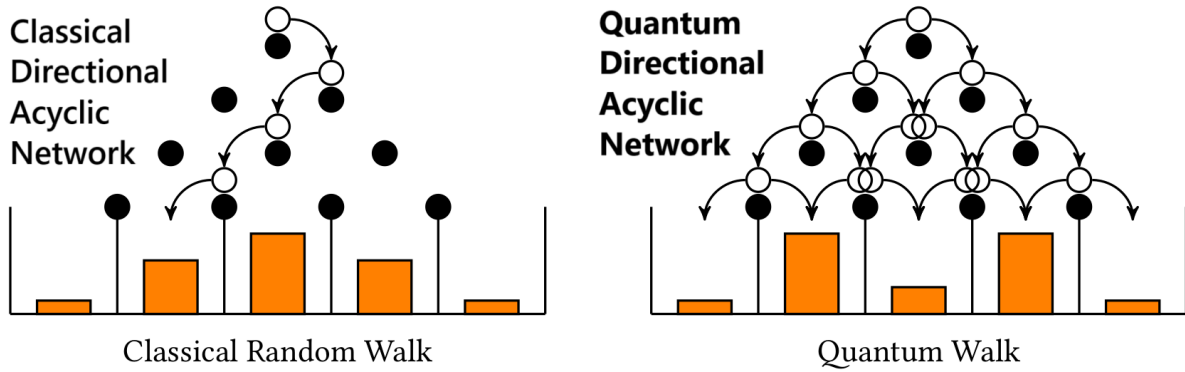


Figure 5: Left to right: A visual representation of a daq vs qdag and their equivalent walk distribution in random vs quantum

Classical Random Walks:

Mechanism: A classical random walk moves from node to node with probabilities determined solely by the graph's structure (e.g., uniform probability to each neighbor). Each step is independent of previous steps.

Distribution: After many steps, the probability distribution over nodes tends toward a stationary distribution, which often reflects the graph's degree distribution (nodes with more connections have higher probabilities). This distribution is typically spread out across many nodes.

Routing: In routing, a random walk can be used as a simple algorithm to explore the network, but it's inefficient because it doesn't prioritize finding short paths. It's akin to randomly exploring a maze.

Quantum Walks:

Mechanism: A quantum walk utilizes quantum superposition and interference. The walker exists in a superposition of states, being at multiple nodes simultaneously. The evolution is governed by unitary operators that update the superposition based on the graph's adjacency matrix. Interference between different paths can lead to constructive or destructive interference, boosting or suppressing the probability of reaching certain nodes.

Distribution: The probability distribution after a quantum walk can be very different from a classical random walk. Interference can lead to concentration of probability at specific nodes, even those not directly well-connected. This is a manifestation of quantum speedup.

Routing: Quantum walks offer the potential for significantly faster routing algorithms. By leveraging quantum interference, they could be designed to find optimal or near-optimal paths more quickly than classical random walks or even many deterministic classical algorithms, especially in large and complex networks.

6.1 The Connection to the XOR Gate Network:

The XOR gate network when interpreted as a quantum circuit acting on qubits representing the walker's position, is essentially a way to define the unitary evolution of a quantum walk. The structure of the XOR gates and their connections would dictate how the walker's superposition evolves as it "walks" through the network. The resulting probability distribution after measurement would be the outcome of this quantum walk, and it's expected to be dramatically different from a classical random walk on the same network due to the effects of quantum interference. In Summary: Graph network routing can be viewed as a problem of finding optimal probability distributions over the nodes. Classical random walks yield a distribution often reflective of the network's basic structure, while quantum walks can lead to very different, potentially more advantageous, distributions thanks to quantum interference. The XOR gate network is a tool to engineer the specific quantum walk and thus control the resulting probability distribution. The efficiency of quantum routing algorithms is directly linked to the ability to achieve this efficient concentration of probability on the desired target nodes.

Assumptions and Clarifications:

XOR Gate Representation: We'll assume the XOR gates are acting on qubits, performing a controlled-NOT (CNOT) operation. A classical XOR gate wouldn't be relevant in a quantum walk context.

Networked Structure: The "splitting" implies that the output of the first XOR gate becomes the input to two more. We need to specify how these are connected. A simple interpretation is that the output qubit is copied (which is impossible in standard quantum mechanics without measurement) or that the output is used as control for two separate target qubits. We'll explore both.

Entanglement: The phrasing "entangled networked circuit" suggests that the qubits are entangled before the XOR gates are applied. This is crucial. If they are initially unentangled, the XOR gates may create entanglement, but the outcome will be different.

Closed vs. Open System: A closed system means no interaction with the environment; an open system allows for decoherence (loss of quantum information). Decoherence significantly alters probability distributions in quantum walks.

Scenario 1: Output qubit copied (impossible, but illustrative) This scenario is physically impossible in quantum mechanics due to the no-cloning theorem. However, it's conceptually easier to understand initially:

Initial State: Let's assume two entangled qubits, $|\psi\rangle = |00\rangle + |11\rangle$ (Bell state, for example).

First XOR: The first XOR gate acts on one of the qubits. The effect depends on the way it's wired. If it acts as a control on the other qubit, the state might remain unchanged. If the output feeds back into itself, it could lead to complex behaviour.

Splitting (impossible copy): The output is "copied" to two more XOR gates. This is where the impossibility hits. You can't copy a qubit.

Second and Third XORs: Even if we ignored the copy issue, the subsequent XOR gates would further

manipulate the entangled state.

Measurement: Measuring the qubits would yield a probability distribution over the possible outcomes (e.g., $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$). This distribution depends heavily on the specific wiring and the initial entangled state.

Scenario 2: Output qubit controls two subsequent XORs (more realistic) This is a more realistic model:

Initial State: Again, we start with an entangled state like $|00\rangle + |11\rangle$.

First XOR: This XOR gate acts as a control, affecting two other qubits (initially in $|0\rangle$ or some superposition). The state becomes more complex. For example, if the first qubit controls two other qubits: the entangled state becomes something like a 3-qubit state: $|000\rangle + |111\rangle$ (assuming the control is not affecting the second qubit). If the first XOR changes the first qubit, the effect would propagate to the others.

Second and Third XORs: These XOR gates further transform the state, potentially creating more entanglement.

Measurement: Measuring all three qubits would yield a probability distribution that is very difficult to calculate analytically without specifying the exact connections and initial state. The distribution reflects the interference patterns created by the quantum walk through the network.

Closed vs. Open System:

Closed: In a closed system, the probability distribution would be solely determined by the unitary evolution of the quantum state. Interference effects would be fully preserved.

Open: In an open system, decoherence would introduce randomness. The probability distribution would tend toward a classical distribution (e.g., equal probabilities for all outcomes) as the degree of decoherence increases. The interference patterns crucial to quantum computation would be lost.

Predicting the exact probability distribution requires a precise specification of the circuit's topology (how the XOR gates are connected), the initial entangled state of the qubits, and whether the system is closed or open. Simulations using quantum computing software packages (like Qiskit or Cirq) would be necessary for accurate calculations in all but the simplest cases. Analytical solutions are generally very complex for even modest-sized networks.

a simple scenario which can emerge in a quantum network consisting of a XOR gate (which, in the quantum context, is a controlled-NOT or CNOT gate) splitting into two more CNOT gates can be represent the qubits' states using Dirac notation. The initial state can be written as: $|\psi\rangle = |0\rangle + |1\rangle$ where $|0\rangle$ and $|1\rangle$ are complex amplitudes such that $||^2 + ||^2 = 1$. The CNOT gate's action on two qubits $|a\rangle$ and $|b\rangle$ is given by: $\text{CNOT}(|a\rangle |b\rangle) = |a\rangle |a+b\rangle$ where $+$ denotes addition modulo 2 (XOR). Applying the gates sequentially:

First CNOT (q0 controls q1): $|\psi\rangle \rightarrow |000\rangle + |110\rangle$

Second CNOT (q1 controls q2): $|000\rangle + |110\rangle \rightarrow |000\rangle + |111\rangle$

Third CNOT (q1 controls q2): $|000\rangle + |111\rangle \rightarrow |000\rangle + |110\rangle$ (notice this is the same as after the second CNOT, because the second and third CNOT gates are essentially acting on the same set of qubits, which will cause the third gate to negate the effect of the second, thus restoring the state to that post the second CNOT)

Probability Distribution: After applying the three CNOT gates, the final state is: $|\psi_{final}\rangle = |000\rangle + |111\rangle$ The probability of measuring each state is :

$$P(|000\rangle) = ||^2$$

$$P(|111\rangle) = ||^2$$

Closed vs. Open System:

Closed System: In a truly closed quantum system, the probabilities $||^2$ and $||^2$ remain unchanged over time. The system evolves unitarily according to the gates applied.

Open System: In an open system, interactions with the environment (decoherence) can introduce errors. This could lead to changes in $||^2$ and $||^2$, altering the probability distribution. For instance, spontaneous emission or other noise sources might cause transitions between $|000\rangle$ and $|111\rangle$, making the probabilities deviate from

$||^2$ and $||^2$. The exact effect depends on the nature and strength of the interaction with the environment.

7 Entangled Light Resources, SPDC and CV

7.1 Current State of the Art in SPDC

Spontaneous parametric down-conversion (SPDC) remains a foundational technique for generating entangled photon pairs, essential for quantum communication, computing, and sensing applications. Recent advancements by the groups led by Anton Zeilinger and Kwiat research groups have significantly improved the quality, brightness, and control of SPDC sources. These developments include the fabrication of highly efficient nonlinear crystals, engineering of phase-matching conditions, and integration with photonic circuits, enabling scalable and robust entanglement generation. Such progress has facilitated the creation of high-fidelity entangled states suitable for complex quantum networks and experimental tests of fundamental quantum mechanics.

7.2 Continuous Variable Quantum Resources performed in Valentina Parigi's Group

Valentina Parigi's research group specifically focuses on the exploitation of continuous variable (CV) quantum resources, leveraging squeezed and entangled light states generated via SPDC processes. Their work emphasizes the generation, manipulation, and characterization of CV entangled states, which are particularly advantageous for quantum information protocols that benefit from deterministic state generation and high data rates. Techniques such as homodyne detection and quantum tomography are employed to analyze these states, paving the way toward practical CV quantum communication and computation platforms. The group's efforts contribute to the broader endeavor of integrating CV quantum resources into scalable quantum technologies.

In our research at Maynooth We investigated the use of BBO and ppKTP crystals, analyzing the entangled states with polarimeters by studying their emission spectra and polarization characteristics such as Stokes parameters and representation on the Poincaré sphere. This data, combined with quantum correlations from the photon pairs, enables us to perform quantum tomography using polarization-controlling optics in a fibered network.

We study the autocorrelated polarizations of single SPDC photons, illustrating our ability to prepare polarization-entangled single photon pairs that violate Bell's inequality. Our research utilizes motorized polarization controllers to select the Bell basis, thereby enhancing our control over entangled light. We conclude that SPDC in a packaged PPKTP module is a versatile and reliable source for single photons at 810 nm.

Moreover, two application approaches are explored: First, we investigate difference frequency generation involving single photons and classical pump fields to create a two-photon entanglement-as-resource node in a quantum network. Second, we discuss quantum network optimization, using prior mapping of the network to determine the network Fiedler vector. This allows us to order our source photons into an optimal polarization state for efficient distribution in complex networks.

For squeezed polarization states, the Poincaré sphere geometrically represents polarization states of light, allowing for state mapping from the EPR source (Figure 1). By employing quasiprobability distributions, we visualize quantum states in phase-space, culminating in the construction of the Wigner function from our polarization measurements.

Quantum state tomography generates the Wigner function from the squeezed polarization state by analyzing measurements across various polarization settings. This yields a phase-space representation that elucidates quantum properties such as squeezing, entanglement, and non-classical features, requiring detailed continuous measurements and data reconciliation for complete state reconstruction.

8 SPDC approach at Maynooth

Various approaches have emerged for single photon preparation. With the advent of quantum key distribution protocols in the 1980s, single photon sources garnered increased interest.

This study focuses on generating degenerate single photon pairs at 808 nm using type-II parametric down conversion with periodically poled potassium titanyl phosphate (KTP) as the nonlinear medium. We analyze the emission spectra of different KTP chips and discuss their characteristics. Using two-photon interference setups, we gain insights into the spectra of degenerate PDC from observed interference patterns.

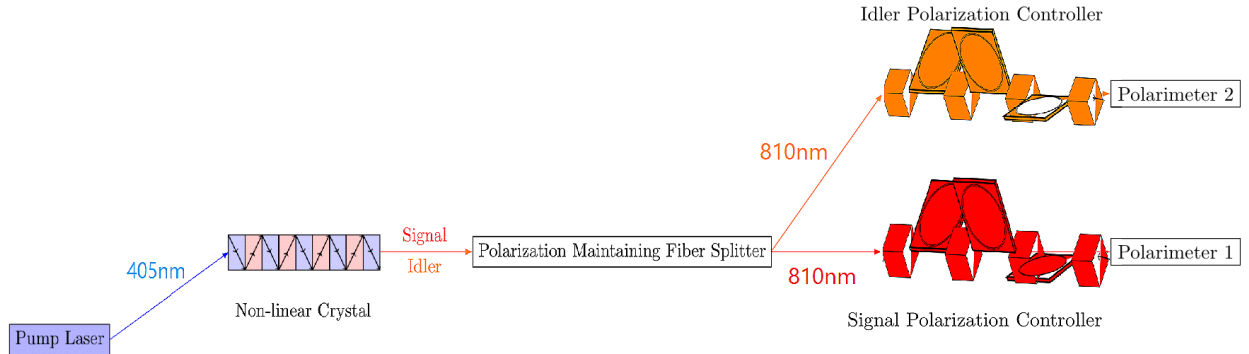
Our efforts demonstrate the viability of preparing polarization-entangled single photon pairs that violate Bell's inequality, affirming PDC in PPKTP as a reliable source of single photons at 808 nm. This paper propounds two practical approaches: (1) exploring difference frequency generation to produce a two-photon entanglement resource, and (2) optimizing quantum networks with polarization states mapped to the network's Fiedler vector for improved distribution.

9 Experimental Setup

We utilize the Poincaré sphere to evaluate the produced polarization states. We construct density matrices from standard polarization measurements, applying quantum state tomography techniques to generate the Wigner function, which lays out phase-space representations of the polarization-entangled states.

In order to generate, transmit and manipulate optical quantum states we needed to develop a versatile setup that can perform polarization control and metrology on entangled photons.

for the sake of testing we chose to make the system fibered.



Polarization Entangled Photon Source with Polarization Control and Metrology

The main advantage here is the closed nature of the optics from the external lab environment. The overall size of the network is also smaller and the dispersion challenges provided by polarization maintaining fiber allow us to examine the system from the view that this will be an inherently lossy and hence motivates one to develop new techniques of quantum tomography with noisy measurements.

10 Squeezed Light and Optical Kerr Effect

Squeezed light is thus a valuable resource in photonic quantum technology that allows us to surpass the standard quantum limit in certain measurements. This paper discusses the generation of polarization-squeezed entanglement, focusing on the generation of two-mode squeezed states entanglement from a non-linear crystal in the process of squeezing. We also examine the utilization of the GKP code, which encodes quantum information in continuous-variable systems, to provide a framework for using polarization-squeezing

in manipulating quantum information in infinite-dimensional spaces. While the GKP code primarily focuses on continuous variables like position and momentum, we show that the principles of encoding and stabilizing quantum information can be extended to other types of quantum states, including polarization-encoded states.

Squeezed light results from reducing the uncertainty in one observable while increasing the uncertainty in its conjugate observable, as dictated by the Heisenberg uncertainty principle [21]. Nonlinear interactions such as the optical Kerr effect can be utilized to produce squeezed light in free space or in optical fibers [22].

The optical Kerr effect describes the change in the refractive index of a material in response to light intensity, I . In the context of polarization squeezing, the focus is on the nonlinearity in optical crystals such as BBO and ppKTP in nodal cavities in a network but may also focus on Kerr nonlinearities in active optical fibers, both of which enable the creation of polarization-squeezed states in a network [23].

10.1 Polarization Squeezing

Polarization squeezing involves manipulating the polarization states of light to achieve squeezing, particularly through the generation of two-mode squeezed states [24]. By splitting light into two orthogonal polarization axes in a polarization beam-splitter either in free space or in optical fiber and combining the resulting squeezed beams, polarization squeezing can be achieved [25].

In the generation of polarization-squeezed entangled light, the quantum states of the two orthogonal polarization modes, typically designated as horizontal (

$$x$$

) and vertical (

$$y$$

), can be modeled as quantum harmonic oscillators characterized by their respective annihilation and creation operators. Specifically, the annihilation operators

$$\hat{a}_x$$

and

$$\hat{a}_y$$

describe the destruction of a photon in the

$$x$$

- and

$$y$$

-polarized modes, respectively, while the corresponding creation operators

$$\hat{a}_x^\dagger$$

and

$$\hat{a}_y^\dagger$$

describe the addition of a photon into these modes. Similarly, when considering the second set of modes, possibly representing different spatial or frequency modes labeled as

$$b$$

, we define

$$\hat{b}_x$$

and

$$\hat{b}_y$$

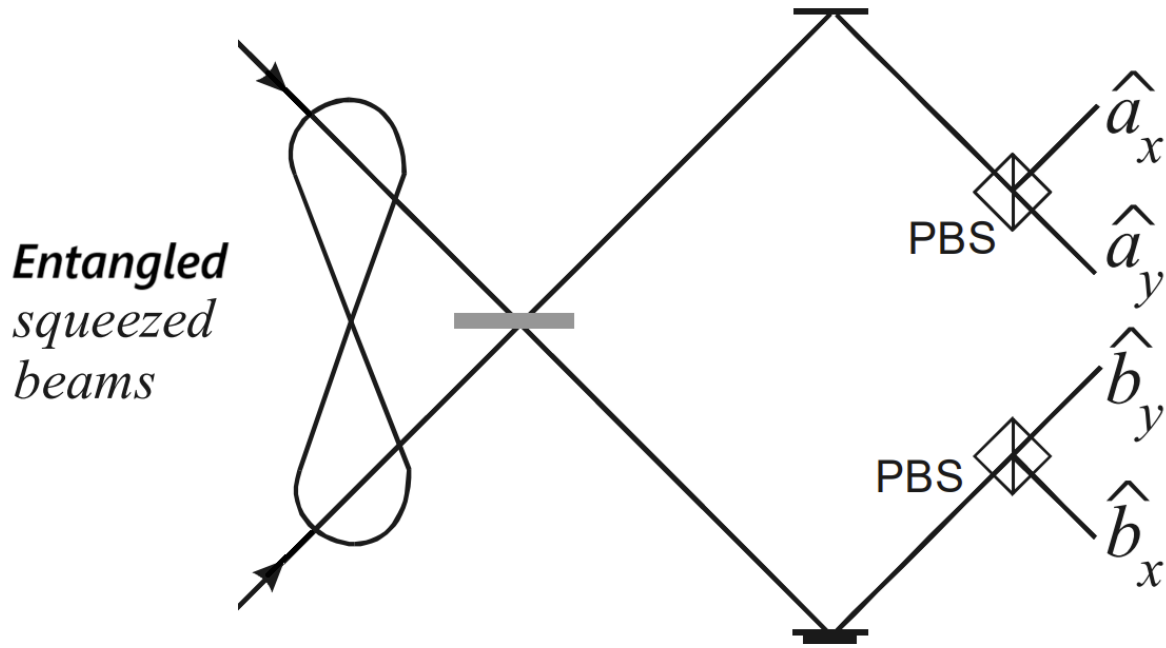


Figure 6: Schematic of a typical polarization entangled squeezed light source

with their adjoint operators

$$\hat{b}_x^\dagger$$

and

$$\hat{b}_y^\dagger$$

. These operators satisfy the canonical bosonic commutation relations:

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \quad [\hat{a}_i, \hat{a}_j] = 0, \quad (16)$$

$$[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}, \quad [\hat{b}_i, \hat{b}_j] = 0, \quad (17)$$

where

$$i, j \in \{x, y\}$$

. The entanglement and polarization squeezing are characterized by the correlations between these operators, often expressed via quadrature operators such as

$$\hat{X}_a^{(i)} = \frac{1}{\sqrt{2}}(\hat{a}_i + \hat{a}_i^\dagger), \quad \hat{Y}_a^{(i)} = \frac{1}{i\sqrt{2}}(\hat{a}_i - \hat{a}_i^\dagger), \quad (18)$$

and analogously for the $\hat{b}$ -modes. Manipulating these operators through nonlinear interactions enables the generation of polarization-squeezed entangled states, which are pivotal for quantum information protocols involving continuous variables and quantum communication.

In optical fiber consider an optical fiber with two orthogonal polarization axes. When light propagates through this fiber, it splits into two identical squeezed beams along these polarization axes. Combining these beams in a Stokes measurement results in polarization squeezing. The degree of squeezing, measured in decibels, can reach significant levels without the need for frequent adjustments [26].

Traditionally, polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. However, recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain. Researchers have achieved long-term stability with more than 5 dB of squeezing, making it applicable to various quantum applications [27].

Such squeezed light sources are essential for entanglement generation and quantum error correction. Particularly, fiber-coupled squeezed light sources with high squeezing levels and stable photon parity in high-photon-number regimes are highly desirable. For large-scale fault-tolerant quantum computing, a squeezing level exceeding 65 percent is necessary to produce time-domain multiplexed entangled states such as two-dimensional cluster states. However, achieving high-quality squeezed light in fiber-coupled systems remains a significant challenge.

Recent advancements have demonstrated the development of a novel optical fiber-coupled quantum light source operating at telecommunications wavelengths. By integrating this source with optical fiber components, continuous-wave squeezed light with more than 75 percent quantum noise suppression over a bandwidth exceeding 6 THz has been generated within an all-fiber system. This achievement signifies the first realization of a broadband, fiber-compatible squeezed light source capable of supporting large-scale optical quantum computing in a stable, maintenance-free configuration. Such a device paves the way for the development of rack-sized, scalable, and robust optical quantum computers, leveraging existing optical communication infrastructure.

These findings are documented in the December 22, 2021, publication in *Applied Physics Letters*, which was designated as an “Editor’s Pick.” Supported in part by the Japan Science and Technology Agency (JST) Moonshot RD Program, this work represents a significant milestone toward practical, large-scale fault-tolerant optical quantum networks.

Background and Technical Progress

Global efforts are underway to develop universal quantum computers using nonlinear quantum optics. While superconducting circuits have demonstrated operations with approximately 100 qubits, achieving fault-tolerance requires scaling to around one million qubits—a formidable challenge. Efforts to increase qubit numbers through device integration and parallelization are ongoing for certain platforms. Conversely, optical quantum computing offers a promising alternative by utilizing time-domain multiplexing and measurement-induced quantum operations, which enable the construction of large-scale entangled states without proportional increases in physical hardware, the network itself by extension fills the gap left by the limits of physical hardware.

Specifically, time-domain multiplexing involves dividing continuous light into discrete time segments, encoding quantum information in separated pulses, thus enabling scalable qubit numbers along the temporal axis. Furthermore, generating large-scale entangled states—such as two-dimensional cluster states—can be achieved with just a handful of squeezed light sources, optical delay lines, and beam splitters. This approach circumvents the need for extensive integration or large-scale parallelization, offering a practical pathway toward universal quantum computation with manageable equipment.

Quantum error correction schemes based on photon parity and continuous variables, by Valentina Parigi et al in research on CV quantum networks at telecommunication wavelengths has also revealed that even a single nonlinear crystal can create many-body bosonic modes that can be treated as a networked cluster state.

Using this one can also borrow stabilization techniques from the quantum computing literature to further enhance the robustness of fiber optical quantum networks. This brings about some familiar stabilization techniques, involving surface codes, along with new interpretations involving synchronization used in classical networks.

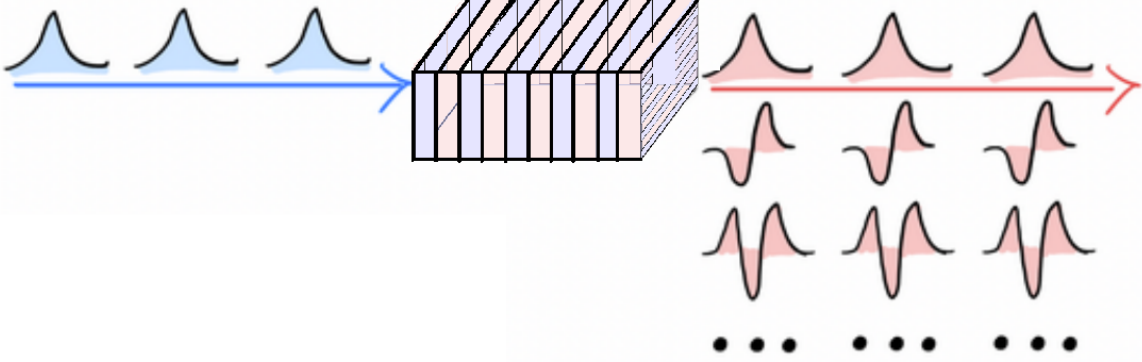


Figure 7: Schematic of a CV entanglement source as used by Parigi et al in developing networked cluster states using bosonic modes

11 Quantum Stabilization: The GKP Code

The Gottesman-Kitaev-Preskill (GKP) code is a quantum error-correcting code that encodes quantum information into an infinite-dimensional space using continuous-variable systems [28]. In this code, a qubit is encoded in a two-dimensional subspace of an infinite-dimensional oscillator space by leveraging the properties of continuous observables and their corresponding operators.

The basic idea behind the GKP code is rooted in the properties of conjugate continuous observables, such as position q and momentum p , which satisfy the Heisenberg uncertainty principle preventing their exact simultaneous measurement [29]. However, the GKP code overcomes this limitation by introducing two operators, $\hat{S}_q$ and $\hat{S}_p$, which are periodic functions of q and p . These operators commute with each other, allowing for their joint measurement and the simultaneous encoding of quantum information.

The GKP states offer advantages in fault-tolerant quantum computing and quantum error correction by enabling the manipulation of quantum information in continuous-variable systems [30]. Their use of conjugate observables and periodic operators provides a novel approach to encoding quantum information in infinite-dimensional spaces, paving the way for advancements in quantum information processing.

The operators are defined as follows:

$$\hat{S}_q = e^{i2\sqrt{\pi}\hat{q}} \quad (19)$$

$$\hat{S}_p = e^{-i2\sqrt{\pi}\hat{p}} \quad (20)$$

11.1 Encoding Quantum Information

By stabilizing the qubit subspace using the operators, the GKP code encodes quantum information in a structured manner within an infinite-dimensional oscillator space.

To examine the dynamics of polarization-entangled states using GKP-encoded qubits, we can represent the total Hamiltonian of the system as:

$$\hat{H} = \hat{H}_0 + \hat{H}_{int}, \quad (21)$$

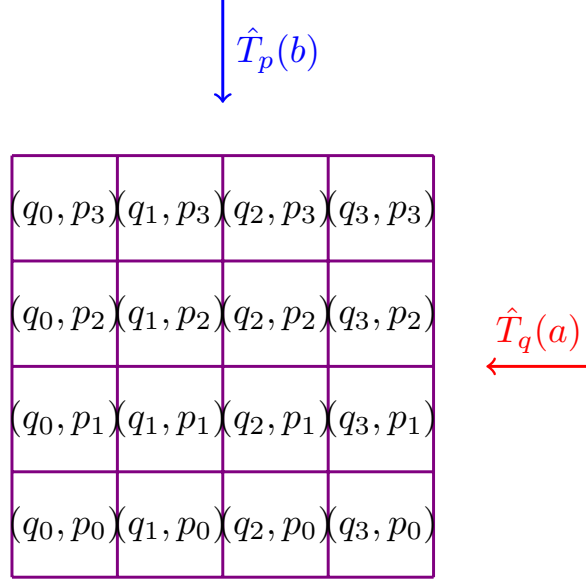


Figure 8: An associated labeled grid network diagram using the GKP states: The GKP states can be visualized as an infinite, periodic lattice structure characterized by the spacing $\sqrt{\pi}\sigma$ in position space, resulting in a periodicity in both q and p . If we denote the lattice points by (q_n, p_m) , we find that $q_n = n\sqrt{\pi}\sigma$, $p_m = m\sqrt{\frac{\pi\hbar}{\sigma}}$, (where n, m are positive and real).

where $\hat{H}_0$ is the Hamiltonian governing the individual oscillators and $\hat{H}_{int}$ represents the interaction terms that couple the qubits.

The time evolution of these states can be described using the Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (22)$$

Using the phase operators $\hat{S}_q$ and $\hat{S}_p$, we can analyze how the GKP states evolve over time under phase rotations and examine the resulting characteristics of the entangled qubits.

The GKP states can be visualized as an infinite, periodic lattice structure in the phase space [?], characterized by the spacing $\sqrt{\pi}\sigma$ in position space, resulting in a periodicity in both q and p . If we denote the lattice points by (q_n, p_m) , we find that:

$$q_n = n\sqrt{\pi}\sigma, \quad (23)$$

$$p_m = m\frac{\sqrt{\pi}\hbar}{\sigma}, \quad (24)$$

(where n, m are positive and real).

The GKP states are robust against certain types of errors due in large part to their periodic nature. The logical operators that act on the GKP states can be represented in the phase space as translations:

$$\hat{T}_q(a) = e^{-\frac{ia}{\hbar}\hat{p}}, \quad (25)$$

$$\hat{T}_p(b) = e^{-\frac{ib}{\hbar}\hat{q}}, \quad (26)$$

which further interact with the lattice structure by adding or shifting these grid points.

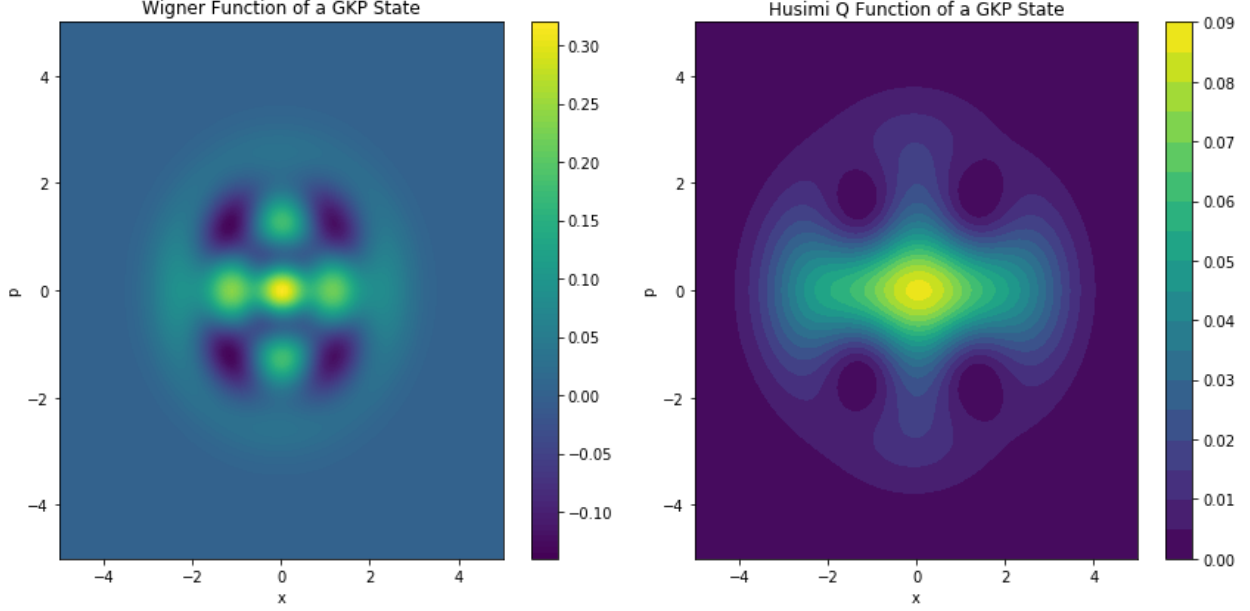


Figure 9: Wigner and Husimi phase space functions of GKP states. Structured representation of quantum information in the phase space offers promising applications in fault-tolerant quantum computation and quantum error correction.

The GKP qubit can be an ancilla or "aid" qubit in stabilizing network structure as is used in some quantum error correction schemes in computation involving GKP encoding.

12 Quantum GKP Encoding

Similarly to quadrature amplitude modulation (QAM) encoding used in classical communication, we may use the quantum GKP encoding to correct loss errors. The basic idea behind the GKP code [28] is that while the exact value of two conjugate continuous observables cannot be measured simultaneously, the two operators

$$\hat{q} = \text{quadrature operator} \quad (27)$$

and

$$\hat{p} = \text{momentum operator} \quad (28)$$

which are periodic functions of $\hat{q}$ and $\hat{p}$, commute with each other and therefore can be measured simultaneously. The GKP code thus encodes a qubit in a two-dimensional subspace of an infinite-dimensional oscillator space. This subspace is stabilised by these two operators, and the GKP state can be visualised as an infinite, periodic grid structure in the (q, p) phase space. For the GKP code based on a square lattice, which we will consider here and in our simulation (describing the square lattice of oscillators as a Kuramoto Quantum Oscillator system), the standard basis states are given as:

$$|0\rangle = \sum_{n=-\infty}^{\infty} |q = 2n\sqrt{\pi l}\rangle \quad (29)$$

Similarly, the GKP X basis logical states are:

$$|+\rangle = \sum_{n=-\infty}^{\infty} e^{i\pi np/2} |q = 2n\sqrt{\pi l}\rangle \quad (30)$$

We can see that the grid corresponding to the basis state $|0\rangle$ is shifted by Δq along the $\hat{p}$ quadrature with respect to $\hat{q}$. Hence, by measuring the two stabilisers, which amounts to measuring both $\hat{q}$ and $\hat{p}$ quadratures of the GKP state modulo $\sqrt{\pi l}$, we can detect and correct any small shifts (of size smaller than Δq) in both quadratures, in a way that does not reveal the encoded logical information.

The two GKP stabilisers can, in fact, be measured using additional GKP ancilla modes through a Steane error-correction process. Application of a two-mode operation between the GKP data mode and the GKP ancilla can transfer the information about the noise from the data qubit onto the ancilla in such a way that the logical information is not revealed. Hence, measuring the ancilla and applying a feedback displacement based on the measurement outcome enables GKP quantum error correction.

13 Phase-Insensitive Amplification (PIA) on GKP Qubits

In the considered architecture, quantum information encoded in the GKP qubits is sent through the repeater chain as follows. After Alice performs the encoding operation, she applies the phase-insensitive amplification and sends the GKP qubit through the lossy channel towards the first GKP repeater. The repeater performs GKP correction first in $\hat{q}$ and then in $\hat{p}$ quadrature. After that, it again applies the phase-insensitive amplification and sends the state to the next repeater. In this way, the encoded qubit can effectively be transmitted to Bob. In our model, we consider two sources of imperfections apart from loss in the communication channel. Firstly, we assume a finite photon in-coupling and out-coupling efficiency η_0 , which quantifies the efficiency of transferring the photon from the fiber to the repeater and back into the fiber. Hence, the total transmissivity of the lossy channel between two neighbouring repeaters separated by the distance L is:

$$T(L) = \exp\left(-\frac{L}{L_0}\right) \quad (31)$$

where L_0 is the attenuation length of the channel. Here, we assume transmission at telecom frequency at which $L_0 = 22$ km. The second imperfection we consider is the finite amount of GKP squeezing. Under finite squeezing, the GKP grid does not consist of delta functions but of Gaussian peaks with an overlying envelope function, such that the peaks in the Wigner function decay to zero height in the limit of large quadrature values. The standard deviation of these finitely squeezed Gaussian peaks is given by σ_{GKP} , and the amount of squeezing can also be quantified by comparing σ_{GKP} to the standard deviation of a Gaussian peak of a coherent state given by σ_{coh} . Hence, squeezing expressed in dB can be defined as:

$$\text{Squeezing (dB)} = 10 \log_{10} \left(\frac{\sigma_{\text{coh}}}{\sigma_{\text{GKP}}} \right) \quad (32)$$

In our analysis, similarly to [28], we consider a conservative error model that allows us to describe a finitely squeezed GKP qubit as an ideal GKP state subjected to a random displacement according to a probability distribution parameterised by σ_{GKP} . Since both finite GKP squeezing and the channel noise lead now to random displacement errors, we can reliably approximate the repeater performance by considering perfect error correction using infinitely squeezed ancillas, which, however, is now performed on the data qubits subjected to an effective communication channel. This effective channel now includes not only the noise coming from η defined in Eq. (4) but also from non-zero σ_{GKP} . That is, we consider an approximation in which the noise from finite squeezing can be effectively incorporated into the channel and combined with the noise due to photon loss. This approximation enables us to construct a simple analytical model through which we can efficiently evaluate repeater performance, including optimisation over repeater spacing. We validate this analytical model against a numerical Monte Carlo simulation.

14 Self-Stabilizing Qubits

The concept of self-stabilizing qubits involves using polarization-squeezed entanglement as resources for stabilizing quantum information encoded in continuous variables. Here, we discuss the implementation of the Gottesman-Kitaev-Preskill (GKP) quantum error correction code for error correction in continuous-variable quantum systems.

14.1 GKP Encoding

The GKP encoding scheme encodes quantum information in a two-dimensional subspace of an infinite-dimensional oscillator space by exploiting the commutation relations of two conjugate continuous observables. The GKP code stabilizes the encoded qubit against small displacements by measuring two stabilizer observables. The GKP basis states are given as follows:

$$|0\rangle = \sum_{n=-\infty}^{\infty} |q = 2n\sqrt{\pi l}\rangle \quad (33)$$

The GKP X basis logical states are:

$$|+\rangle = \sum_{n=-\infty}^{\infty} e^{i\pi np/2} |q = 2n\sqrt{\pi l}\rangle \quad (34)$$

15 GKP Quantum Error Correction Methods applied to Networking or Fault-Tolerance

Error correction in GKP qubits involves interaction between data and ancilla modes, followed by measurement to detect errors without revealing encoded information.

an auxiliary GKP qubit introduced into a quantum error correction or encoding scheme can assist in maintaining the stabilizer structure while expanding the accessible state space. Unlike the primary logical qubits that encode the information in quantum computing, the aid qubit is not directly measured during the error detection process. Instead, it serves as a resource to facilitate more robust interactions and operations, helping to preserve the error-detecting properties of the stabilizer code.

This design allows the system to detect and potentially correct errors more effectively by expanding the overall state space, providing additional degrees of freedom for encoding and error management. The aid qubit's unmeasured status ensures that it does not collapse the quantum state prematurely, thereby maintaining coherence and enabling more resilient quantum interaction encoding schemes.

This network illustrates that an error (shift) occurs on the data mode, which interacts with an ancilla mode via a two-mode operation. Measuring the ancilla provides syndrome information for error detection and correction, all while preserving the encoded quantum information.

16 Simulation of GKP in Phase Space using a Grid of Kuramoto Oscillators

The GKP-encoded qubits from two Entanglement Resources, Rho and Sigma, are sent through a series of PM fibers, a combiner, and a splitter by performing GKP correction and phase-insensitive amplification at each repeater station to combat losses in the communication channel. Imperfections such as finite photon

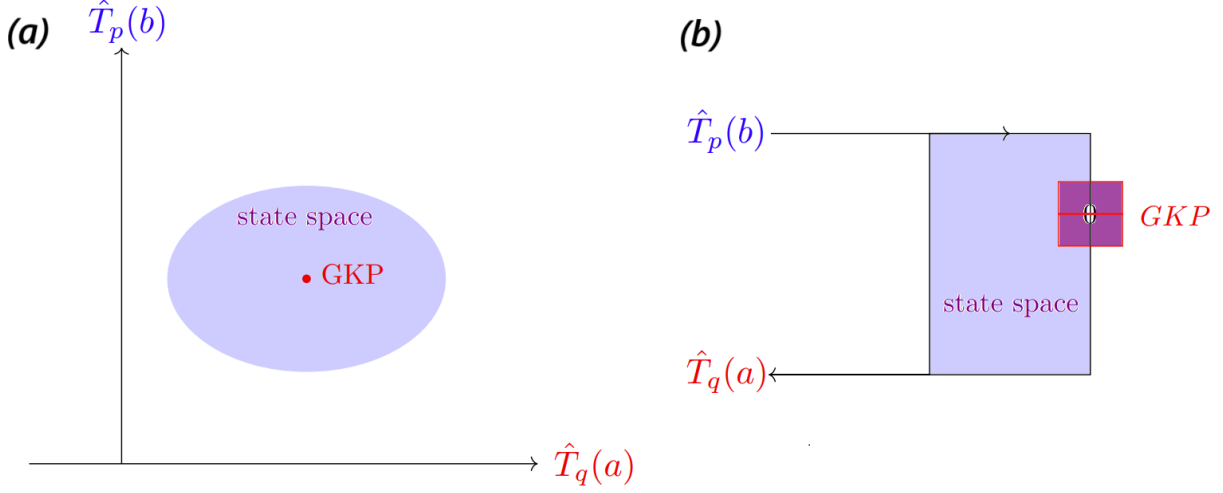


Figure 10: An auxiliary GKP qubit (red) introduced into a quantum error correction or encoding scheme can assist in maintaining the stabilizer structure while expanding the accessible state space (purple) in fig (a). Unlike the primary logical qubits that encode the information in quantum computing, the aid qubit is not directly measured during the error detection process in fig (b). Instead, it serves as a resource to facilitate more robust interactions and operations, helping to preserve the error-detecting properties of the stabilizer code.

in-coupling efficiency and finite GKP squeezing are considered in the model. The transmissivity of the lossy channel between repeaters separated by distance L is given by:

$$T(L) = \exp\left(-\frac{L}{L_0}\right) \quad (35)$$

where L_0 is the attenuation length of the channel. The amount of squeezing in dB is calculated as:

$$\text{Relative Squeezing (dB)} = 10 \log_{10} \left(\frac{\rho_{\text{GKP}}}{\sqrt{2}} \right) \quad (36)$$

In our scenario, we associate the phase of each Kuramoto oscillator with the phase of the GKP states that are transmitted through the PM fiber. The model can be described as a network graph of oscillators using the following equations.

Network Graph of Quantum Harmonic Oscillators as a Reduced Density Matrix

Overall, it is possible to represent a network with given adjacency and network Laplacian matrices as a reduced cavity density matrix. From this, it is possible to compute an operational meaning of the trace distance between the EPR pair as the source density matrix and the network to which the density matrix is operationally coupled. First, let us create a reduced cavity density matrix from a network Laplacian. Given a network graph G :

$$M(G)_{(i,j),k} = \begin{cases} 1 & \text{if } v_k = v_i \\ -1 & \text{if } v_k = v_j \\ 0 & \text{otherwise} \end{cases} \quad (37)$$

We get the following formula to retrieve a density matrix given a graph Laplacian L :

$$\rho(G) := \frac{L(G)}{\text{Tr}[L(G)]} \quad (38)$$

which can be expressed in terms of Z as:

$$\tilde{\rho}(G) := \frac{e^{-\tau L(G)}}{Z} \quad (39)$$

where

$$Z = \text{Tr} \left[e^{-\tau L(G)} \right] \quad (40)$$

Oscillator Node Dynamics

Each Kuramoto oscillator (representing a GKP state) evolves according to the following equation:

$$\frac{d\phi_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\phi_j - \phi_i) - \gamma \phi_i \quad (41)$$

where:

- ϕ_i is the phase of the i -th oscillator,
- ω_i is the natural frequency of the i -th oscillator,
- K is the coupling strength between the oscillators,
- N is the total number of oscillators,
- γ represents a damping factor accounting for channel losses.

Transmissivity Incorporation

Incorporate the effect of transmissivity $T(L)$ into the oscillator dynamics, reflecting how these oscillators' coupling is affected by the distance L :

$$K(L) = K_0 \cdot T(L) = K_0 \cdot \exp \left(-\frac{L}{L_0} \right) \quad (42)$$

where K_0 is the original coupling strength without distance attenuation.

16.1 Inclusion of Squeezing

The effect of relative squeezing can alter the natural frequencies of the oscillators. A squeezed oscillator can be modeled as having its natural frequency adjusted based on the relative squeezing:

$$\omega'_i = \omega_i + R_i \quad (43)$$

with the squeezing term given by the formula:

$$R_i = -\frac{\kappa}{2} \cdot \text{RelativeSqueezing}(dB) \quad (44)$$

where κ is a scaling factor reflecting how squeezing influences the frequency of the oscillators.

High-Dimensional Entanglement

If we have multiple distances (repeaters) and encodings, we can model this in a higher-dimensional phase space: These logical operators interact with the lattice architecture of GKP states, facilitating the addition or shifting of grid points within the lattice framework. Consequently, the action of these operators can be conceptualized in terms of oscillatory behaviors, analogous to the dynamics observed in higher-order networks of Kuramoto oscillators.

In the context of GKP surface codes, we define a higher-dimensional phase space that accommodates multiple distances and encoding strategies. The dynamics of this higher-order Kuramoto system can be expressed as follows:

$$\frac{d\phi_{i,j}}{dt} = \omega'_{i,j} + \frac{K(L_j)}{N} \sum_{k=1}^N \sin(\phi_{k,j} - \phi_{i,j}) - \gamma\phi_{i,j} \quad (45)$$

Here, $\phi_{i,j}$ represents the phase of the i -th oscillator situated at the j -th repeater, where each oscillator is tied to the intricate structure of the GKP lattice.

Importantly, this higher-order Kuramoto model not only captures the synchronization dynamics among oscillators affiliated with nodes (panel (a)) but also extends to the synchronization of topological signals within simplicial complexes (panel (b)). In this vein, oscillators represent not just nodes but also higher-dimensional simplices—such as links and triangles—that characterize a type of surface code in a GKP configuration.

This framework unveils an interplay within topological signals of varying dimensions, revealing their capacity to coexist and engage in non-trivial coupling. Such dynamics can lead to simultaneous explosive transitions, akin to the percolation thresholds observed in classical networks. These phenomena underscore the intricate connections between complex systems, where topology governs the emergent behavior, showcasing how GKP states and their logical operators can serve as powerful models for exploring synchronization and phase transitions in multifaceted systems.

16.2 Experimental Transmission through a PM Fiber System

The use of polarization squeezing for self-stabilizing qubits in continuous-variable quantum systems shows promise for quantum error correction and communication applications. By incorporating finite squeezing and channel imperfections into the analysis, we can optimize tomography performance in an operationally realistic scenario.

In the context of polarization squeezing, the focus is on the nonlinearity in optical crystals such as BBO and ppKTP in nodal cavities in a network, but may also focus on Kerr nonlinearities in active optical fibers. Both of these enable the creation of polarization-squeezed states in a network.

Consider an optical fiber with two orthogonal polarization axes. When light propagates through this fiber, it splits into two identical squeezed beams along these polarization axes. Combining these beams in a Stokes measurement results in polarization squeezing. The degree of squeezing achieved can be quantified in decibels (dB). Researchers have reported more than 5 dB of directly measured squeezing using robust setups without the need for frequent adjustments.

16.2.1 Robust All-Fiber Setup

Traditionally, polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. Recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain using GKP coded states.

The proposed experiment can leverage a polarization-squeezed state generated from a BBO Spontaneous Parametric Down-Conversion (SPDC) source of light

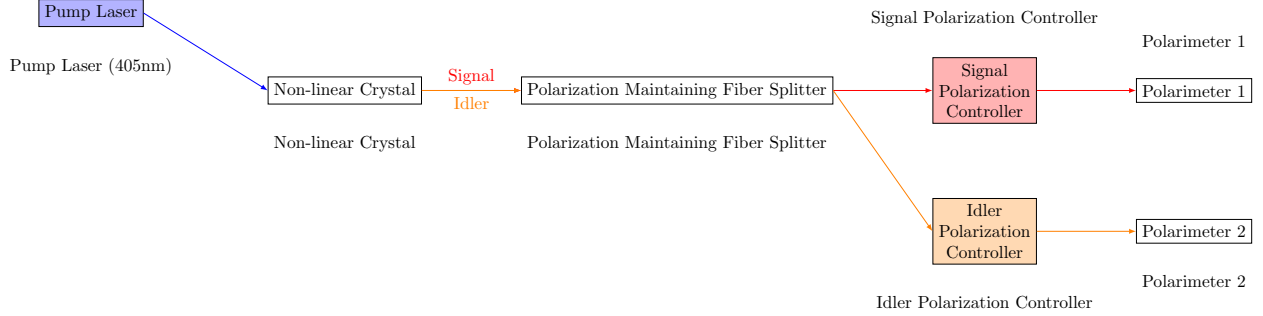


Figure 11: Polarization Squeezed Entangled Light Source in a Robust All-fiber setup with polarization control and characterization

This setup is incorporated into an optical fiber interferometer. The characteristics of the interferometer will allow for the measurement of the polarization states and enable the exploration of the effects of polarization squeezing within the fiber system.

The dynamic logical operators, represented as translations in phase space, can be seen as driving forces within the Kuramoto model. Their interaction with the lattice structure, analogous to adding or shifting grid points, affects the synchronization dynamics of the oscillators. This synchronization, in turn, is directly related to the error correction properties of the GKP states. Emergence of Complex Behavior

The higher-order Kuramoto model unveils a fascinating aspect of GKP error correction: the co-existence and co-evolution of topological signals of different dimensions. This intricate interplay can lead to "simultaneous explosive transitions," reminiscent of percolation thresholds observed in critical behaviors within classical networks. These transitions signify a sudden shift in the system's dynamics, potentially leading to the emergence of robust error correction properties.

The higher-order Kuramoto model effectively bridges the gap between complex systems phenomena and the intricate dynamics of GKP state error correction. By simulating the synchronization of topological signals on a simplicial complex, it captures the emergence of collective behavior and its relationship to the stability and resilience of these states. This approach offers a novel and powerful perspective on understanding the complex interplay of logical operators, surface codes, and error correction within GKP systems.

Future research could explore the detailed mapping between the dynamics of the higher-order Kuramoto model and specific error correction protocols for GKP states. Investigating the impact of different system parameters, such as coupling strength and network topology, on the synchronization behavior and error correction performance would be particularly insightful. This deeper understanding could pave the way for the development of more efficient and robust error correction strategies for fault-tolerant quantum computation based on GKP states.

16.3 Superactivation of Quantum Channels

A particularly fascinating phenomenon is **superactivation**, where two quantum channels—each individually having zero quantum capacity—can, when used together, activate one another to achieve a non-zero capacity.

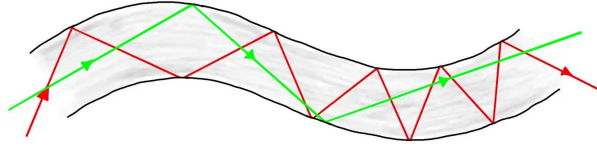


Figure 12: Bosonic modes i.e. Light rays travelling through an optical fibre can take different paths; since each path is associated to a specific phase and/or polarization, not knowing the path taken amounts to an effective dephasing noise and/or polarization extinction

Formally, this implies:

$$\text{If channels } \mathcal{N}_1, \mathcal{N}_2 \text{ are individually zero-capacity, then } Q(\mathcal{N}_1 \otimes \mathcal{N}_2) > 0,$$

highlighting the non-trivial interplay between channels.

16.4 Entanglement-Binding and Entanglement-Breaking Channels

If one considers a channel that is entanglement-breaking, and mixes it with an unknown channel (with unknown isometries), the superactivation effect suggests that the other channel must be *entanglement-binding* in some way. This supports the conjecture that certain hidden channels possess intrinsic binding properties preventing entanglement distillation, as discussed in Mark Wilde's work.

a channel that is *entanglement-breaking*, cannot preserve any entanglement with other systems. Such channels are a subset of *quantum channels* that destroy all entanglement passing through them, effectively transmitting only classical information.

An intriguing aspect arises when these channels are combined with unknown or hidden channels (referred to as *mystery channels*). The phenomenon of **superactivation** suggests that if a superactivation occurs with an entanglement-breaking channel and an unknown channel, then the unknown channel must be *entanglement-binding*. These *entanglement-binding* channels are characterized by their intrinsic inability to distill entanglement, supporting the conjecture that they serve as the "hidden" resources preventing classical simulation and enabling quantum capacity under certain conditions.

17 Emergent Memory in General Networks

Networks can exhibit forms of intrinsic or emergent memory, wherein collective properties arise from the interactions among simpler components such as neurons, leading to the capacity to store and retrieve information not explicitly embedded in individual units. This phenomenon underpins many cognitive and computational functions in biological and artificial systems, allowing for complex information processing beyond the capabilities of individual components [4].

17.1 Definition of Emergent Memory

Emergent memory refers to the ability of a network to retain information over time, originating from the internal dynamics and interactions within the network rather than from any single element. Unlike simple storage mechanisms, emergent memory arises through collective behavior, leading to phenomena such as attractor states, persistent activity, or non-Markovian dynamics [5]. Formally, if we denote the state of the network at time t as $\mathbf{X}(t)$, then the emergent memory manifests as the dependence of $\mathbf{X}(t)$ on the history $\mathbf{X}(t' < t)$, often characterized by long-range correlations:

$$\text{Memory} \sim \int_{-\infty}^t K(t - t') \mathbf{X}(t') dt' \quad (46)$$

where $K(\tau)$ is a kernel function encoding the memory decay or persistence [?].

17.2 Types of Networks Exhibiting Emergent Memory

Various network architectures demonstrate emergent memory through different mechanisms:

17.2.1 Neural Networks

Biological and artificial neural networks process and store information via spiking activity and synaptic plasticity mechanisms such as long-term potentiation (LTP) and long-term depression (LTD) [?]. These networks can exhibit *spatiotemporal memory*, maintaining stimulus-specific information over several seconds:

$$\Delta w_{ij} = \eta \text{pre}_i \text{post}_j \quad (47)$$

where w_{ij} is the synaptic weight between neurons i and j , and η is the learning rate [8].

17.2.2 Stochastic Networks

In stochastic or probabilistic networks, memory emerges from the dynamics of state transitions. When the transition rates between states are finite, the system exhibits a non-Markovian process where the transition probability depends on the elapsed time since the last transition [6]. The master equation describing state probability $P_i(t)$ can be written as:

$$\frac{dP_i(t)}{dt} = \sum_j [W_{ji}P_j(t) - W_{ij}P_i(t)] \quad (48)$$

where W_{ij} are the transition rates, and finite transition times induce a memory effect.

17.2.3 Large-Scale Brain Networks

Brain networks composed of multiple interconnected regions utilize functional connectivity to retain information [7]. Synchronization of activity across regions enables the network to perform cognitive functions like memory retrieval, with the collective dynamics supporting emergent memory:

$$\text{FC}_{ij}(t) = \langle x_i(t)x_j(t) \rangle \quad (49)$$

where $x_i(t)$ and $x_j(t)$ are activity signals from regions i and j , and $\langle \cdot \rangle$ denotes averaging over time.

17.3 Mechanisms Underlying Emergent Memory

Several mechanisms facilitate emergent memory within networks:

Spiking Activity In neural networks, the pattern and timing of spikes can encode and sustain information, creating a fading memory that persists over relevant timescales [9]:

$$\text{Fading Memory} \sim \sum_{k=0}^{\infty} \alpha^k \mathbf{u}(t - k) \quad (50)$$

where α is a decay factor and $\mathbf{u}(t)$ is the input stimulus.

Transition Dynamics In stochastic systems, the finite duration of transitions between states introduces a form of kinetic hysteresis, whereby the system’s past influences its future [1]:

$$\text{Memory Effect} \propto \tau_{\text{transition}} \quad (51)$$

with $\tau_{\text{transition}}$ being the average transition time.

Functional Connectivity In large-scale networks, the interplay between regions enables the retention and manipulation of information, effectively supporting cognitive functions such as attention and working memory [10].

17.4 Implications for Quantum Networks

Emergent memory is a fundamental feature that enhances the computational and adaptive capabilities of quantum computing. It allows for complex temporal processing, pattern recognition, and decision-making. Recent studies have explored phenomena such as emergent oscillations in strongly driven networks [1] and the role of kinetic hysteresis in neural dynamics [2]. Understanding these mechanisms deepens our insight into both biological cognition and the development of adaptive quantum systems.

18 Quantum Memories and Their Certification in Quantum Networks

Quantum memories are vital for quantum networks, enabling storage and retrieval of quantum information over time. A key challenge is verifying their truly quantum nature—specifically, whether they preserve quantum entanglement during storage.

Recent advances, such as the measurement-device-independent quantum memory test developed by Rosset, Buscemi, and Liang, provide a faithful and measurement-device-independent method to certify quantum memories without making assumptions about the measurement devices. This test relies on a *single-character* protocol, where a trusted state preparation device (Abby) acts as both sender and receiver at different times, allowing the estimation of *time-like entanglement* and certifying that the memory preserves quantum correlations.

In the context of quantum channels, if a quantum memory acts as an *entanglement-breaking channel*, it ceases to maintain entanglement and can only transmit classical information. Therefore, certifying the quantum nature of memories involves verifying that the channel is *not* entanglement-breaking.

18.1 Visualization and Analysis Techniques

Graphical methods, such as computing the transformation matrix $X(\chi)$, facilitate the visualization of quantum states and the assessment of channel properties, including the preservation of entanglement. These techniques are instrumental in understanding the structure of quantum memories and their behavior within a network.

18.2 Quantum Graph Network Memories in Superactivation Channels

The interplay between quantum memories and channels—particularly entanglement-breaking and entanglement-binding channels—can influence the overall capacity of quantum networks. For example, the presence of entanglement-binding channels within a network may enable superactivation effects, where the combined use

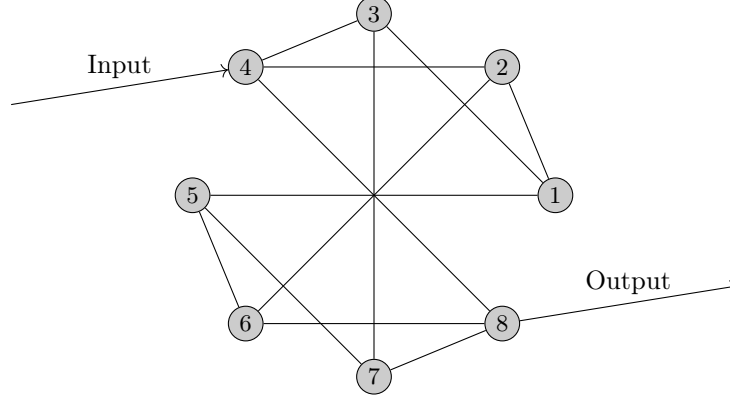


Figure 13: *

- (i) Quantum network memory represented as a DAG, illustrating the entangled connections within the memory system.

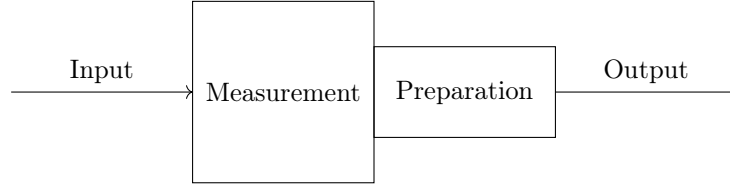


Figure 14: *

- (ii) Circuit depicting measurement followed by state preparation (corresponding to entanglement-breaking channels) such a memory is not genuinely quantum.

of multiple channels or memory devices results in non-zero quantum capacity, despite the individual channels being limited or classical in nature.

Beyond the conventional view of quantum memories as static storage devices, recent research suggests that quantum networks themselves can exhibit memory-like functionalities arising from their intrinsic dynamics. In particular, phenomena such as hysteresis—commonly observed in classical magnetic systems and neural networks—can be extended into the quantum domain, enabling quantum networks to act as emergent memories rooted in their collective behavior.

18.2.1 Quantum Networks as Hysteretic Systems

Hysteresis refers to the dependence of a system's current state on its history, often manifesting as a lag between input stimuli and the system's response. In classical systems, hysteresis is characterized by a loop in the output-input relation, such as in magnetic materials:

$$M(H) = M_0 \tanh \left(\frac{H + \lambda M}{H_c} \right), \quad (52)$$

where

M

is the magnetization,

H

is the applied magnetic field,

λ

is a coupling parameter, and

$$H_c$$

is the coercive field.

Analogously, in quantum networks, collective interactions among qubits or oscillators can produce a form of phase hysteresis. For example, networks of coupled quantum oscillators or spins can exhibit bistabilities and memory effects due to their intrinsic nonlinear dynamics. These emergent properties can be described by mean-field equations akin to classical hysteresis models, but with quantum corrections:

$$\langle \hat{\sigma}_z \rangle = \tanh [\beta (\Omega + J \langle \hat{\sigma}_z \rangle)], \quad (53)$$

where

$$\hat{\sigma}_z$$

is the Pauli-

$$z$$

operator,

$$\Omega$$

is an external field,

$$J$$

is the interaction strength, and

$$\beta = 1/(k_B T)$$

is the inverse temperature. Such self-consistent equations can lead to multiple stable solutions, enabling the quantum network to "remember" prior configurations through phase states.

18.2.2 Phase Memory in Kuramoto-Type Quantum Oscillator Networks

Kuramoto models, originally classical, have been extended into the quantum regime to study synchronization phenomena among quantum oscillators. In these models, phase synchronization emerges from the interplay between oscillators' natural frequencies and their couplings, often resulting in multistability and hysteresis in the phase dynamics:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad (54)$$

where

$$\theta_i$$

are the phase operators associated with each oscillator, and their expectation values can be considered in the quantum framework. When quantum fluctuations and dissipation are included, the phase dynamics can retain memory of previous states, akin to classical phase hysteresis, thus facilitating the formation of a quantum "phase memory" (see, e.g., [?], Phys. Rev. Lett.).

18.2.3 Quantum Network-Based Memory via Hysteresis

In quantum networks, such hysteretic or phase-based memory effects can be formalized through the evolution of the many-body density matrix

$$\rho$$

governed by a nonlinear or feedback-influenced Hamiltonian

$$H(\rho)$$

:

$$\frac{d\rho}{dt} = -i[H(\rho), \rho] + \mathcal{L}(\rho), \quad (55)$$

where

$$\mathcal{L}(\rho)$$

represents dissipative Lindblad processes. The nonlinear dependence of

$$H(\rho)$$

on

$$\rho$$

can induce multistability and hysteresis, effectively encoding information in the system's collective phase or state. Such a quantum network can serve as a form of memory without dedicated quantum storage elements, leveraging its intrinsic dynamical properties.

18.2.4 Emergent Memory and Intrinsic Hysteresis as a Resource

This perspective aligns with recent notions of dynamical quantum memories [Liu et al., 2021, Nature Physics], where the memory is not stored in static states but is an emergent property of the network's history-dependent dynamics. Furthermore, the analogy with neural networks—where synaptic weights and network activity encode memories—can be extended to quantum networks, with phase relations or collective states acting as the "weights" that preserve information over time.

Harnessing intrinsic hysteresis and phase memory in quantum networks opens avenues for robust, self-sustaining quantum memories that leverage their collective dynamics.

Understanding the nature of quantum memories and their certification in this context is crucial for designing robust quantum networks. Moreover Ensuring that memories are not entanglement-breaking preserves the quantum advantage, while the development of device-independent certification protocols enhances trustworthiness in practical implementations. Moreover, the study of superactivation and entanglement-binding channels informs strategies for harnessing hidden quantum resources within complex network architectures.

18.3 Graphical Methods for Quantum State Visualization

For quantum tomography and visualization, one can compute the transformation matrix $X(\chi)$, which maps an operator basis to a representation that interprets quantum distances or other metrics. Such matrices facilitate understanding the structure of quantum states and channels.

18.4 Quantum Homodyne Detection and Local Oscillators

Recent theoretical insights suggest that when Alice and Bob's local oscillators are modeled as distinct operators W and Z , the **Holevo information** across the channel can diminish or even vanish. Conversely, if the local oscillators are classically synchronized, they function effectively as a single super-operator, enabling quantum information sharing.

Moreover, if these oscillators update conditionally based on measurement outcomes, they can themselves carry quantum information, implying a form of shared quantum resource. This raises questions about potential *variational methods* for quantum hypothesis testing, possibly leveraging the Wigner function of quantum harmonic oscillators in phase space analysis.

18.5 Superoperator Monotonicity and Information Bounds

Superoperators $\mathcal{X}$, which are trace-preserving maps on operators, are known to *not* increase the Holevo information. This property can be shown via Lindblad operators, leading to the **monotonicity** of certain entropy measures, such as the von Neumann entropy $S(\rho)$.

In particular, the *strong subadditivity* of the von Neumann entropy, as discussed in Preskill's exercises, underpins that superoperators serve as upper bounds on the quantum information encoded in states, processes, or ensembles:

$$\chi(\mathcal{X}(\rho)) \leq \chi(\rho),$$

where χ denotes the Holevo quantity.

18.6 Implications for Quantum Networks and Synchronization

Superoperators can be conceptually modeled as networks of optical elements—mirrors, spatial light modulators (SLMs), and local oscillators—connected in a circuit-like architecture. Synchronization of these operators (nodes) can be envisioned via path integral formulations over the entire configuration space, effectively treating the entire quantum network as a superoperator.

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20 Quantum Networks as Density Matrices

The density matrix, also known as the density operator, is an alternative representation of the state of a quantum system, replacing the wavefunction. While describing a quantum system using the density matrix is equivalent to using the wavefunction, it offers practical advantages for certain time-dependent problems, particularly relaxation and nonlinear spectroscopy in the condensed phase .

20.1 Definition of the Density Matrix

The density matrix is defined as the outer product of the wavefunction with its conjugate:

$$\rho(t) \equiv |\psi(t)\rangle\langle\psi(t)| \quad (56)$$

This implies that if you specify a state $|x\rangle$, then $\langle x|\rho|x\rangle$ gives the probability of finding a particle in the state $|x\rangle$. Its name derives from the observation that it plays the quantum role of a probability density. If you think of the statistical description of a classical observable obtained from moments of a probability distribution P , then ρ plays the role of P in the quantum case:

$$\langle A \rangle = \int AP(A)dA \quad (57)$$

$$= \langle\psi|A|\psi\rangle = \text{Tr}[A\rho] \quad (58)$$

where $\text{Tr}[\dots]$ refers to tracing over the diagonal elements of the matrix:

$$\text{Tr}[\dots] = \sum_a \langle a|\dots|a\rangle \quad (59)$$

The last expression is obtained as follows. If the wavefunction for the system is expanded as:

$$|\psi(t)\rangle = \sum_n c_n(t)|n\rangle \quad (60)$$

the expectation value of an operator is:

$$\langle\hat{A}(t)\rangle = \sum_{n,m} c_n(t)c_m^*(t)\langle m|\hat{A}|n\rangle \quad (61)$$

Also, from Equation 56 we obtain the elements of the density matrix as:

$$\rho(t) = \sum_{n,m} c_n(t)c_m^*(t)|n\rangle\langle m| \equiv \sum_{n=m} \rho_{nm}(t)|n\rangle\langle m| \quad (62)$$

We see that ρ_{nm} , the density matrix elements, are made up of the time-evolving expansion coefficients. Substituting into Equation 61 we see that:

$$\langle \hat{A}(t) \rangle = \sum_{n,m} A_{mn} \rho_{mn}(t) = \text{Tr}[\hat{A}\rho(t)] \quad (63)$$

In practice, this makes evaluating expectation values as simple as tracing over a product of matrices.

20.2 Interpretation of the Density Matrix Elements

The density matrix elements ρ_{nm} contain important information about the quantum system. The diagonal elements ($n = m$) give the probability of occupying a quantum state:

$$\rho_{nn} = c_n c_n^* = p_n \geq 0 \quad (64)$$

For this reason, diagonal elements are referred to as populations. The off-diagonal elements ($n \neq m$) are complex and have a time-dependent phase factor:

$$\rho_{nm} = c_n(t) c_m^*(t) = c_n c_m^* e^{-i\omega_{mn}t} \quad (65)$$

Since these describe the coherent oscillatory behavior of coherent superpositions in the system, they are referred to as coherences.

21 Applications of the Density Matrix

The density matrix becomes a particularly important tool when dealing with mixed states, which are statistical mixtures in which we have imperfect information about the system and need to perform statistical averages to describe quantum observables. For mixed states, calculations with the density matrix are greatly simplified. Given that you have a statistical mixture and can describe p_k , the probability of occupying quantum state $|\psi_k\rangle$, evaluation of expectation values is simplified with a density matrix:

$$\langle \hat{A}(t) \rangle = \sum_k p_k \langle \psi_k(t) | \hat{A} | \psi_k(t) \rangle \quad (66)$$

$$\rho(t) \equiv \sum_k p_k |\psi_k(t)\rangle \langle \psi_k(t)| \quad (67)$$

$$\langle \hat{A}(t) \rangle = \text{Tr}[\hat{A}\rho(t)] \quad (68)$$

Evaluating the expectation value is the same for pure or mixed states.

21.1 Relating general graph theory to the density matrix description

By considering the density matrix of a many-body quantum network, we can extract valuable information about the system's graph representation. Specifically, we can construct graphs using the network Laplacian derived from the density matrix picture. The network Laplacian captures the connectivity and interactions between the constituent particles, allowing us to analyze the network's properties and study phenomena such as synchronization and entanglement.

To construct the graph, we start by defining a network adjacency matrix based on the relationships between the particles. This adjacency matrix encodes the connections between the particles, indicating whether they

are directly interacting or not. From this adjacency matrix, we can derive the network Laplacian matrix, which characterizes the overall dynamics and behavior of the quantum network.

First, let us create a reduced cavity density matrix from a network Laplacian. Given a network graph G :

$$M(G)_{(i,j),k} = \begin{cases} 1 & \text{if } v_k = v_i \\ -1 & \text{if } v_k = v_j \\ 0 & \text{otherwise} \end{cases}$$

We get the following formula to retrieve a density matrix given a graph Laplacian, L :

$$\rho(G) := \frac{L(G)}{\text{Tr}[L(G)]}$$

which can be written in terms of Z as:

$$\tilde{\rho}(G) := \frac{e^{-\tau L(G)}}{Z}$$

where

$$Z = \text{Tr} \left[e^{-\tau L(G)} \right]$$

Using the formula for the reduced cavity density matrix derived from the network Laplacian, we can obtain a density matrix representation of the many-body system. This density matrix provides a powerful tool to analyze the system's quantum correlations and study its entanglement properties. Moreover, the density matrix description allows us to compute operational quantities such as the trace distance between the source density matrix (e.g., a node that generates the source EPR pair) and the network to which it is operationally coupled.

22 Integration of EPR Resources as Nodes in a scalable Quantum Network

Implementing nodes that can act as EPR sources and network nodes (repeaters, clients, measuring devices and memories) on simple experimental circuitry would be the obvious next phase in development of this technology in tandem with computer simulation using tensor networks for real world applications in similar vein to how we have applied the toy-model of metaheuristic Boltzmann Machines to the problem of ring reconstruction, which could be used for polymer ring simulations.

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23 Part 2: Building a Quantum Optical Network

24 Quantum Optical Circuits

Quantum optical circuits utilize photonic systems to implement quantum information processing tasks. They are essential for scalable quantum computing and communication.

These circuits rely on components such as beam splitters, phase shifters, and single-photon sources. They exploit quantum interference and entanglement of photons.

24.1 Components and Operations

- **Beam Splitter:**

$$\hat{B}(\theta) = \exp [\theta(a^\dagger b - ab^\dagger)] \quad (69)$$

where

$$a^\dagger, b^\dagger$$

are creation operators.

- **Phase Shifter:**

$$\hat{P}(\phi) = \exp(i\phi\hat{n}) \quad (70)$$

with

$$\hat{n}$$

being the number operator.

24.2 Quantum Interference as an optical Gate

The Hong-Ou-Mandel effect illustrates two-photon interference, with the probability of both photons exiting different ports given by:

$$P_{\text{diff}} = \frac{1}{2} (1 - |\langle \psi_1 | \psi_2 \rangle|^2) \quad (71)$$

we can represent this as our most fundamental optical entanglement circuit:

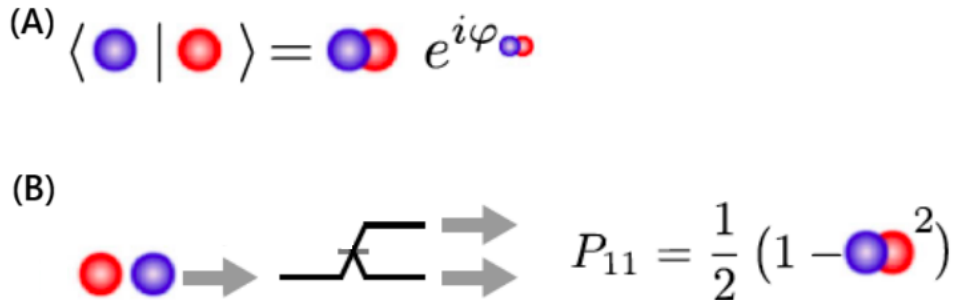


Figure 15: (B) Schematic of a passive fiber HOM circuit used for polarization entanglement state (A) containing horizontal and vertical states in superposition before entering the polarizing beamsplitter

An extension of the HOM interferometer for use in converting polarization entanglement into time-binned entanglement is the use of the Franson interferometer circa 1989

With stabilization in mind we can consider using fiber loops as the 2 arms of the Franson interferometer circuit - later these will be fiberised Sagnac loops which are the nodes we will consider when modeling emergent and induced synchronization throughout an optical entanglement network.

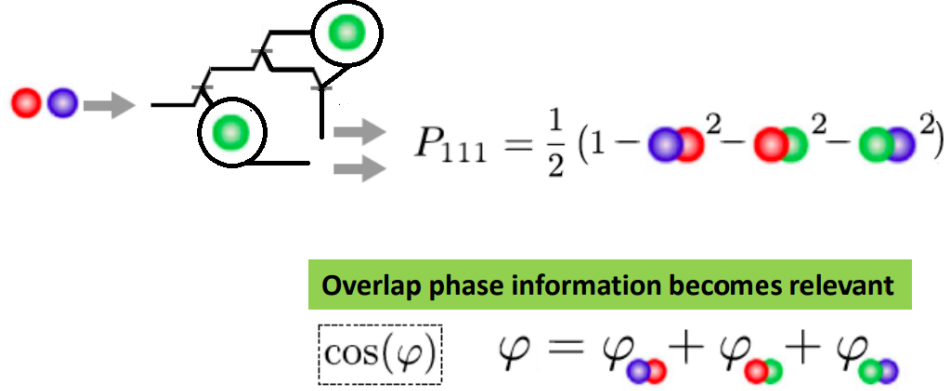


Figure 16: (B) Schematic of a passive fiber HOM circuit used for polarization entanglement state (A) containing horizontal and vertical states in superposition before entering the polarizing beamsplitter

Where the phase overlap becomes important these then means hardware nodes themselves may be considered as being phase-entangled when the network has achieved synchronization. This is due to the phase overlap that occurs in the interferometer arms, the Sagnac loops in our case.

25 Applications

Quantum optical circuits are used for implementing quantum gates, state preparation, and quantum teleportation.

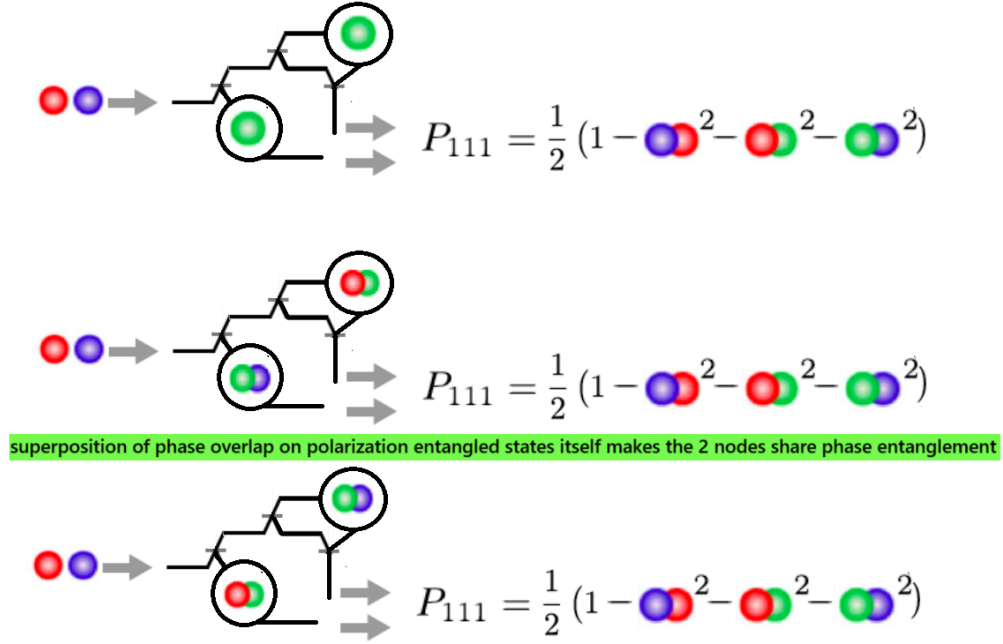


Figure 17: superposition of phase overlap on polarization entangled states itself makes the 2 nodes share phase entanglement

Entanglement Generation Entanglement is a quintessential quantum resource enabling quantum communication and computation.

26 Basics of Entanglement as Synchronization

A two-qubit state

$$|\psi\rangle$$

is entangled if it cannot be written as a product state:

$$|\psi\rangle \neq |\phi\rangle_A \otimes |\chi\rangle_B \quad (72)$$

26.1 Bell States

Maximally entangled states, known as Bell states, are:

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \quad (73)$$

Quantum networking then will naturally emerge from network growth and placing more nodes to stabilize the network. this will leverages quantum properties such as entanglement and superposition to enable secure communication, distributed quantum computing, and advanced sensing.

Nonlinear effects present in classical networking will emerge in their own way in such a system and we can borrow a lot that has been developed from graph theory to come to solutions needed to stabilize and synchronize quantum networks, focusing on both hardware implementations and systemic approaches.

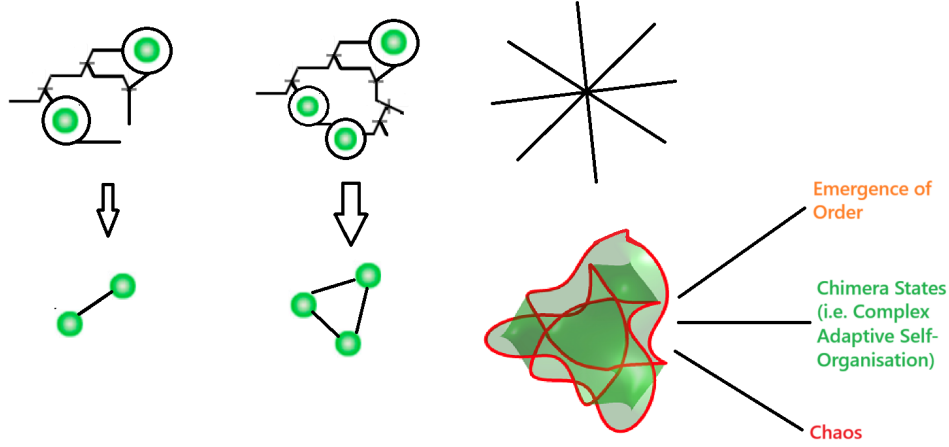


Figure 18: phase entangled network growth creates the need for general models of network dynamics that can work scale free and create predictions and descriptions of their emergent behaviors

27 State of the Art in Quantum Network Stabilization and Synchronization

Recent advances in quantum networks focus on overcoming challenges related to coherence, stability, and scalability. Classical synchronization models, such as the Kuramoto model [1], have been extended to quantum systems, providing insights into emergent synchronization phenomena in quantum oscillators [2]. These models underpin strategies for passive and active stabilization methods in quantum communication channels.

Hardware-based stabilization approaches leverage polarization-maintaining fibers, electro-optic modulators, and passive interferometric devices such as Sagnac loops [3, 4]. These techniques aim to preserve quantum states over long distances with minimal loss and phase drift.

On the systemic level, network topology optimization through graph rewiring algorithms has demonstrated potential in enhancing robustness and coherence [5]. The quantum Kuramoto model, a quantum analogue of the classical synchronization framework, offers a promising theoretical foundation for understanding collective dynamics in large-scale quantum networks [2].

Compared to classical fiber-optic networks, quantum networks face unique challenges, including maintaining entanglement fidelity and quantum coherence over extended distances. Technological solutions such as quantum memories, repeaters, and integrated photonic circuits are being developed to address these issues, with recent experimental demonstrations showing significant progress [5].

Overall, the integration of systemic models, hardware stabilization techniques, and advanced network design continues to push the frontier of scalable and stable quantum communication networks.

28 Network Stabilization via Classical Synchronization

Achieving stable quantum networks requires synchronization techniques that ensure coherence and timing precision across distributed nodes. Synchronization can be modeled using classical and quantum frameworks, with many methods inspired by classical network synchronization theories.

28.1 Synchronization Modeled as Networks of Oscillators

Classically, synchronization phenomena are often modeled using networks of coupled oscillators [1]. The Kuramoto model is a fundamental representation:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad (74)$$

where:

- θ_i is the phase of the i -th oscillator,
- ω_i is its natural frequency,
- K is the coupling strength,
- N is the total number of oscillators.

This model captures the transition from incoherence to synchronization as K increases.

28.2 Quantum Network Nodes as Oscillators

In quantum networks, nodes such as quantum harmonic oscillators can be modeled similarly, with phase and amplitude dynamics influencing coherence [2]. The coupling of these nodes can be engineered to achieve desired synchronization states, vital for quantum communication fidelity.

29 Approaches to Test the Model

Two primary approaches are considered to validate and analyze the synchronization models:

29.1 Hardware Integration

Implementing synchronization techniques using physical hardware components.

29.1.1 Polarization Control and Stabilization

Quantum channels often employ polarization-encoded qubits. Stabilization involves maintaining polarization states against environmental perturbations.

(a) Polarization-maintaining (PM) fiber and Electro-Optic Modulators (EOMs) PM fibers preserve polarization states [3]. EOMs allow dynamic control of polarization, enabling active stabilization:

$$P(t) = P_0 + \Delta P \sin(\omega t + \phi), \quad (75)$$

where $P(t)$ is the polarization at time t , with modulation parameters tailored to counteract disturbances.

(b) Passive Fiber Sagnac Loops as Stabilizer Nodes Using Sagnac interferometers, passive stabilization of polarization can be achieved [4]. These loops act as "stabilizer nodes" that automatically compensate phase drifts.

Figure 19: Schematic of a passive fiber Sagnac loop used for polarization stabilization.

29.2 Systems Approach

Rewiring and network topology optimization to promote stability and synchronization through systemic design.

29.2.1 Graph Network Rewiring Techniques

Applying graph theory and rewiring algorithms to optimize network structure for emergent stability [5].

The goal is to modify the network adjacency matrix A to enhance synchronization:

$$A' = \text{Rewire}(A), \quad (76)$$

based on criteria such as maximizing algebraic connectivity or minimizing path length.

29.2.2 Rewiring Optimization

Using algorithms like simulated annealing or genetic algorithms to find optimal configurations:

$$\text{maximize } \lambda_2(A') \quad \text{subject to constraints}, \quad (77)$$

where λ_2 is the Fiedler Value, i.e. second smallest eigenvalue of the Laplacian matrix, related to network robustness and synchronizability.

This value is itself embedded in the Fiedler Vectors, core to any method of graph optimization to enhance synchronization.

It is necessary therefore to discuss the relationships between the Fiedler vector, as well as other constants of importance in graph network optimization such as the Cheeger constant. This is necessary to describe threshold behaviors in networks of quantum resource nodes represented as Quantum Directed Acyclic Graphs (QDAGs). We can explore how these mathematical constructs can be applied to understand the dynamics of quantum information flow in QDAGs, especially in the context of routing problems with bottlenecks in graph signal processing, and how new eigenvalue-based rewiring techniques can optimize entanglement distribution between nodes in a QDAG.

30 Quantum Directed Acyclic Graph Network Rewiring Optimization

Quantum Directed Acyclic Graphs (QDAGs) are a versatile framework in entanglement networking that encode relationships between entanglement resources, nonlinear oscillators elements, i.e., qubit sources, memories, repeaters, clients etc., in a non-cyclic manner [21]. Understanding the underlying mathematical structures governing QDAGs allows for optimization using algorithms developed from classical graph theory and network science [23] to help understand and optimize quantum information flow. Central to this discussion are the Fiedler vector and the Cheeger constant [5], which provide insight into the connectivity and bottlenecks in a network, two details which may not always be mutually exclusive as this research hopes to elucidate. A research focus is to examine implications for quantum states on the graph and when the Fiedler eigenbasis is changed. This involves examining dynamical aspects of quantum systems represented by graphs, properties such as stability, connectivity as they are rewired for optimization. To accomplish this we start at the foundations of the Fiedler vector, in theorem form, to form conceptual bridges with the field

of dynamical quantum graphs. Also demonstrated is the use of the Fiedler vector to perform a novel type of graph rewiring for optimization of Quantum DAG's. This method is dubbed QFERN, standing for Quantum Fiedler Eigenbasis for Rewiring Networks. It represents a method, implimented as a workflow and then as an algorithm, that utilizes specifically the Fiedler vector and eigenbasis, to rewire or reconfigure quantum networks to optimize the bottlenecks in such networks, as represented by spectral gaps, and effective resistances in such graphs of the entanglement between nodes.

31 Mathematical Foundations

31.1 Fiedler Vector

The Fiedler vector $\mathbf{v}$ is derived from the second smallest eigenvector of the Laplacian matrix $\mathbf{L}$ of a graph. The Laplacian is given by:

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

where $\mathbf{D}$ is the degree matrix, and $\mathbf{A}$ is the adjacency matrix. The Fiedler vector captures essential connectivity properties of the graph, indicating how tightly coupled the vertices are [21].

31.2 Cheeger Constant

The Cheeger constant h quantifies the connectivity of a graph and is defined as:

$$h = \min_{S \subset V} \frac{|\partial S|}{|S|}$$

where $|\partial S|$ counts the edges crossing the cut defined by S and V is the set of vertices. The Cheeger constant is crucial for understanding the bottleneck behaviors in QDAGs [24].

32 Relation between Fiedler Vector and Cheeger Constant

The Fiedler vector and Cheeger constant are related through spectral graph theory. Specifically, the Cheeger inequality provides a connection:

$$\frac{h^2}{2} \leq \lambda_2 \leq 2h$$

where λ_2 is the second smallest eigenvalue of the Laplacian matrix. A larger Cheeger constant reflects a better connectivity, aligning with a lower value of the Fiedler vector's components [23].

32.1 Fiedler Vector and Dirichlet Energies

The Fiedler vector, denoted as $\mathbf{f}_2 \in \mathbb{R}^n$, corresponds to the second smallest eigenvector of the Laplacian matrix of a graph, often termed the Fiedler matrix. This vector provides essential insights into the connectivity properties of the graph, particularly in terms of cluster membership [23]. Specifically, the elements of $\mathbf{f}_2$ can be interpreted as indicators of node membership to each of the two clusters that arise from a spectral clustering process.

The associated eigenvalue λ_2 of $\mathbf{f}_2$ encapsulates the Dirichlet energy of the system, providing a measure of the smoothness of the function represented by the Fiedler vector across the graph. This eigenvalue plays a

critical role in quantifying the overall "bottleneckedness" of the graph, as described by the Cheeger constant, which, in turn, reflects the minimum ratio of the edge cut to the volume of the smaller subset of vertices within the graph.

We aim to optimize the bottleneck width through a novel form of diffusion-based graph rewiring characterized by a search matrix $\mathbf{A}_{\text{soft}}$, which closely resembles the original adjacency matrix $\mathbf{A}$ but minimizes the bottleneck size. This optimization is guided by spectral derivatives, allowing us to express the adjustments in the graph structure as follows:

$$\mathcal{L}_{\text{Fiedler}} = \mathbf{A}_{\text{soft}} - \mathbf{A} + \alpha \frac{\lambda_2}{2} \nabla \mathbf{A}_{\text{soft}} \lambda_2 \equiv \text{Tr} \nabla \mathcal{L}_{\text{mi}} \lambda_2$$

To elucidate the relationship between the Fiedler vector and the structure of the graph, we employ the relationship:

$$T \nabla \mathbf{A}_{\text{soft}} \mathcal{L}_{\text{mi}} = \text{diag}(\mathbf{f}_2 \mathbf{f}_2^T) \mathbf{1}_1^T - \mathbf{f}_2 \mathbf{f}_2^T$$

The expression for λ_2 can thus be seen as representing the spectral gap or bottleneck size, indicating how effectively the clusters are separated in the process of optimization.

As we manipulate the underlying graph structure, the evolution of the Fiedler vector and its corresponding eigenvalue λ_2 plays a crucial role in guiding the optimization of the graph to achieve a balance between connectivity and separability of the clusters present.

A proposed algorithm to accomplish this requires the following, general recipe for graphs:

32.2 Graph Processing Steps

- **Random Directed Acyclic Graph Generation:** We use NetworkX to create a random directed acyclic graph (DAG).
- **Fiedler Vector Calculation:** We compute the Fiedler vector from the Laplacian matrix of the original adjacency matrix. The Fiedler vector is the second smallest eigenvector of the Laplacian, which is critical for our adjustments.
- **Gradient Calculation:** The gradient is derived based on the outer product of the Fiedler vector $(-|f_{\text{iedler_vector}}[i] * f_{\text{iedler_vector}}[j]|)$. This allows us to adjust the adjacency matrix in a manner reflecting the graph's structural properties.
- **Adjustment of Asoft:** The Asoft matrix is adjusted using the gradient while maintaining non-negativity and symmetry. The adjustment is controlled by the alpha parameter.
- **Laplacian Calculation and Plotting:** The code calculates the Laplacian for both the original adjacency matrix and the Asoft, then plots them for visual comparison as in Fig 1.

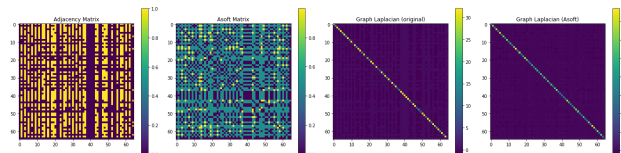


Figure 20: Graph Adjacency and Asoft Matrices and their respective Graph Laplacians

33 Quantum Directed Acyclic Graphs (QDAGs)

QDAGs are employed to model quantum circuits. The vertices represent quantum gates, and the edges signify the quantum states' directed flow. The Fiedler vector evaluates the potential bottlenecks in these flows [21].

33.1 Thresholds in QDAGs

In the context of QDAGs, we define critical bottlenecks as thresholds for qubit entanglement and coherence loss. These thresholds can be derived from the Cheeger constant, treating it as a bottleneck index, thus linking classical graph properties with quantum behaviors, such as bottlenecks leading to effective resistances to the transfer of entanglement in a network.

33.2 Effective Resistances in QDAGs

In the context of Quantum Directional Acyclic Graphs (QDAGs), the effective resistance between two nodes u and v can be quantitatively analyzed using the properties of the graph's Laplacian matrix. The effective resistance $R_{u,v}$ is mathematically expressed as:

$$R_{u,v} = (e_u - e_v)^T L^+ (e_u - e_v), \quad (78)$$

where L^+ denotes the pseudoinverse of the Laplacian matrix L . This formulation emphasizes the importance of the Laplacian in characterizing the resistance in a network by capturing the influence of the graph's topology on the flow between nodes.

To deepen our understanding of effective resistance in the framework of QDAGs, we can invoke the relationship between effective resistance and the spectral characteristics of the graph. In particular, we can connect the effective resistance to the eigenvalues associated with the Fiedler vector, which provides insight into the graph's connectivity. This relationship can be formulated as:

$$R_{u,v} = \sigma_i > 2 \sum_{i=1}^n \frac{1}{\lambda_i} f_i f_i^T, \quad (79)$$

In this equation, σ_i represents the weights assigned to the eigenvalues λ_i , and f_i corresponds to the components of the Fiedler vector tied to the nodes of the graph. The Fiedler vector, being the eigenvector associated with the second smallest eigenvalue of the Laplacian, plays a pivotal role in describing the connectivity of the network [21].

This connection between the structure of QDAGs, their spectral properties, and the effective resistance highlights how the Fiedler vector encapsulates the underlying entanglement flow characteristics within the network.

Optimizing the network Laplacian based on the Fiedler vector can lead to more robust designs of quantum networks that minimize error rates and maximize entanglement distribution in bottlenecked networks, i.e., with only a single entanglement resource and many clients, repeaters, memory nodes, etc.

34 QFERN Method for Graph Modification wrt bottlenecks and effective resistances

34.1 Overview

Given a directed acyclic graph (DAG), we aim to modify its adjacency matrix based on the spectral properties of its Laplacian matrix to produce a new matrix, termed A_{soft} . The algorithm sequentially computes gradients based on the Fiedler vector and applies them to adjust A_{soft} .

Modification of Adjacency Matrix Generate a random DAG with

$$n$$

nodes. Compute the adjacency matrix

$$A$$

of the graph. Compute the Laplacian matrix

$$L = D - A$$

. Calculate the eigenvalues and eigenvectors of

$$L$$

. Identify the Fiedler vector from the eigenvectors. Initialize

$$A_{\text{soft}} = A$$

. Foreach

$$(i, j)$$

in the nodes If $i \neq j$ Compute gradient based on Fiedler vector:

$$\text{gradient}[i, j] = -f[i] \cdot f[j]$$

EndIf EndFor Update

$$A_{\text{soft}} \leftarrow A_{\text{soft}} + \alpha \cdot \text{gradient}$$

Ensure

$$A_{\text{soft}}$$

is non-negative and symmetric.

34.2 Optimization Scheme

The optimization scheme aims to minimize the effective resistance across the network while maximizing the Cheeger constant, reflecting the network's bottlenecks. In this subsection, we will outline the procedure for this optimization process, including the necessary calculations and evaluations.

34.2.1 Objective Function

The optimization problem can be formulated as:

$$\min_G R_{\text{eff}}(G) \quad \text{s.t.} \quad \max_G h(G) \quad (80)$$

where R_{eff} is the effective resistance of the graph G , and $h(G)$ is the Cheeger constant of the graph.

34.2.2 Cheeger Constant Calculation

The Cheeger constant $h(G)$ is defined as:

$$h(G) = \min_{S \subseteq V} \frac{|\partial S|}{\min\{|S|, |V \setminus S|\}} \quad (81)$$

where V is the set of vertices in the graph, $|S|$ is the size of the set S , and $|\partial S|$ is the number of edges crossing the boundary of S (i.e., edges that have one endpoint in S and the other in $V \setminus S$).

The initial Cheeger constant is calculated to assess the connectivity of the network before any modifications are made.

34.2.3 Optimization Loop

The optimization scheme operates in an iterative loop as follows:

1. A random edge e is removed from the graph G . The edges are added to the graph using the function: the set of edges connecting nodes in groups A and B.
2. A new edge e' is added from a list of possible edges, ensuring adherence to the graph's structural rules. Let E' be the set of possible edges.

$$E = \{(i, j) | i \in A, j \in B\} \quad (82)$$

3. The new Cheeger constant $h'(G)$ is calculated as per Equation (2) and printed.

This loop continues until a stopping criterion is met, such as a maximum number of iterations or no improvement observed in the Cheeger constant.

34.2.4 Objective Evaluation

During the optimization process, we dynamically adjust the edges and evaluate configurations that yield better Cheeger constants while simultaneously analyzing the effective resistances:

$$R_{\text{eff}}(G) = \sum_{u, v \in V} R_{uv} \quad (83)$$

where R_{uv} is the effective resistance between nodes u and v .

The effective resistance

$$R(u, v)$$

between two nodes

$$u$$

and

$$v$$

is calculated using the eigenvalues

$$\lambda_i$$

and the Fiedler vector

$$f$$

obtained from the Laplacian's eigenvalue decomposition:

$$R(u, v) = \sum_{i=1}^n \frac{f_u f_v}{\lambda_i} \quad (84)$$

where

$$f_u$$

and

$$f_v$$

are components of the Fiedler vector corresponding to nodes

$$u$$

and

$$v$$

The optimization scheme examines how modifications in the edge structure can lead to favorable changes in both the Cheeger constant and effective resistance.

34.2.5 Final Visualization

Upon completing the QFERN optimization iterations, the effective resistances are recalculated to analyze the impact of edge modifications on network properties. Visualization techniques, such as graph plotting and heat maps, are employed to represent the changes in effective resistances and Cheeger constants across the iterations.

The resulting visualizations can help in understanding the effects of edge modifications and how they influence the overall connectivity and efficiency of the network.

We can compare the graph structure before and after the Adjacency to Asoft graph rewiring modifications to check for bottlenecks as follows:

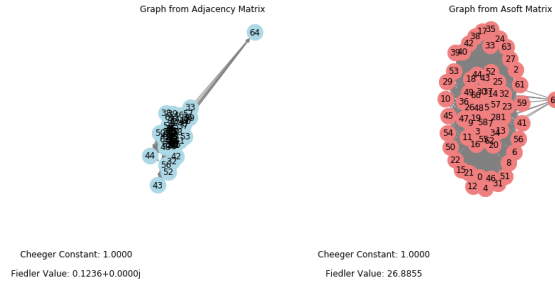


Figure 21: Graph network structure of Adjacency Matrix versus Asoft Matrix

Finally, the effective resistances are visualized (Fig 3) using a color map on the graph. Nodes are colored based on their average effective resistance, allowing for an intuitive understanding of the graph's connectivity properties.

In summary, the code demonstrates a systematic approach to optimizing a graph based on fundamental graph theory concepts, specifically focusing on the Cheeger constant and effective resistance metrics.

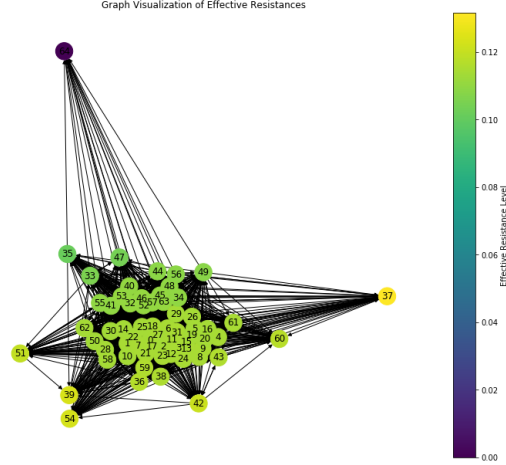


Figure 22: effective resistances of pair entangled states in a network

35 Utilizing QFERN Results to Identify and Stabilize Desynchronization Regions

Building upon the spectral analysis provided by the QFERN method, it is possible to leverage the relationship between the graph’s Laplacian eigenvectors—particularly the Fiedler vector—and synchronization phenomena modeled by the Kuramoto framework. Specifically, regions of the network where the effective resistance is notably high or where the Fiedler vector exhibits significant variations can be interpreted as zones of de-synchronization or weak connectivity.

35.1 Spectral De-synchronization Detection via Kuramoto Model Relations

In the Kuramoto model, the stability of phase synchronization across oscillators is intimately connected to the network’s Laplacian spectrum. The Fiedler vector, associated with the second smallest eigenvalue (the algebraic connectivity), indicates how tightly coupled the network components are. Large differences in the Fiedler vector components between nodes suggest potential phase lag and a propensity toward de-synchronization in those regions.

Synchronization in oscillator networks and spectral graph theoretical tools is rooted in linearized Kuramoto dynamics and network analogies [25]. By analyzing the derived relation:

$$\Delta\theta_{ij} \propto R_{ij} \cdot \omega_{ij} \quad (85)$$

where $\Delta\theta_{ij}$ is the phase difference between nodes i and j , R_{ij} the effective resistance, and ω_{ij} the natural frequency difference, we observe that high effective resistance correlates with increased phase disparities. Consequently, regions with elevated effective resistance—identified via the spectral analysis—serve as indicators of potential desynchronization zones.

35.2 Locating Desynchronization Regions Using the QFERN Method

The procedure involves:

1. Computing the Fiedler vector f from the Laplacian’s eigen-decomposition.

2. Mapping the effective resistance R_{uv} between nodes u and v using the spectral relation (outlined in the appendix):

$$R_{uv} = \sum_{i=1}^n \frac{(f_u - f_v)^2}{\lambda_i} \quad (86)$$

where λ_i are Laplacian eigenvalues.

3. Identifying clusters or regions where R_{uv} exceeds a certain threshold, indicating weak coupling and potential de-synchronization.

These high-resistance regions often correspond to nodes or subgraphs where phase coherence is fragile.

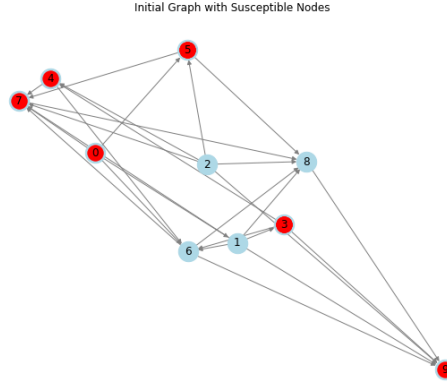


Figure 23: fragile network with nodes susceptible to de-synchronization

36 Network Hardware that can Induce Stabilization

Once a de-synchronization region is identified, a strategic intervention can be designed by adding a new node—conceptually a "stabilizer"—aimed at reinforcing connectivity and restoring synchronization in an entanglement distribution network. This node could be connected to multiple nodes within the high-resistance region, effectively bridging weak links and increasing the local algebraic connectivity.

The criteria for placing this stabilizer node include:

- Connecting to nodes with the highest R_{uv} values.
- Ensuring that the addition reduces the local effective resistance, as verified through spectral recalculation.
- Adjusting the weights of these new edges (via the soft adjacency matrix A_{soft}) to optimize the spectral gap and enhance robustness.

Which results in a graph that has grown by 1 node but has used that node as the stabilizer:

This approach aligns with the spectral graph theory principle that augmenting the network with nodes or edges to increase the algebraic connectivity λ_2 effectively stabilizes the synchronization dynamics modeled by the Kuramoto framework.

We will explore in detail what this quantum network stabilizer hardware can look like in the experimental Sagnac Interferometry part of this thesis. For the QFERN Method we take it for granted that such a stabilizer node simply exists.

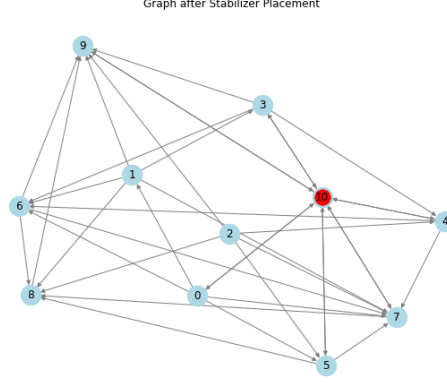


Figure 24: stabilized network

37 Conclusion

This paper presented a mathematical framework connecting the Fiedler vector, Cheeger constant, and thresholds in Quantum Directed Acyclic Graphs. Using this framework as a means of locating regions in a quantum network where the effective resistance is minimized can be used to study regions of de-synchronization in a network. Future research should focus on applying these insights to improve quantum circuit design and fault tolerance, as well as exploring the implications of this optimization on broader quantum network architectures and how to improve network synchronization by mapping a network using the QFERN method.

By integrating the spectral insights from the QFERN method with the Kuramoto model's stability analysis, one can systematically detect regions prone to de-synchronization and implement targeted network modifications. The addition of a stabilizer node, strategically placed within these high-resistance, weakly connected regions, provides a theoretical mechanism to enhance network coherence, ultimately leading to more resilient quantum and classical network architectures.

The spectral and resistance-based analysis within the QFERN framework validates the conditions outlined in [25] for synchronization stability as a function of algebraic connectivity. It substantiates the spectral norm condition of the Laplacian $\|L^\dagger \omega\|_{E,\infty} < 1$ as outlined in the appendix.

Appendix: Spectral Conditions for Synchronization Stability

This appendix provides a detailed spectral and resistance-based analysis that corroborates the conditions outlined in the corollary regarding the stability of synchronized states in coupled oscillator networks, particularly within the Kuramoto model framework.

The corollary 7.5 in [25] states that for a connected, acyclic graph $G(V, E, A)$, a phase-cohesive and locally exponentially stable equilibrium manifold exists if and only if the spectral norm condition

$$\|L^\dagger \omega\|_{E,\infty} < 1$$

is satisfied, where: - $L^\dagger$ is the Moore-Penrose pseudoinverse of the Laplacian matrix L , - $\omega \in \mathbb{R}^n$ is the vector of natural frequencies orthogonal to the all-ones vector, i.e., $\mathbf{1}^T \omega = 0$.

This criterion aligns with the cutset condition and ensures that the phase differences remain bounded, guaranteeing synchronization.

Spectral Decomposition and Effective Resistance Interpretation

Leveraging spectral graph theory, the spectral norm $\|L^\dagger \omega\|_{E,\infty}$ can be expressed via the eigen-decomposition of L . Let $V = [f^{(1)}, f^{(2)}, \dots, f^{(n)}]$ be the orthonormal eigenvectors of L , with eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$. Since the graph is connected and acyclic, $\lambda_1 = 0$ corresponds to the trivial eigenvector $\mathbf{1}$.

The spectral condition (58) can be written as:

$$V \operatorname{diag} \left(0, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n} \right) V^T \omega_E < 1,$$

where ω_E is the component of ω projected onto the eigenbasis. This inequality indicates that the weighted projection of ω onto the inverse eigenvalues must remain below one, emphasizing the role of the algebraic connectivity λ_2 .

Furthermore, the spectral norm can be bounded as:

$$\|L^\dagger \omega\|_{E,\infty} \leq \frac{\max_i |f^{(i)T} \omega|}{\lambda_2},$$

highlighting that larger λ_2 (i.e., better network connectivity) reduces the potential for desynchronization caused by frequency mismatches.

Effective Resistance and Network Robustness

The effective resistance R_{uv} between nodes u and v is given by:

$$R_{uv} = \sum_{i=2}^n \frac{(f_u^{(i)} - f_v^{(i)})^2}{\lambda_i},$$

which encapsulates the ease of current (or information) flow between nodes and reflects the network's robustness. Regions with high effective resistance correspond to small spectral gaps λ_2 and significant variation in the Fiedler vector $f^{(2)}$, indicating potential points of vulnerability where synchronization may fail.

QFERN Framework and Spectral Bounds

The QFERN approach provides a framework that supports the sufficiency of the spectral condition. It demonstrates that increasing the algebraic connectivity λ_2 or decreasing the maximum effective resistance R_{uv} enhances the synchronization stability margin.

Specifically, the spectral bounds derived through QFERN confirm that the condition

$$\|L^\dagger \omega\|_{E,\infty} < 1$$

is both necessary and practically tight, especially in topologies where the extremal eigenvalues are well characterized, as discussed by Dörfler and Bullo (2013) [25].

38 Systemic Review of Technical Challenges and Values in Quantum Network Solutions

Quantum networks are at the forefront of quantum information science, integrating various technologies such as repeaters, memories, measurement nodes, and stabilizers. The complexity arises from the competition

among these technologies within a shared network infrastructure. This review explores the potential for network-intrinsic stabilization, the application of models like the quantum Kuramoto model, and comparisons with classical fiber networks.

39 Stabilization Technologies in Quantum Networks

Quantum networks often require a multitude of specialized devices:

- **Quantum Repeaters:** Extend communication distances by entanglement swapping.
- **Quantum Memories:** Store quantum states reliably.
- **Measurement Nodes:** Perform quantum measurements for control and error correction.
- **Stabilizers:** Maintain coherence and phase stability across the network.

39.1 Competing Technologies and Roles

Graphical representation of network components:

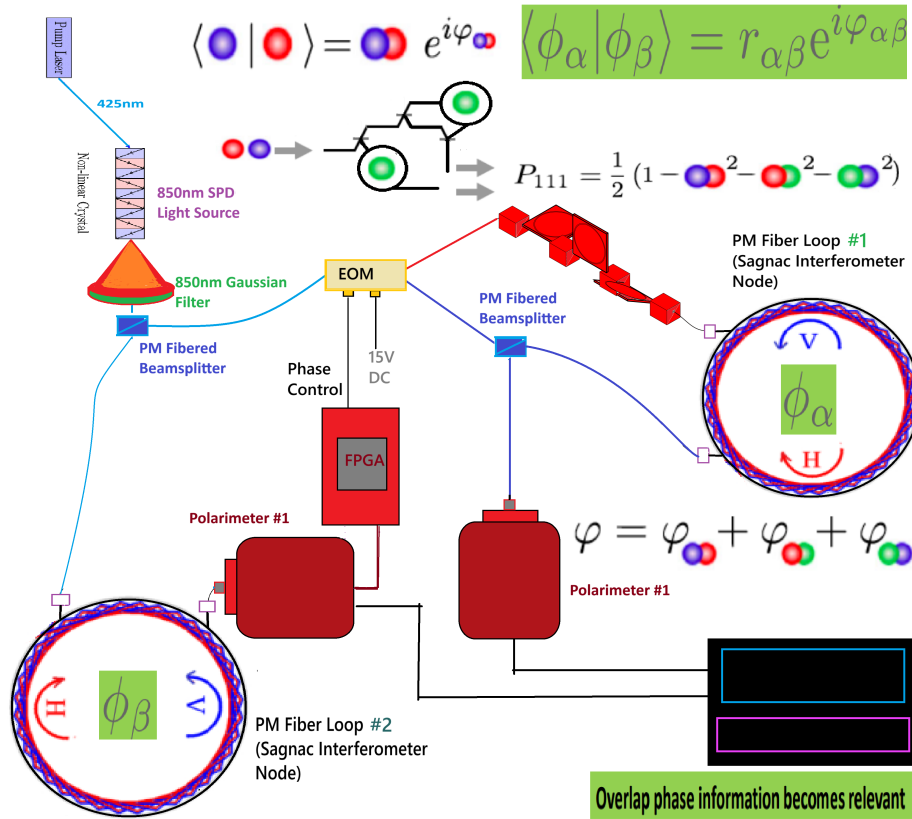


Figure 25: Distribution of technological roles in a quantum network

40 Network Self-Stabilization and Synchronization

40.1 Can the Network Achieve Stabilization Internally?

The notion of leveraging the network's inherent properties for stabilization and synchronization is compelling. In classical systems, feedback mechanisms and passive configurations can induce stability.

40.2 Quantum Kuramoto Model

The classical Kuramoto model describes synchronization phenomena:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) \quad (87)$$

where θ_i is the phase of oscillator i , ω_i its natural frequency, and K the coupling strength.

40.2.1 Quantum Extension

The quantum Kuramoto model adapts this to quantum phases or states:

$$\frac{d\hat{\rho}_i}{dt} = -i[\hat{H}_i, \hat{\rho}_i] + \sum_{j \neq i} \mathcal{L}_{ij}(\hat{\rho}_j) \quad (88)$$

where $\hat{\rho}_i$ is the density matrix of node i , $\hat{H}_i$ its Hamiltonian, and $\mathcal{L}_{ij}$ a Lindblad-type coupling representing the network interactions.

40.3 Emergent Stability

Analyzing the quantum Kuramoto model reveals conditions under which the network exhibits spontaneous synchronization, contributing to stability without active stabilization devices.

41 Comparison with Classical Fiber Networks

41.1 Passive Optical Networks (PONs)

Certain classical configurations like star or ring topologies are inherently more stable:

- **Star Networks:** Centralized stability due to shared hub.
- **Ring Networks:** Redundancy offers inherent fault tolerance.

41.2 Key Performance Indicators

Signal-to-Noise Ratio (SNR) is critical:

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad (89)$$

which impacts polarization control and distribution within fibers.

42 Quantum Signal Quality Metrics

42.1 Quantum Analog of Power Ratios

In classical optics, the power ratio between signals is crucial. For quantum signals, we consider metrics like **Quantum Signal-to-Noise Ratio** (QSNR) and **Adjacent Power Ratio** (APR):

$$\text{QSNR} = \frac{\langle \hat{P}_{\text{signal}} \rangle}{\langle \hat{P}_{\text{noise}} \rangle} \quad (90)$$

42.2 Thresholds for Quantum Networks

The fidelity F of quantum state transmission depends on the noise:

$$F = \langle \psi_{\text{ideal}} | \hat{\rho}_{\text{actual}} | \psi_{\text{ideal}} \rangle \quad (91)$$

with a threshold fidelity F_{th} necessary for quantum communication viability.

43 Cost and Reliability Considerations

43.1 Inertia and Natural Stability

The network's inertia can be analyzed via stability margins derived from the quantum Kuramoto model, indicating the extent to which the network maintains coherence "for free."

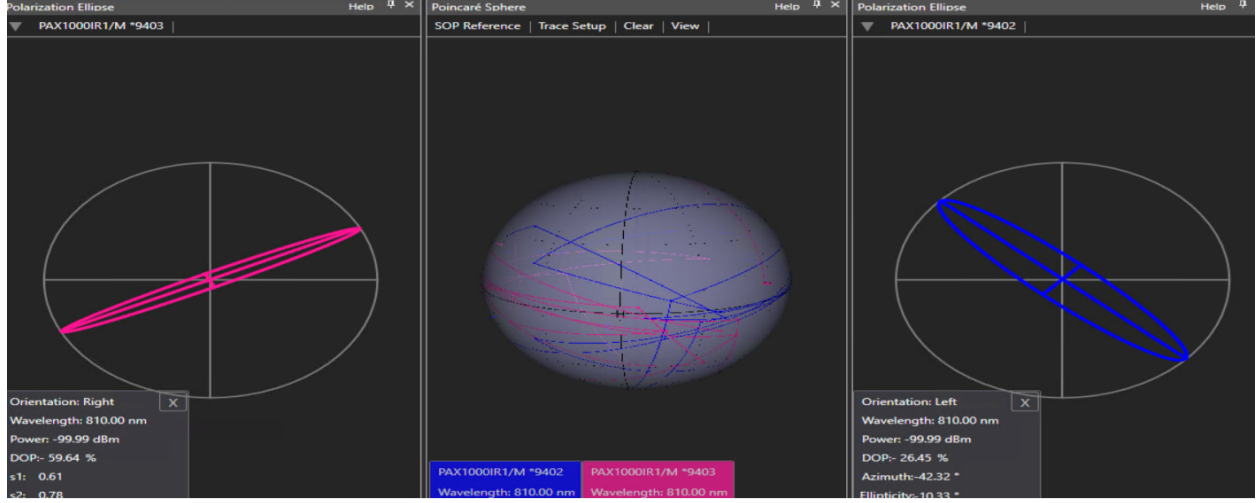
43.2 Trade-offs

Balancing cost and reliability involves optimizing network topology, device placement, and control protocols to leverage natural stability mechanisms.

Representing Polarization for Use in Quantum Optics

44 Introduction to the Poincaré Sphere

The Poincaré sphere provides a three-dimensional geometric representation of polarization states of light. It allows visualization of both amplitude and phase information of polarization. The north and south poles correspond to linear polarization states (e.g., horizontal and vertical), while the equator represents circular polarization states (e.g., right-circular and left-circular).



The Poincaré sphere is mathematically related to the Bloch sphere through transformations involving the Stokes vectors.

45 Stokes Vectors and Polarization Parameters

Stokes vectors are a set of four parameters that describe the polarization state of light:

$$\mathbf{S} = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}$$

where:

- S_0 represents the total intensity of light.
- S_1 , S_2 , and S_3 describe the polarization state.

45.1 Relationship to Pauli Spin Matrices

The connection between the Pauli matrices and Stokes parameters is given by:

$$S_0 = |\langle \psi | \sigma_0 | \psi \rangle|^2 \quad (92)$$

$$S_1 = |\langle \psi | \sigma_x | \psi \rangle|^2 - |\langle \psi | \sigma_y | \psi \rangle|^2 \quad (93)$$

$$S_2 = 2 \operatorname{Re} [\langle \psi | \sigma_x | \psi \rangle^* \langle \psi | \sigma_y | \psi \rangle] \quad (94)$$

$$S_3 = 2 \operatorname{Im} [\langle \psi | \sigma_x | \psi \rangle^* \langle \psi | \sigma_y | \psi \rangle] \quad (95)$$

These equations link the polarization state represented via Pauli matrices to the measurable Stokes parameters.

46 Application of Stokes Vectors in Quantum Optics

Stokes vectors are particularly useful in describing optical quantum states. They can be easily manipulated using optical components such as nonlinear crystals, wave plates, and polarization-maintaining fibers. These fibers can be set to specific measurement bases, including:

- Random basis
- Bell basis
- Novel measurement bases for quantum networks

46.1 Measurement Bases

The general formulas for measurements in different bases are:

46.1.1 Random Measurement Basis

$$P_{\text{random}} = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{S}_{\text{random}} = P_{\text{random}} \mathbf{S}$$

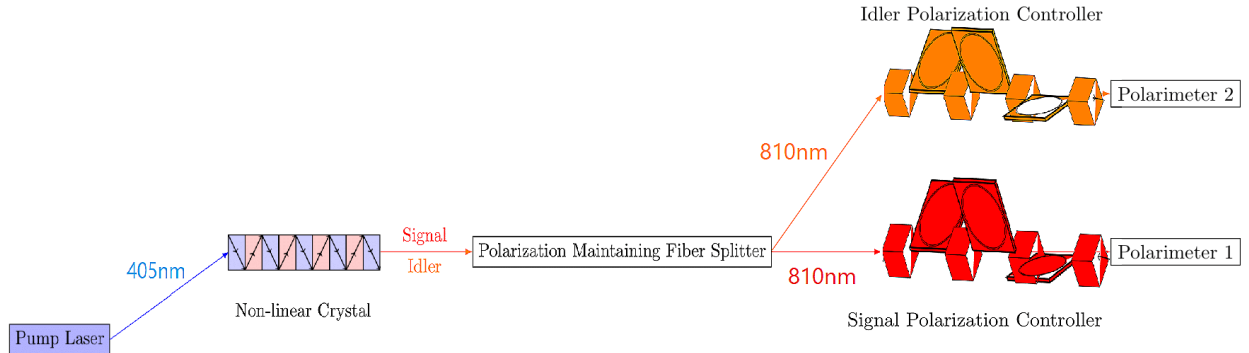
46.1.2 Bell Measurement Basis

$$P_{\text{Bell}} = \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{S}_{\text{Bell}} = P_{\text{Bell}} \mathbf{S}$$

Here, $\mathbf{S}$ is the original Stokes vector, while $\mathbf{S}_{\text{random}}$ and $\mathbf{S}_{\text{Bell}}$ are the post-measurement Stokes vectors in the respective bases.

47 Experimental Non-Linear Optics Test Bench for Quantum State Polarization Measurements

To generate, transmit, and manipulate optical quantum states, a versatile setup was developed. For testing purposes, the system was fibered.



The main advantage of this configuration is the closed optical environment, which minimizes external interference. The compact size and the use of polarization-maintaining fibers address dispersion challenges, enabling the examination of systems that are inherently lossy—thus motivating the development of quantum tomography techniques suitable for noisy measurements.

48 Benchmarks and Performance Verification

48.1 Experimental Benchmarks

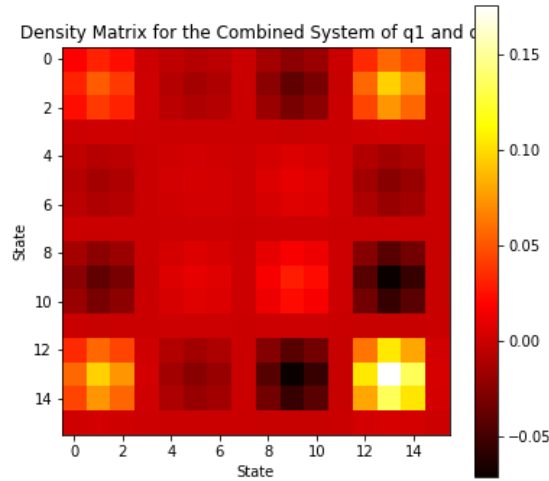
Reconstructed Density Matrix of Quantum Entanglement Source Node

Full state tomography with polarization optics involves characterizing the quantum state of a light field using different polarization bases, such as Horizontal (H), Vertical (V), Left-diagonal (L), Right-diagonal (R), Diagonal (D), and Antidiagonal (A) projected states. The Stokes parameters S_i can be obtained from measurements in these bases, and the density matrix ρ of the quantum state can be reconstructed using the measured data.

0.5 The equation for reconstructing the density matrix ρ from Stokes vectors $\mathbf{S}$ using the Pauli spin matrices σ_i is given by:

$$\rho = \frac{1}{2} \left(I + \sum_{i=1}^3 S_i \sigma_i \right)$$

where I is the identity matrix and σ_i are the Pauli spin matrices



0.5

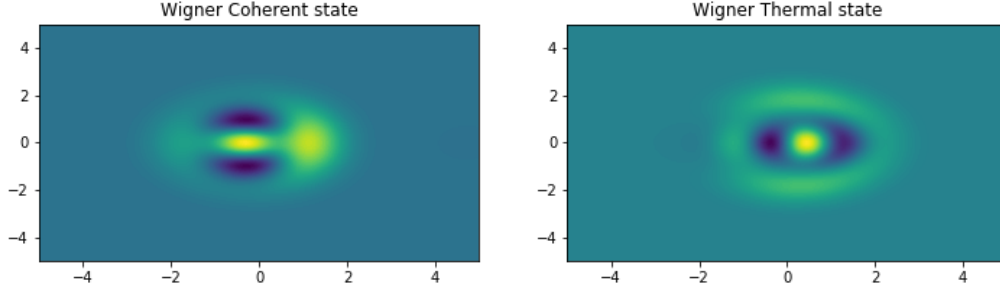
Quantum Polarimetry in Phase-Space using Functional Representation

Quantum state tomography can be applied to generate the phase-space functions from the polarization measurements with different polarization settings and analyzing the obtained data to form a density matrix.

Employing both the Wigner and Husimi Q Function representations to give us a picture of the time evolution of the state from target to recovery.

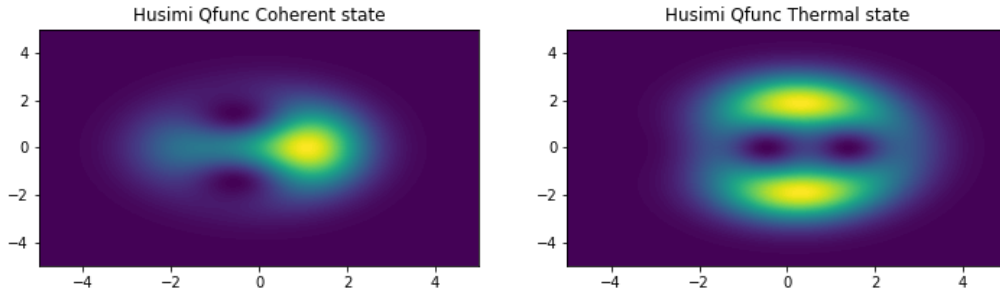
The Wigner function in bra-ket notation is given by:

$$W(x, p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} \langle x - \frac{\xi}{2} | \rho | x + \frac{\xi}{2} \rangle e^{\frac{2ip\xi}{\hbar}} d\xi$$



The Husimi Q function is another quasi-probability distribution function in phase space that provides a smoothed version of the Wigner function. It is expressed in bra-ket notation as:

$$Q(\alpha) = \frac{1}{\pi} \left| \int_{-\infty}^{\infty} \langle x | \rho | x \rangle e^{-\frac{(\alpha-x)^2}{2}} dx \right|^2$$



In these expressions, ρ represents the density matrix of the quantum state. The notation $\langle x | \rho | x \rangle$ denotes the expectation value of the density matrix in the position basis. The variable α in the Husimi Q function still represents a point in phase space.

Both functions are important tools in quantum mechanics for studying the behavior of quantum states in phase space.

49 Fuchs-van de Graaf Inequalities and Their Applications in Quantum State Benchmarking

The Fuchs-van de Graaf inequalities establish a fundamental relationship between two key measures of quantum states: the trace distance and the fidelity. These inequalities serve as a benchmark for comparing quantum states, providing bounds that relate their distinguishability and similarity.

The trace distance quantifies how easily two quantum states can be distinguished, while the fidelity measures how close they are to each other. The inequalities particularly show that the square of the trace distance is bounded in terms of the fidelity, offering a robust framework for assessing quantum state similarity—especially relevant when considering error bounds in practical quantum systems.

49.1 Trace Distance

The trace distance between two quantum states, represented by density matrices ρ and σ , is defined as:

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (96)$$

where $\|\cdot\|_1$ denotes the trace norm.

49.2 Fidelity

The fidelity between two states is a measure of their similarity, given by:

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2, \quad (97)$$

or equivalently, as the squared trace norm of the product of their purifications:

$$F(\rho, \sigma) = |\sqrt{\rho} \sqrt{\sigma}|_1^2. \quad (98)$$

50 Fuchs-van de Graaf Inequalities

These inequalities provide bounds on the fidelity in terms of the trace distance:

$$1 - F(\rho, \sigma) \leq D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)^2}. \quad (99)$$

A tighter lower bound, especially when one of the states is pure, is given by:

$$1 - \sqrt{1 - F(\rho, \sigma)^2} \leq D(\rho, \sigma). \quad (100)$$

It is noteworthy that the inequality can be saturated in cases where one state is pure, i.e.,

$$F(\rho, \sigma)^2 = 1 - D(\rho, \sigma)^2. \quad (101)$$

51 Applications and Benchmarking Framework

51.1 Calculating and Relating Measures

1. **Calculate the Trace Distance and Fidelity:** Determine these measures between the actual quantum state and a target or ideal state.
2. **Apply Fuchs-van de Graaf Bounds:** Use the inequalities to relate the computed trace distance and fidelity, ensuring consistency of the measurements.
3. **Incorporate Error Bounds:** When experimental or theoretical uncertainties exist, integrate error bounds into the analysis. For example, if the fidelity measurement has an error ϵ , the corresponding bounds on the trace distance can be derived using the inequalities.
4. **Establish Benchmarking Criteria:** Given a fidelity or trace distance threshold, the inequalities enable estimation of the corresponding measure, facilitating performance assessment of quantum operations or systems.

51.2 Example

Suppose a two-qubit gate in a quantum computer has an experimentally measured fidelity $F \approx 0.99$. Using the inequality:

$$D(\rho, \sigma) \geq 1 - \sqrt{1 - F^2}, \quad (102)$$

we find:

$$D(\rho, \sigma) \geq 1 - \sqrt{1 - 0.99^2} \approx 0.14. \quad (103)$$

This indicates that the states produced by the actual and ideal gates are distinguishable with a probability at least of 0.14. If there is an error ϵ in the fidelity measurement, the corresponding uncertainty in the trace distance can be estimated via these inequalities.

52 Significance and Implications

Incorporating the Fuchs-van de Graaf inequalities into quantum benchmarking allows us to:

- **Assess Quantum State Similarity:** Quantify how close an experimentally realized quantum state is to a desired target.
- **Evaluate Quantum Operations:** Benchmark the performance of quantum gates by comparing their output states to ideal states.
- **Consider Realistic Errors:** Account for noise and imperfections in experimental settings.
- **Develop Robust Quantum Technologies:** Use the bounds to guide the design of reliable, fault-tolerant quantum systems.

This framework provides a critical link between theoretical measures and practical experimental parameters, enabling more precise control and validation of quantum devices.

53 Polarimetry Measures applied to Quantum Network Benchmarking

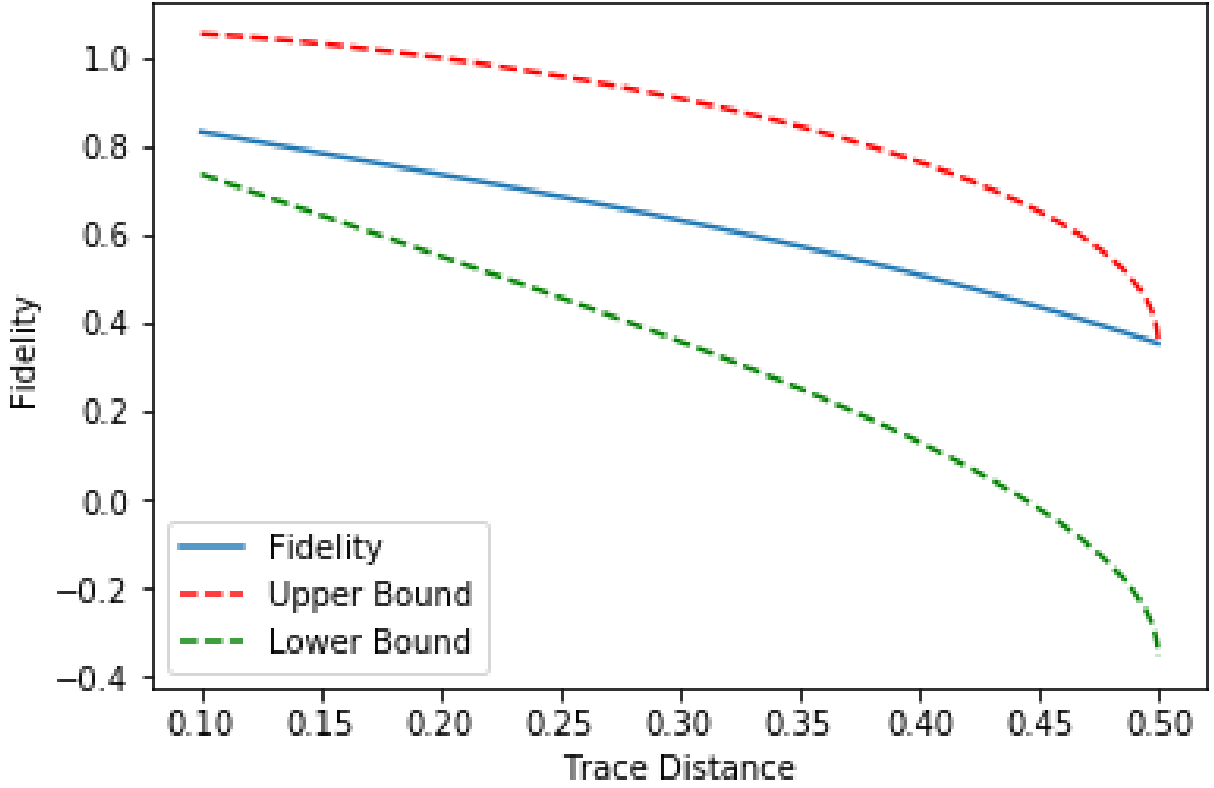
The expected fidelity vs trace distance measures are plotted against the Fuchs-Van de Graaf bounds [3]:

53.0.1 Upper bound:

The upper bound of the fidelity, denoted as F_{upper} , is given by: $F_{\text{upper}} = \sqrt{1 - F(\rho, \sigma)}$, where $F(\rho, \sigma)$ represents the fidelity between the states ρ and σ . In terms of the Trace distance this is then: $F_{\text{upper}} = \sqrt{1 - TD}$, where TD represents the trace distance between the two quantum states.

53.0.2 Lower bound:

The lower bound of the fidelity, denoted as F_{lower} , is given by: $F_{\text{lower}} = \sqrt{F(\rho, \sigma)}$, where $F(\rho, \sigma)$ represents the fidelity between the states ρ and σ . In terms of the Trace distance this is then: $F_{\text{lower}} = \sqrt{TD}$.



From our point of view, the upper bound can serve as a measure of the error level of a quantum channel. A higher upper bound indicates a smaller overlap between the transmitted quantum states and potential repeater's states. Therefore, a higher upper bound ensures a lower probability of information leakage due to error or de-synchronization of the measurement setup, i.e. by problems with coincidence polarimetry.

The lower bound can provide a measure of the potential failure of a quantum communication system, wherein de-synchronization has effectively collapsed the exchange of entanglement, i.e. by depolarization of the channel if we are using the physical variable of polarization density (Coulombs/meter) as our entanglement observable and/or measuring the polarization extinction ratio (PER). The PER measurement is defined as the ratio of the power in a principal polarization state to the power in an orthogonal polarization state after propagation through a device or system, expressed in dB. [4] .

54 Conclusions

- Quantum state generation and tomography is possible using off-the-shelf polarization optics components
- The trace distance vs fidelity benchmark with error bounds can be used to interpret the state reconstruction wrt polarimetry measures i.e. polarization extinction ration (PER) in dB.
- polarization density matrices can be used to model a quantum network as a directional acyclic graph.

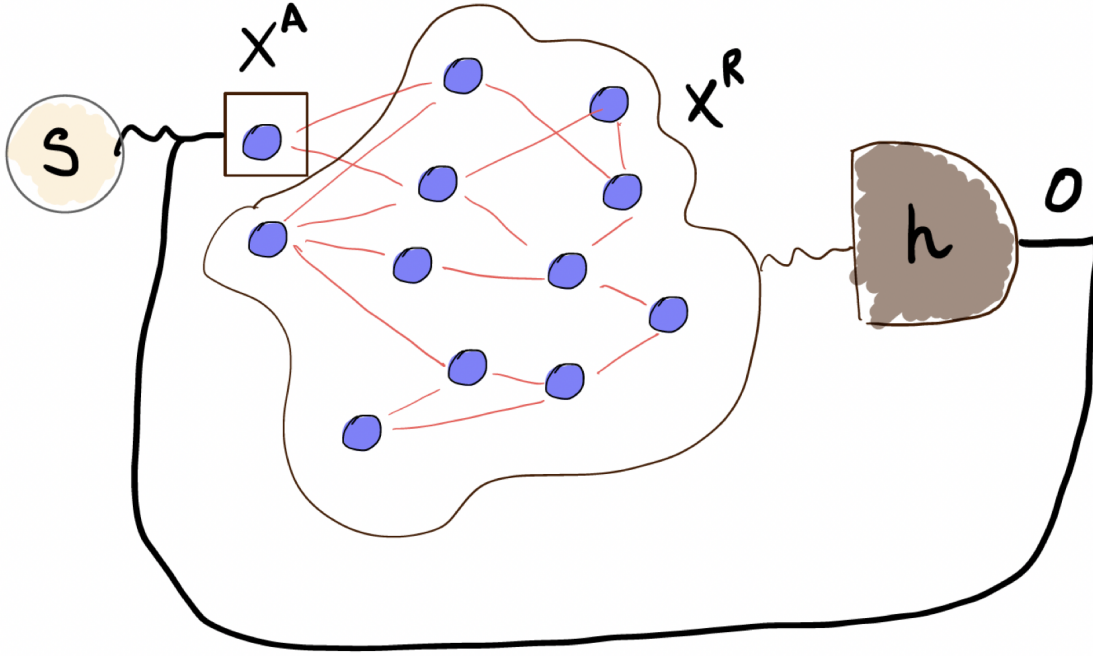


Figure 26: CV networks (such as those developed by Parigi et al) using a non linear crystal EPR source S , with a stabilization node (X_A) that enter into a network reservoir (X_R) is then subject to a detector in continuous time-binned "listen" 0-state mode (h). This then emphasizes the need for a continuous sensitive sensor system that can in principle work around some of the limits of the network if quantum repeaters and/or memory nodes are limited.

55 Detection Techniques in Quantum Networks

High-efficiency, low-noise single-photon detectors are vital for quantum communication. Technologies include avalanche photodiodes (APDs), superconducting nanowire single-photon detectors (SNSPDs), and transition-edge sensors (TES). SNSPDs, in particular, offer efficiencies exceeding 90

Detection schemes for continuous variable state quantum networks can involve time-correlated single-photon counting (TCSPC), which enables precise temporal resolution necessary for time-bin encoding and synchronization in CV quantum networks. Advances in detector technology continue to push the limits of sensitivity and timing resolution, directly impacting the performance and security of quantum networks.

56 Future Directions and Challenges

While significant progress has been made in developing key systems for fibered quantum networks, several challenges remain. These include achieving long-lived, high-fidelity quantum memories compatible with telecom wavelengths; developing scalable, integrated quantum photonic platforms; and implementing robust synchronization techniques over large-scale networks.

Emerging research focuses on hybrid systems combining different physical platforms, such as solid-state emitters coupled to photonic circuits, and quantum error correction protocols to mitigate loss and decoherence. Overcoming these hurdles will be crucial for realizing practical, global quantum communication networks.

57 Network-Integrated Continuous Single Photon Detection Architecture

High-efficiency, low-noise single-photon detectors are critical for heralding entanglement and performing quantum measurements.

The current state-of-the-art systems include:

- **Superconducting Nanowire Single-Photon Detectors (SNSPDs):** Offer detection efficiencies above 90%, low timing jitter, and low dark counts.
- **Transition Edge Sensors (TES):** Provide photon-number resolution, beneficial for complex quantum protocols.
- **Microwave Kinetic Inductance Detectors (MKIDs):** superconducting photon detectors that work by changing the resonant frequency of a superconducting LC circuit when photons strike it. This change is caused by Cooper pair breaking, which temporarily alters the inductor's inductance, thus shifting the resonant frequency. MKIDs are advantageous for astronomical observations because they can simultaneously measure photon arrival times, energy, and can be easily multiplexed into large arrays for increased sensitivity. Ideal for energy and temporal resolution.

a single detector combining the precise energy measurement of a Transition-Edge Sensor (TES) and the ultrafast timing of a Superconducting Nanowire Single-Photon Detector (SNSPD) has not been described as a standard technology. Instead, researchers achieve similar goals by using multiple devices, such as arrays of SNSPDs for photon-number resolution, or combining different detector types to leverage their complementary strengths, such as integrating SNSPDs with other photon counting devices like Microwave Kinetic Inductance Detectors (MKIDs) or even silicon-based technology to create hybrid systems.

TES detectors measure the total energy deposited by photons, which then triggers a large temperature rise that is detected as a change in the sensor's resistance. SNSPDs operate differently, detecting a single photon by breaking Cooper pairs and creating a resistive hotspot in a superconducting nanowire, which is then read out with high-speed electronics. MKIDs use a microwave probe signal is used to monitor the resonant frequency shifts. Each detector in an MKID array has a unique resonant frequency, allowing for frequency-domain multiplexing to read out thousands of detectors simultaneously How the Goals Are Achieved

57.1 Complementary Strengths

The intrinsic advantage of TES is its excellent energy resolution, which allows for photon-number resolution, while SNSPDs offer superior speed and timing resolution for detecting individual photons.

MKIDs are essentially superconducting microresonators with a high quality factor (Q). MKIDs provide both precise arrival times for photons, down to the microsecond level. They can measure the energy of incoming photons, which helps to determine their wavelength.

By interleaving multiple SNSPD pixels on a single chip, systems can achieve photon-number resolution similar to TES, but at much higher speeds than traditional TES.

Combining different detector technologies allows for a broader range of capabilities. For example, the integration of SNSPDs with metasurfaces and other optical components to create photon-counting reconstructive spectrometers that use SNSPDs for photon detection.

Instead of a single physical device, systems are designed to include both types of detectors, allowing them to work together to achieve the desired functionality, such as a TES for energy resolution and an SNSPD for timing measurements. An MKID can be useful as a happy medium between the two.

Further Integration of these detectors with fiber-coupled systems ensures minimal loss and high-speed operation.

The key technology required for all of these detectors is superconducting materials that can be etched into detector circuits, usually interdigitated, and operate either in DC or AC mode. The latter of which requires systems integration within microwave components owing to the resonance frequency of many of the transitions in superconductors lying in this range.

by far the most important detector component to consider that matches the complimentary strengths of superconducting single-photon detectors relies on the Josephson Junction, formed when an insulating layer is formed between 2 superconducting layers so that quantum tunneling of the cooper-pair can occur in resonance. In the electric field of a single photon this resonance can be disturbed.

Cryogenic cooling is required in the milliKelvin range for most superconducting detectors however developments during the PhD have showed promise of using new materials that employ intrinsic Josephson junctions within Cuprate superconductors (BSCCO and TBCCO) that can in principle work around the liquid Nitrogen range.

This is a highly significant step for scalability as it means that dilution refrigeration the form factor of a cabinet can be replaced with desktop even data center shelf sized components using 90K miniature cryocoolers that have a long history of development and use in the field of infrared cameras and detectors.

In short it means on a KPI front we have in view a solution to 1 scalability of quantum networks in data centers, 2 the bridging of optical to microwave in quantum comms, 3 the use of off-the-shelf cryogenics as well as well as 4 the ability to employ semiconductor-based solutions to quantum optics that can be operable with photonics which also require cooling.

These KPIs are as always dependent on the experimental testing of josephson junctions for use in quantum networks

Experimental Testing of Intrinsic Josephson Junctions (IJJs) as combined Detectors and Entanglement Resources for Quantum Telecoms

This document explores a potential setup we may use involving the integration and testing of BSCCO and TBCCO High-TC Josephson Junction Arrays, cryocooled to 90 Kelvin and pumped with a 6Ghz source to generate signal and idler beams in the X-band.

Based on the block diagram below we can treat the Josephson Junction Array as an element in a standard VNA/Phase Estimation setup by integration of a mixing stage in the received signal with the "stored/delayed" Idler signal.

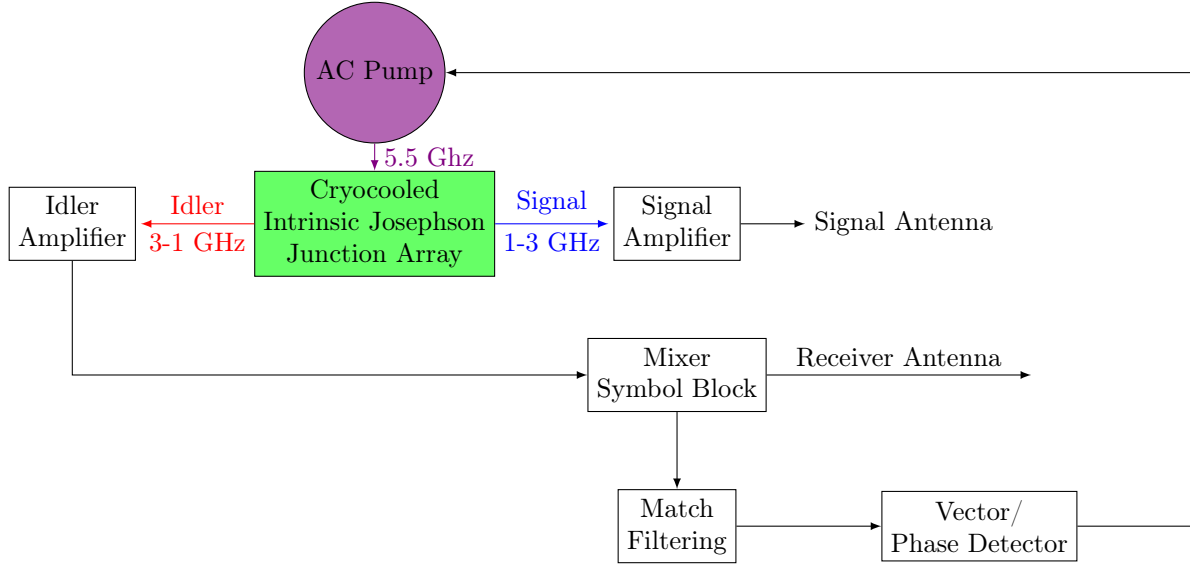
By using a signal antenna that can utilise both polarizations we can transmit the signal in a superposition of polarization states that is then selected when the receiver and idler signals are mixed and passed through matching filters.

The motivations for exploring the use of High-TC ceramics such as YBCO, BSCCO and TBCCO for use in Josephson Junctions are many. for one reason, they offer an attractive resource for entanglement generation which is itself a resource for building quantum networks for sensing and communications wherein we want to make use generally of the microwave band for its resilience to environmental disturbances when compared to optical and Near-Infrared for quantum communications. In terms of specifics, Josephson Junctions made using High-TC superconductors require established and scalable cooling technologies, in some cases requiring no-liquid coolant and operable at liquid N2 temperatures. Ultimately however it is the potential brightness of these entanglement sources that is attractive, in particular BSCCO has a unique property of the crystal structure forming Intrinsic Josephson Junctions, IJJs. This has gained interest from

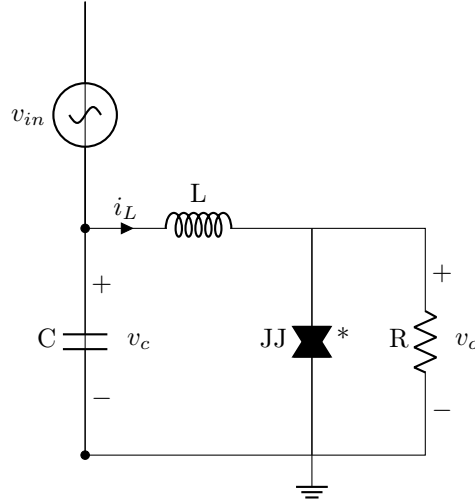
Terahertz optics as a source of coherent THz light when an external DC field is applied. Applying an external AC field however can induce a non-linear effect in Josephson Junctions which is multiplied in an IJJ many times throughout the crystal structure. Generating Josephson Junction Arrays in SQUID and SNAIL type structures is expensive and laborious however BSCCO and TBCCO can effectively generate

nanoscale IJJs for free and in testing BSCCO crystals integrated into bulk elements we can begin to examine just how versatile they may be in terms of an entanglement resource in the microwave band in similar fashion to the field of optics where intrinsically periodically poled non-linear crystals are used to great effect to create entangled beams in the NIR for telecoms.

Below we showcase a proposed test setup for a cryocooled IJJ Array that generates signal and idler microwaves in the X band:



Josephson Junction Equivalent Circuit:



The current through the Junction will form a phase-dependent current, $I \sin \phi$ which is the The Josephson AC supercurrent due to the Cooper pair tunneling through the junction. Since the junction operates at a temperature above absolute zero, there will be a noise current $I(t)$ superimposed on the bias current $I(dc)$.

the intrinsic Josephson junction will also have some inductance due to transient magnetism generated throughout the crystal by vortices forming throughout the lattice.

Intrinsic Josephson junctions are formed in layered superconductors, such as high-temperature superconductors, where the superconducting layers are separated by insulating layers. When a voltage is applied across the layers, vortices can form in the superconducting layers, which can induce transient

magnetic fields. These magnetic fields can lead to the generation of an effective inductance in the Josephson junction. This phenomenon is known as the dynamic inductance of the intrinsic Josephson junction.

58 Round Robin Measurement of Superconductivity Using a 5.5 GHz Source

In this study, we evaluate the microwave surface resistance (R_s) of high-temperature superconducting thin films of $Tl_2Ba_2Cu_2O_{8-x}$ (Tl-2212) using a 10 GHz microwave source. The samples consist of 400 nm thick Tl-2212 films coated on both sides of a 0.5 mm thick $LaAlO_3$ substrate, with a diameter of 50 mm (2 inches). These research-grade superconductors, manufactured by DuPont, have been stored in dry conditions for approximately 15 years, yet exhibit low microwave surface resistance indicative of their high quality.

We also wished to compare the thin TBCCO on LAO films with diodes constructed of BSCCO Superconductor and silver in a NSN Junction Circuit. These were intended to be used in an In-fiber configuration, perhaps in novel cryogenic photonics akin to BSCCO mesa structures being developed for THz applications etc.

At low temperatures ($T < 100$ K), we observed dips in dV/dI at zero bias and anomalous peaks near the bias of $2\Phi_0/e$ (where $2\Phi_0$ is the gap energy of superconducting junction) and at higher bias V_c . The zero-bias dips are supposed to originate from Andreev reflections of electrons and the peaks near $2\Phi_0/e$ from the interference of quasiparticles inside S. We attribute the peaks at V_c to the transition of the superconducting region to the normal state as the current exceeds the critical current I_c of S.

Spintronics with NSN Junction of one-dimensional quantum wires : A study of Pure Spin Current and Magnetoresistance We demonstrate possible scenarios for production of pure spin current and large tunnelling magnetoresistance ratios from elastic co-tunnelling and crossed Andreev reflection across a NSN superconducting junction.

Intrinsic Josephson junctions in BSCCO as potential source of quantum entangled microwaves using a portable Stirling cryocooler setup

It is also possible to use the setup as a detector of photons using the intrinsic Josephson junctions (IJJs) in the BSCCO crystal lattice.

When supplied with an AC source of 10 GHz (close to the characteristic resonance frequency of the BSCCO element) Cooper pairs will oscillate in the CuO superconducting layer and tunnel through the insulator layer. As they tunnel they will split into 2 electrons and, by conservation of momentum, emit 2 entangled microwaves. The energy barrier for tunneling from the superconducting to classical conductive state is fixed at constant temperature below the superconducting phase transition of 110 K. In the presence of an infrared photon, that heats the crystal in a region the superconducting phase will be "switched off" in this region and the electrons will split from Cooper pairs to electrons with separate spins and emit entangled microwaves in a pulse.

The setup depends critically on a Portable Cryocooled Miniature desktop Stirling cryocooler (lowest temperature rated 70 K) built originally by Ricor Inc for testing infrared camera CCDs but repurposed by me to work on 24V DC for cooling semiconductor crystals and superconducting ceramic crystals

X4 BSCCO NSN (Normal metal- Superconductor- Normal metal) Junctions - Made using sintered BSCCO High-Temp Superconductor Crystals at CRANN TCD circa 2014. Contain Intrinsic Josephson junctions for potential use in entangled microwave generation

X2 TBCCO NSN on LAO made by Dupont

X2 RTL SDR USB Dongles - For use in detection of twin microwaves emitted from the crystals. Possible to use software to count for correlations

Custom DC voltage supply to bias the NSN junction

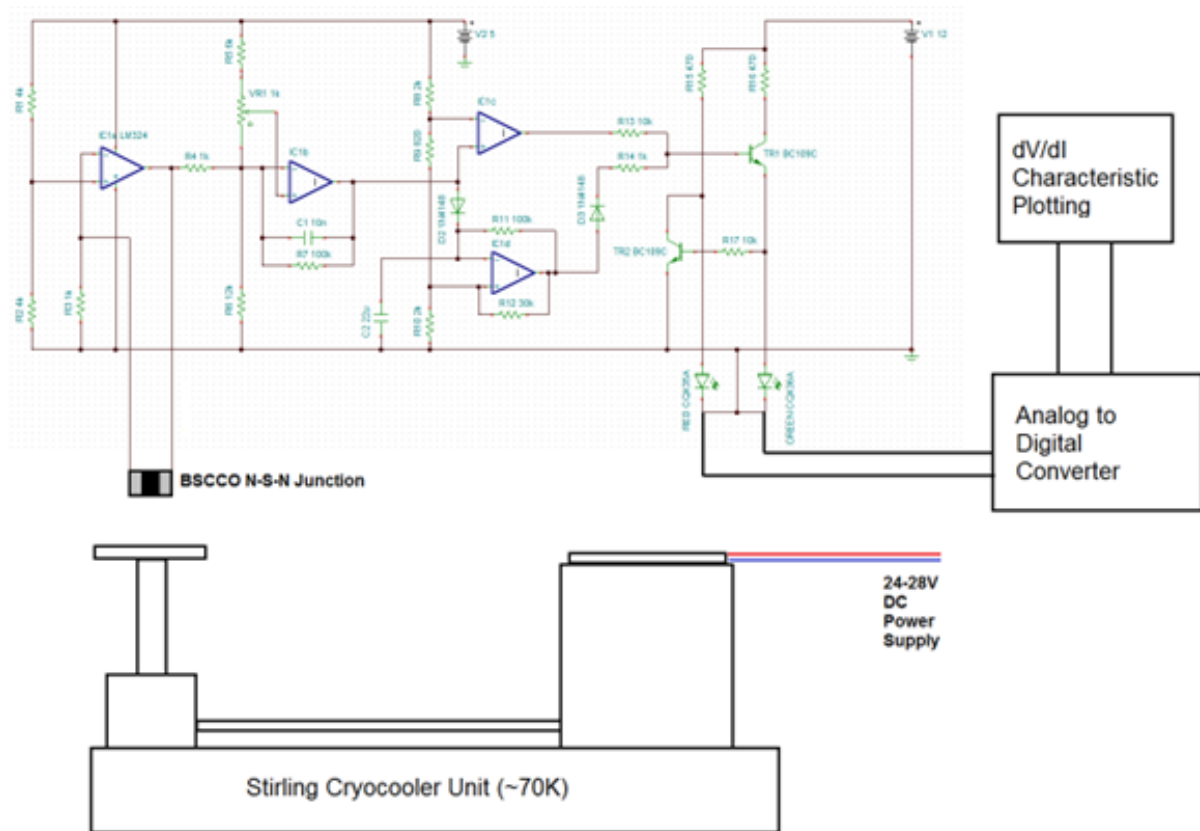


Figure 27: Intrinsic Josephson Junction for use in quantum photonics test setup, involving IV curve tracer and cryogenic cooler rated for use at 90 Kelvin making the system cryogen free

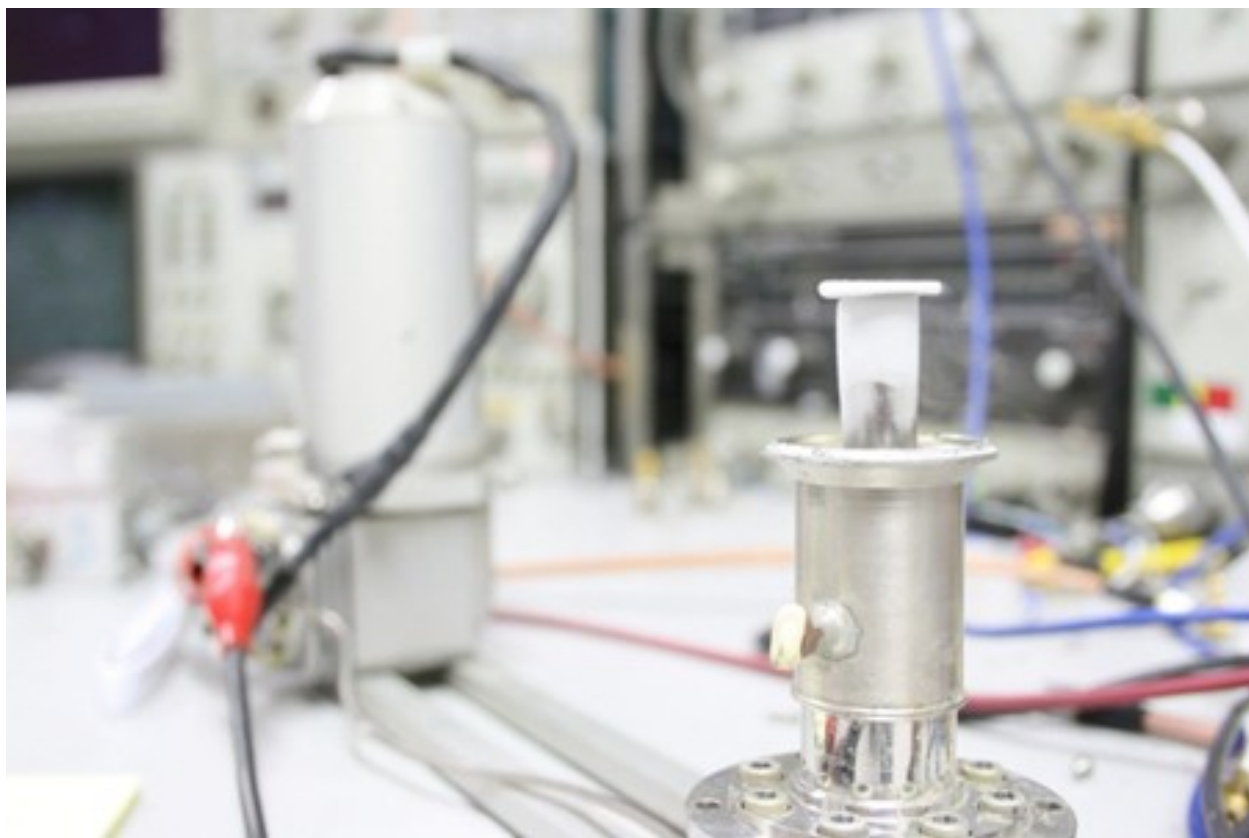


Figure 28: 90 Kelvin cryogenic free mini cryocooler for testing High Temperature Cuprate IJJs

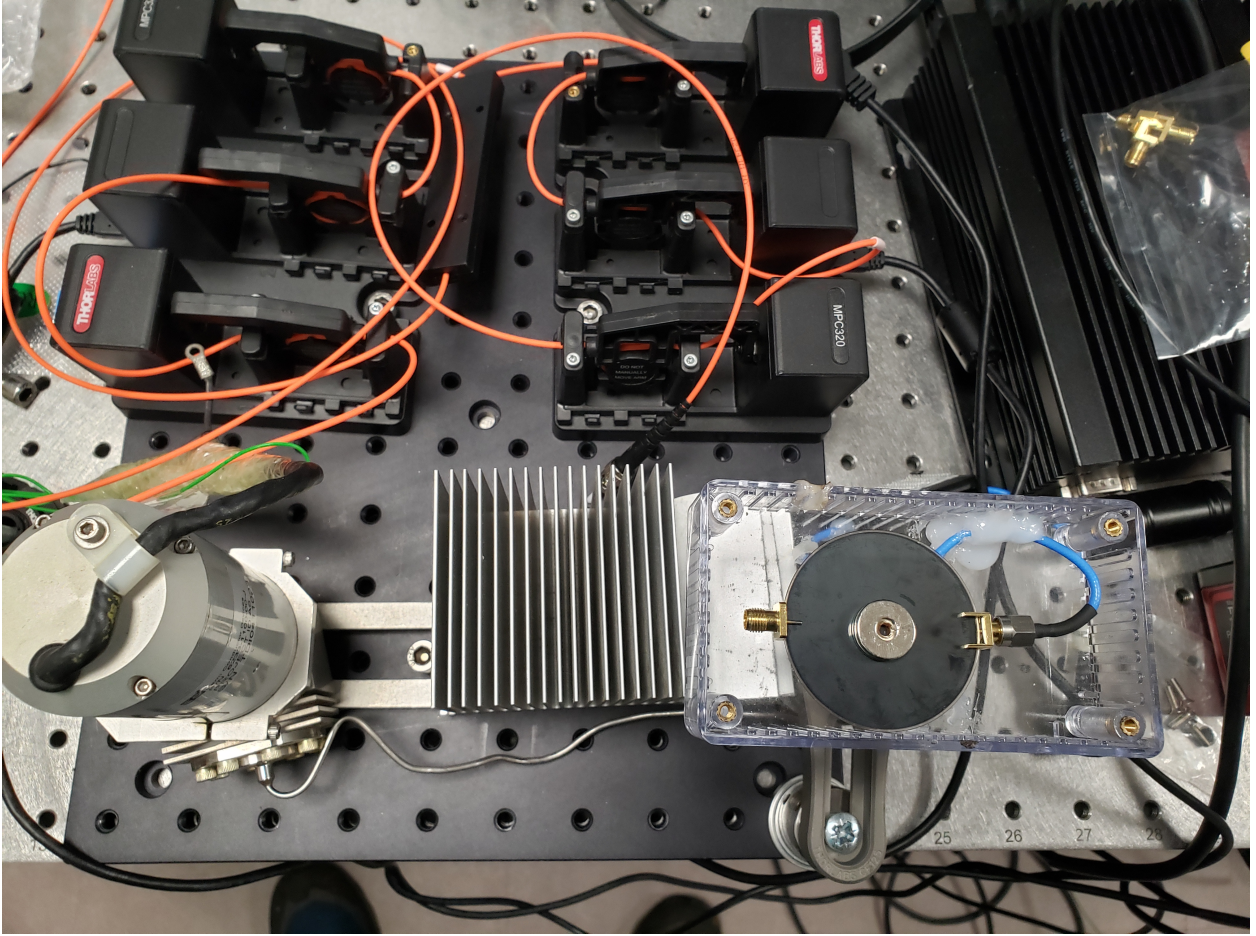


Figure 29: 400nm TBCCO thin film resonator on LAO disk (with SMA connections) substrate for testing on the cold head of 90 Kelvin cryogenic free mini cryocooler integrated on optics bench with polarization maintaining and control fiber optic network integration with quantum optics resources

This Setup is inherently noisy which may not be a bad thing considering starting with a demonstrated exercise between limiting classical sources of noise in the mechanics of the system and observing how this effects the quantum noise of the correlations (or anti-correlations such as they are) may yield some interesting results and be a guidepost of sorts.

58.1 TBCCO Sample Description and Fabrication

The superconducting films were prepared following the methods described by Holstein *et al.* [1], involving deposition techniques that yield high-quality, epitaxial Tl-2212 films. The multilayered structure, with coatings on both sides of the substrate, enhances the effective interaction with the microwave field, facilitating accurate measurements. The thin films are deposited on LAO which has crystal axis matching the TBCCO along with strong structural integrity to the vibrations induced when mounted on the cryogen free stirling cryocooler as shown in fig 27

58.2 Measurement Technique

The measurements were conducted following the round robin protocol outlined in IEC 61788-7 [?], which standardizes the procedure for microwave surface resistance measurements of superconductors at

microwave frequencies. The test setup employed a sapphire resonator designed for a frequency of 5.5 GHz; however, for the current study, the measurements are extrapolated or adapted to 10 GHz.

58.3 Theoretical Background

The surface resistance R_s of a superconductor at microwave frequencies is related to the complex conductivity $\sigma = \sigma_1 + i\sigma_2$, via the following relation:

$$R_s = \frac{\omega\mu_0}{2} \operatorname{Re} \left[\frac{1}{\sigma} \right], \quad (104)$$

where $\omega = 2\pi f$ is the angular frequency, f is the operating frequency (10 GHz), and μ_0 is the permeability of free space.

In the local limit ($\lambda \gg \xi$), the surface resistance simplifies to:

$$R_s \approx \frac{\mu_0^2 \lambda^3 \omega^2}{2\rho}, \quad (105)$$

where λ is the London penetration depth, and ρ is the normal-state resistivity.

58.4 Round Robin Measurement Approach

The round robin protocol involves multiple laboratories measuring the same set of samples to ensure reproducibility and standardization. The measurements at 10 GHz are conducted using a resonator method, where the sample's surface resistance influences the quality factor (Q/Q_0) of the resonator:

$$R_s = \frac{G}{Q_s} - \frac{G}{Q_0}, \quad (106)$$

where G is the geometric factor of the resonator, Q_0 is the unloaded quality factor, and Q_s is the quality factor with the sample inserted.

For a cylindrical resonator, the geometric factor G can be calculated as:

$$G = \frac{\omega\mu_0 \int_V |\mathbf{H}|^2 dV}{\int_S |\mathbf{H}_t|^2 dS}, \quad (107)$$

where $\mathbf{H}$ is the magnetic field distribution, and the integrals are over the resonator volume V and the sample surface S .

58.5 Results and Discussion

Preliminary measurements indicate that the R_s at 10 GHz remains low, consistent with prior observations at 5.5 GHz, and the samples retain their superconducting properties with a critical temperature T_c around 105–107 K. It should be noted that the measured R_s values are effective values ($R_{s, \text{eff}}$) obtained without substrate correction, implying actual R_s may be lower upon appropriate correction.

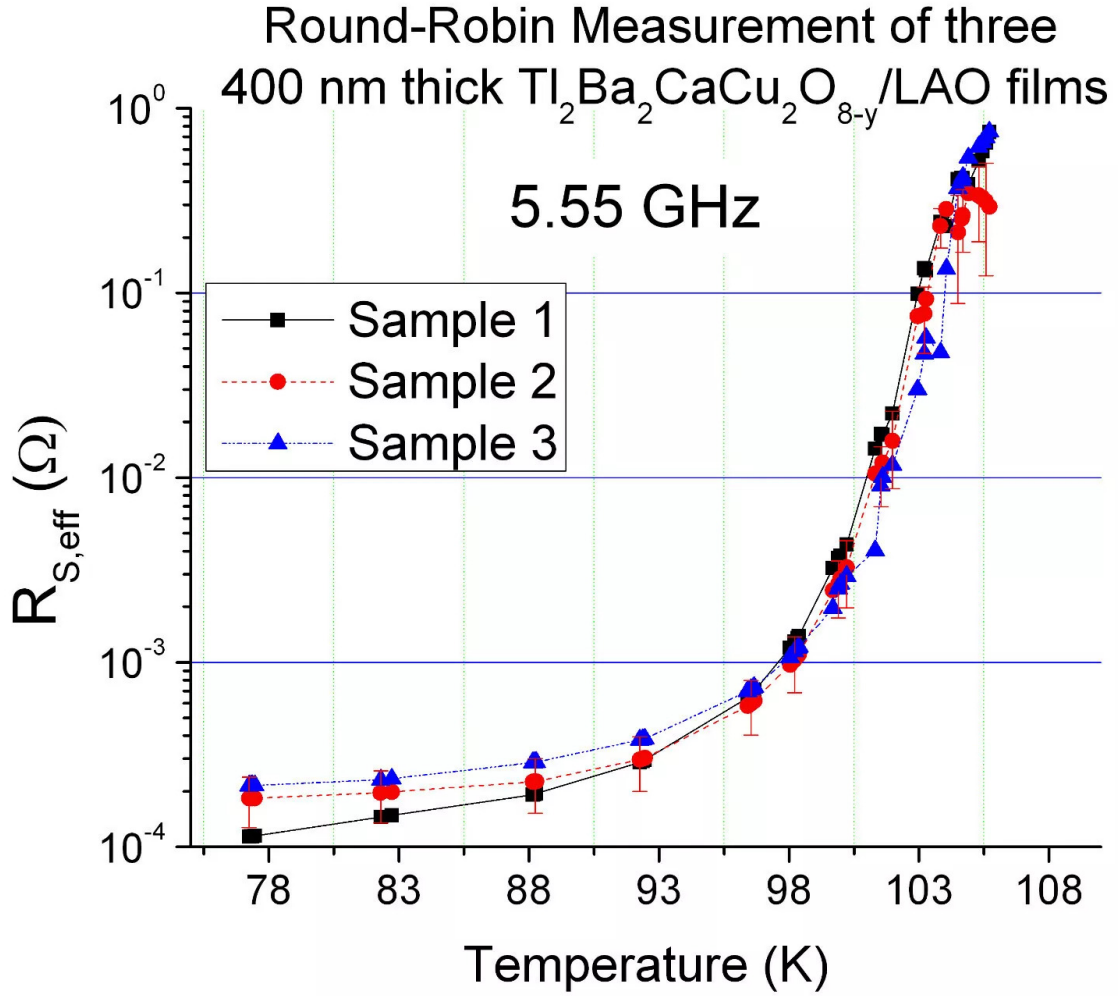


Figure 30: microwave surface resistance measurement on superconducting films using a sapphire resonator. The measurement was performed according to IEC 61788-7 (a standard for superconducting materials), but at a frequency of 5.55 GHz instead of the standard's typical frequency of 6GHz [?] This was done as part of a three-sample round robin, likely to ensure measurement accuracy and repeatability between different labs or systems. The specific measurement technique involves a sapphire dielectric resonator with a copper shield, where high-temperature superconducting (HTS) TBCCO films are used for the endplates

58.6 Conclusion

The round robin measurement method provides a robust means to assess the microwave surface resistance of high-temperature superconductor thin films at 10 GHz. The high quality and stability of the Tl-2212 films over extended storage periods demonstrate their suitability for microwave applications, with the measurement framework established here serving as a baseline for further reproducibility studies.

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59 Quantum Repeaters

Quantum repeaters extend the range of quantum communication by segmenting the channel into shorter links, entanglement swapping, and purification.

- **Entanglement Swapping:** Connects shorter entangled links into a longer one via Bell state measurements.
- **Protocols:** Protocols such as the Duan-Lukin-Cirac-Zoller (DLCZ) scheme utilize atomic ensembles and linear optics for efficient entanglement distribution.

Implementation challenges include maintaining coherence during entanglement swapping and minimizing loss.

60 Open Loop Ends and Future Directions

While significant progress has been made, several open questions remain:

- **Integration Challenges:** Developing fully integrated, chip-scale quantum networks combining sources, memories, detectors, and repeaters remains a key goal.
- **Loss and Decoherence:** Improving fiber and component efficiencies to reach fault-tolerance thresholds.
- **Hybrid Systems:** Combining different physical systems (e.g., atoms, solid-state emitters, and photons) to leverage their unique advantages.
- **Standardization and Scalability:** Establishing protocols and architectures that support large-scale deployment.

Addressing these issues will enable practical, robust, and scalable fibered quantum networks, bringing us closer to widespread quantum communication and distributed quantum computing.

However the need for network stabilization hardware cannot be overstated and interferometry adapted to fiber networks is a key consideration here. A setup involving the modified Franson interferometer, with Sagnac loops replacing the traditional Mach-Zehnder arms, allows us to test network synchronization as viewed through the lens of quantum networking.

61 Polarization Stabilization Using Sagnac Interferometry Nodes in Fiber Networking and Sensing

This work investigates how phase synchronization among fiber network nodes, quantified by the parameter R , the degree of synchronization between nodes in a network from the Kuramoto model and how R between Sagnac interferometer nodes influences the polarization properties of transmitted light between these interferometers. Using concepts from oscillator synchronization, coherence, and polarization of optical fields, we develop a physical intuition connecting the degree of synchronization R , the State of Polarization (SoP), and the Degree of Polarization (DoP).

In fiber networks containing Sagnac interferometers as stabilizer nodes, phase coherence between these nodes leads to transfer of stable polarization states, essential for some forms of information transfer being explored for classical and quantum networking. When

$$R \rightarrow 1$$

, the fields are phase-aligned, producing high DoP, whereas

$$R \rightarrow 0$$

results in incoherent superpositions with decreased polarization purity.

Understanding this connection offers a pathway to optimize polarization entanglement distribution and polarization stabilization in quantum networks.

In the Kuramoto Model for N oscillator nodes in a network the degree of synchronization between nodes is given by the general formula:

$$Re^{i\psi} = \frac{1}{N} \sum_{k=1}^N e^{i\theta_k} \quad (108)$$

where:

[2] θ_k : phase of the k^{th} oscillator

[3] $R \in [0, 1]$: magnitude indicating synchronization.

Considering polarized light is an Electric Field Superposition

$$E(t) = \sum_{k=1}^N |E_k| e^{i(\omega t + \phi_k)} \mathbf{e}_k \quad (109)$$

- $|E_k|$: amplitude
- ϕ_k : phase
- $\mathbf{e}_k$: polarization vector

High R between synchronized polarized light nodes in fiber loops implies ϕ_k are locked, leading to stable polarization; low R causes incoherence and polarization fluctuations.

[4] When phases are aligned (

$$R \rightarrow 1$$

), the superposed electric field is highly coherent, resulting in a well-defined polarization state with high DoP.

[5] When phases are random (

$$R \rightarrow 0$$

), the superposition is incoherent, leading to depolarization or unpolarized light with low DoP.

[6] The superposition can be seen as a weighted combination of polarization states:

$$\mathbf{S}_{\text{super}} \propto R \hat{\mathbf{S}}_A + (1 - R) \hat{\mathbf{S}}_B$$

with R representing the degree of phase coherence.

Define fixed states:

$$\begin{aligned} \mathbf{S}_A &= [1, 1, 0] \quad \text{(Horizontal linear)} \\ \mathbf{S}_B &= [1, 0, 0] \quad \text{(Circular right)} \end{aligned} \tag{110}$$

Normalize:

$$\hat{\mathbf{S}}_A = \frac{\mathbf{S}_A}{\|\mathbf{S}_A\|}, \quad \hat{\mathbf{S}}_B = \frac{\mathbf{S}_B}{\|\mathbf{S}_B\|}$$

Choose R values:

$$\{1, 0.9, 0.7, 0.5, 0.3, 0.1, 0\}$$

Initialize list

superpositions

for each R in R-values Compute superposition

Results of the simulation here

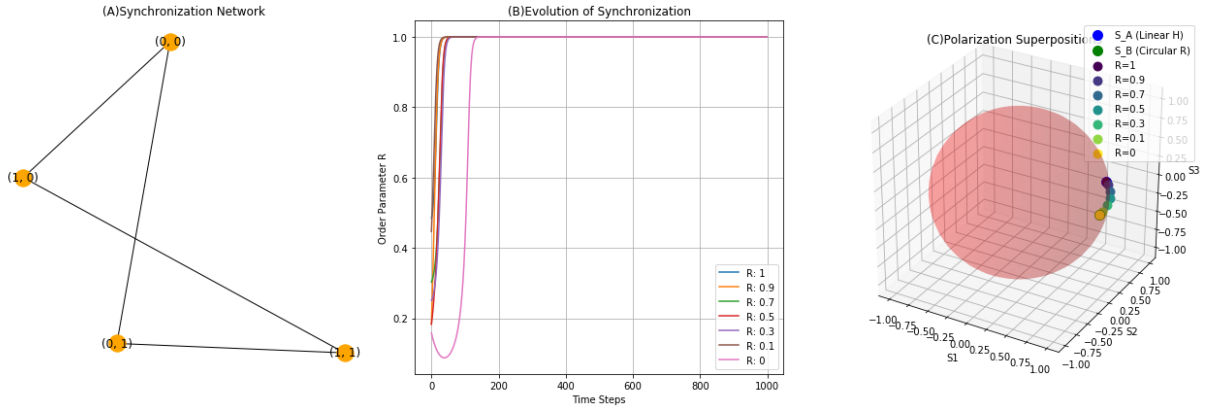
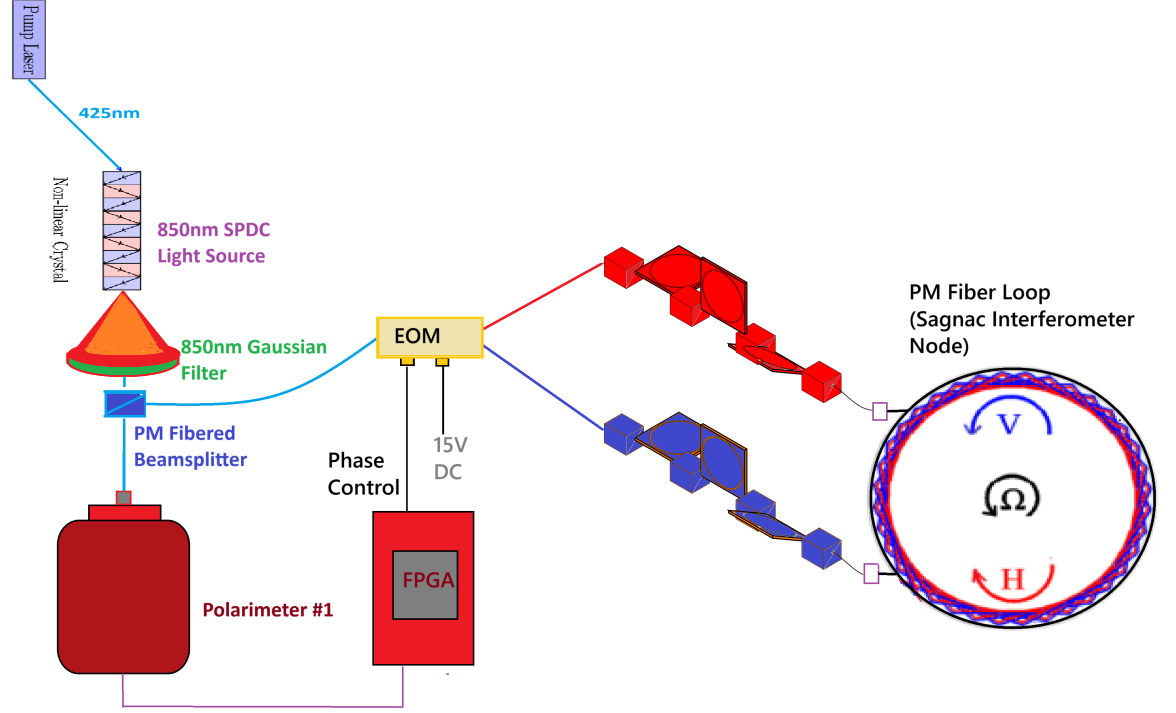


Figure 31: (A) Simulation of a Network of Sagnac Loops in Synchronization, with each loop carrying 2 States of Polarization (B) Evolution of Synchronization over a list of degrees of synchronization, R , in the Kuramoto Model (C) The two fixed polarization states, on the Poincaré sphere. Showing how their superposition changes as R varies from 1 (full coherence) to 0 (no coherence). Illustrating the resulting superposed Stokes vector for different R values. The blue point at the equator (linear horizontal). The green point at the north pole (right circular). As R decreases, the superposition point moves from the S_A position towards a mixture closer to S_B , eventually collapsing towards the middle (less polarized state). When $R=1$, the superposition coincides with S_A ; when $R=0$, it coincides with S_B .

62 Experimental Test Setup



polarization squeezing setups have been sensitive to external fluctuations and required careful tuning. Recent developments have led to a new all-fiber setup that is simple, robust, and stable. This setup consists of passive elements only, making it easy to set up and maintain using coded states. Active tuning is possible however to mitigate external noises, using an FPGA and EOM for phase control (and in principle synchronization between multiple Sagnac nodes in a real network). The proposed experiment can leverage a polarization-squeezed state generated from a BBO Spontaneous Parametric Down-Conversion (SPDC) source of light, incorporated into an optical fiber Sagnac interferometer. The characteristics of the Sagnac interferometer will allow for the measurement of the polarization states and enable the exploration of the effects of polarization squeezing within the fiber. Even environmental effects, both passive such as orientation to Earth's rotation Ω , and/or seismic rotational effects.

63 Polarization Shift Relation to Action Derivatives

Based on the established relation between frequency shifts and the derivative of the action, (i.e. the Doppler effect applied to the Lagrangian) we propose that the polarization shift

$$\Delta P$$

in a fiber network can be modeled as being proportional to the rate of change of the action:

$$\Delta P \propto_P \frac{\partial S}{\partial t}$$

where

$$P$$

is a proportionality constant encapsulating how sensitive the polarization state is to variations in the physical parameters affecting the system. This relation suggests that physical disturbances—such as

stress, temperature fluctuations, or fiber imperfections—that alter the underlying action will directly influence the polarization state of the transmitted light.

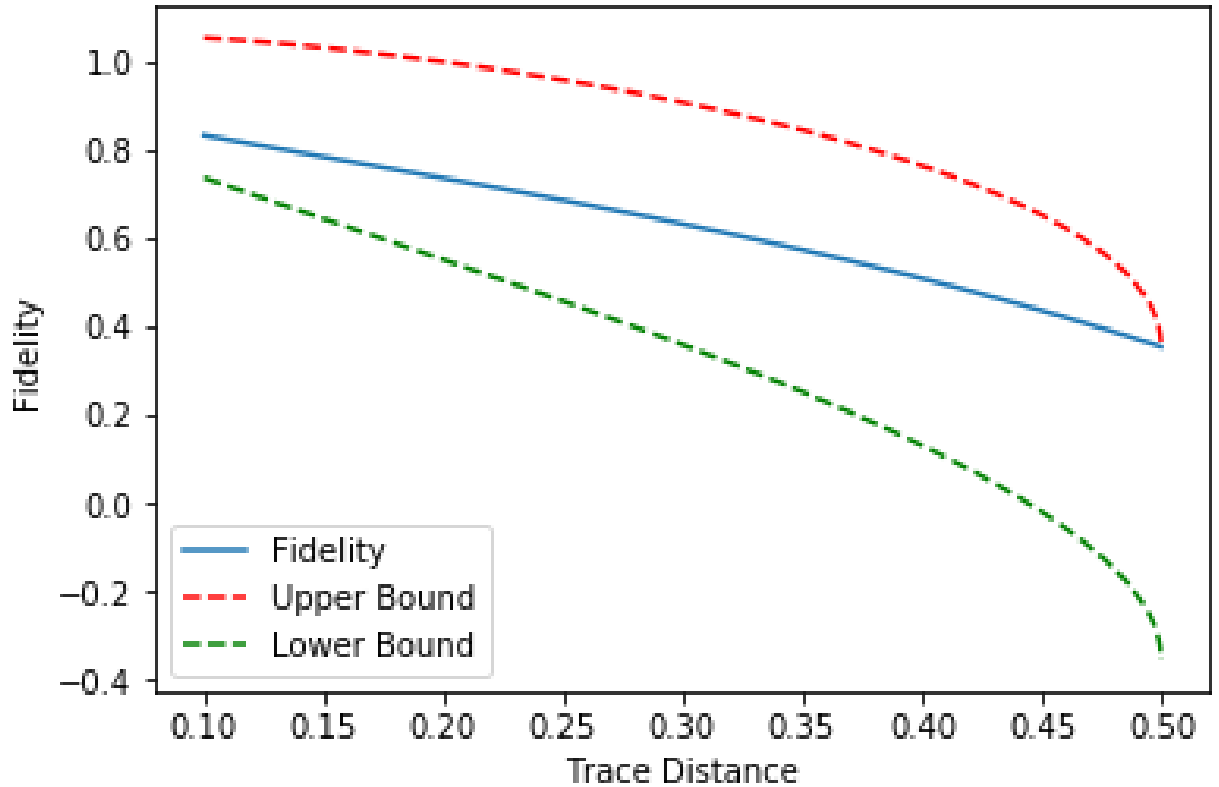
Physical Consequences

In fiber networks, this relation highlights the importance of controlling environmental and physical factors, as they can induce changes in the system’s action, leading to polarization drift or scrambling. Such polarization shifts can degrade the performance of polarization-sensitive applications, including high-speed communications, quantum key distribution, and coherent detection schemes. Understanding and modeling the relation between action derivatives and polarization can help develop more robust fiber systems with enhanced stability against external perturbations.

64 R vs Sop and Dop measurements using a polarimeter and FPGA with and without EOM phase tuning

From our point of view, the upper bound of R vs DoP can serve as a measure of the error level of a quantum channel. a higher upper bound ensures a lower probability of information leakage due to error or de-synchronization of the measurement setup, i.e. by problems with Sagnac node synchronization.

The lower bound can provide a measure of the potential failure of a quantum communication system, wherein de-synchronization has effectively collapsed the exchange of entanglement, i.e. by depolarization of the channel if we are using the physical variable of polarization density (Coulombs/meter) as our entanglement observable and/or measuring the polarization extinction ratio (PER), expressed in dB. [4]
0.5



65 Conclusions

- Quantum Sagnac Interferometry and tomography is possible using off-the-shelf polarization-maintaining optics components
- The R vs Dop can serve as potential networking benchmark to interpret the state reconstruction wrt polarimetry measures i.e. polarization extinction ration (PER) in dB.

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66 Power-Law Based Signalling As a Quantum Signalling Archetype

The most basic signaling we can construct that works for quantum as well as classical light is one based on some sort of power law for signalling. Light signalling can of course be represented by a power law. Light signal transmission, like light in general, obeys the inverse-square law in its propagation through space. That is to say, the intensity of the light, I , is inversely proportional to the square of the distance from the light source.

$$I(r) = \frac{I_0}{r^2}$$

Therefore the light signal gets weaker and weaker as the distance increases.

The meta-heuristic algorithm therefore must contain a function which monotonically decreases its signalling power between each transceiver node with distance, r , under a discrete light absorption coefficient for the physical medium Y . [3]

Under a given physical medium, the Gaussian form of the light intensity is then determined by:

$$I(r) = I_0 e^{-\gamma r^2}$$

We can use this equation of course to find the light absorption coefficient itself:

$$\begin{aligned} \frac{I(r)}{I_0} = e^{-\gamma r^2} &\Rightarrow \ln \left(\frac{I(r)}{I_0} \right) = -\gamma r^2 \Rightarrow \\ \gamma &= \frac{-\ln \left(\frac{I(r)}{I_0} \right)}{r^2} \end{aligned}$$

The generic power law signalling function of is then written to be monotonically decreasing:

$$\beta = \beta_0 e^{-\gamma r^m}$$

Where r is the distance between the different nodes. m is the power law parameter for the signal, for light signals it is $m = 2$. (for signals that use audio, chemical, etc, power laws it will be different) Where β_0 is the emitted signal at $r = 0$

The distance between any two nodes, i and j at $\mathbf{X}_i$ and $\mathbf{X}_j$ respectively, is the Cartesian distance as follows:

$$r_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\| = \sqrt{\sum_{k=1}^d (x_{i,k} - x_{j,k})^2}$$

Where $x_{i,k}$ is the (k) th component of the spatial coordinate $\mathbf{X}_i$ of the (i) th node.

Where d is the number of dimensions.

The time evolution of each individual node, i , over its signalling cycle is the governed by the fact each typical node, i , is coupled to the brightest node it sees, j , by the following equation:

$$X_j = X_i + \left(\beta_0 e^{-\gamma r^2} \right) (X_i - X_j) + \alpha$$

The second term is the signalling term, for most cases in implementation $\beta_0 = 1$

The third term is the randomization term for a possible discrete coupling path which uses a randomization parameter, α , which for most cases in implementation is distributed in the domain of $[0, 1]$.

An interesting way to write this equation is:

$$X_j \left(1 + \beta_0 e^{-\gamma r^2}\right) = \left(1 + \beta_0 e^{-\gamma r^2}\right) (X_i) + \alpha$$

Which when tidied up further reads:

$$(X_i - X_j) = \frac{\alpha}{\left(1 + \beta_0 e^{-\gamma r^2}\right)}$$

We can then define the signalling Activation Function, A:

$$A(X_{i,j}) = \left(1 + \beta_0 e^{-\gamma r^2}\right)^{-1}$$

Which reduces our equation to a more fundamental signal field representation:

$$\Phi(X_{i,j}) = A(X_{i,j}) \cdot \alpha$$

Based on these formulae, we have the corresponding rules:

- Each node in the field will emit and absorb discrete light signals equally.
- The signals are proportional to the intensity of the light emitted, which both decrease in proportion to increasing distance between the nodes.
- For any 2 flashing nodes in the field, the signal absorbed with the highest intensity will induce the strongest coupling
- If there is no signal detected with higher intensity than any one particular node, it will designate itself as being isolated and signal randomly
- The light intensity of any signalling node is determined by the landscape of the field, itself determined by the nature of the activation function itself.

67 Why is this type of signalling Meta-heuristic? - Defining the criteria of the meta-heuristic behavior:

Heuristic algorithms typically intend to find a good solution to an optimization problem by 'trial-and-error' in a reasonable amount of computing time. Here 'heuristic' means to 'find' or 'search' by trials and tallying the hits and misses. There is no guarantee to find the best or optimal solution, though it might be a better or improved solution than an educated guess.

Any reasonably good solution, often suboptimal or near optimal, would be good enough for such problems. Broadly speaking, local search methods are heuristic methods because their parameter search is focused on the local variations, and the optimal or best solution can be well outside this local region. However, a high-quality feasible solution in the local region of interest is usually accepted as a good solution in many optimization problems in practice if time is the major constraint.

Metaheuristic algorithms are higher-level heuristic algorithms. Here, 'meta-' means 'higher-level' or 'beyond', so metaheuristic means literally to find the solution using higher-level techniques, though certain trial-and-error processes are still used. Broadly speaking, metaheuristics are considered as higher-level techniques or strategies which intend to combine lower-level techniques and tactics for exploration and exploitation of the huge space for parameter search.

From the rules derived from the physical parameters used to construct the pre-programmed signalling mechanism we can define that the metaheuristic nature of the signalling algorithm is based the behaviours of exploration and exploitation

Exploration in the metaheuristic algorithm is achieved, in the case of the above signalling algorithm, by the use of randomization which enables the algorithm to have the ability to jump out of any local state of coupling between 2 or more signalling nodes so as to explore the search for more couplings in the network over a potentially larger area.

Randomization is also used for local search around the current state if the signalling are limited to a local region. When the signalling frequency is long, randomization then allows for exploration of the the search space on a larger scale. Fine-tuning the right amount of randomness and balancing local search and global search are crucially important in controlling the performance of the synchronizing metaheuristic algorithm.

Exploitation is the use of local knowledge of the search and solutions found so far so that new search moves can concentrate on the local regions or neighbourhood where the optimality may be close; however, this local optimum may not be the global optimality. Exploitation tends to use strong local information such as gradients, the shape of the mode such as convexity, and the history of the search process.

Observations, using simulations, of the convergence behaviour of common optimisation algorithms suggests that exploitation tends to increase the speed of convergence, while exploration tends to decrease the convergence rate of the algorithm. On the other hand, too much exploration increases the probability of finding the global optimality, while strong exploitation tends to make the algorithm being trapped in a local optimum.

The relationship between exploration and exploitation both can be seen with a simple implementation of the metaheuristic derived above in the context off-sensory exploration and contrasted learning exploitation based on the activation signalling function

The sample code is implemented as:

```

Y=[y(k,1); Y(1:(n-1),1)]; % Learn by contrast - i.e. shift regression
vector and load in new value - exploitation (or endowment)

xhat(k,1)=mean(Y); %Get the mean of the contrastive learning
procedure
end

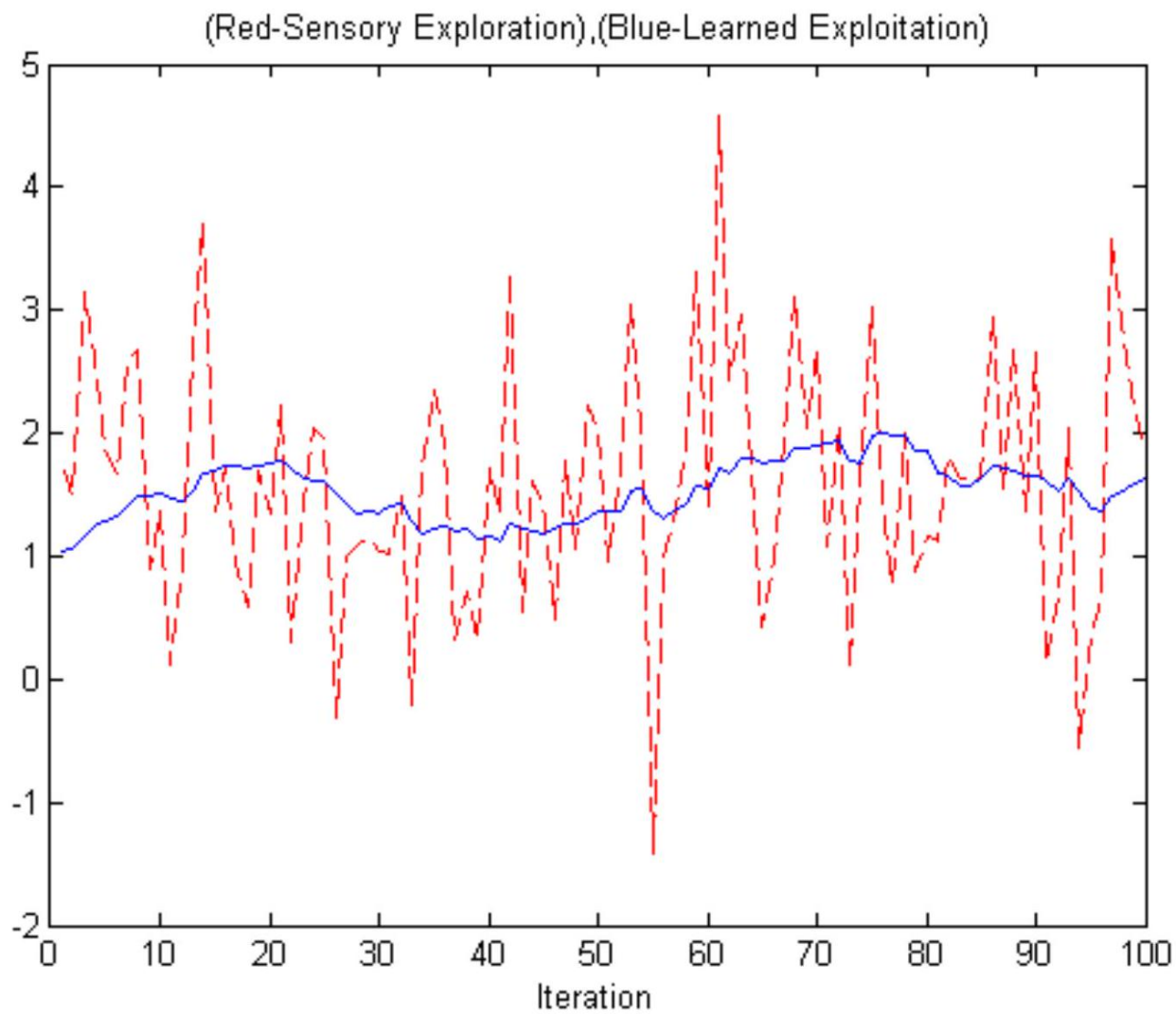
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We plot the exploration and exploitation convergence plots for 100,1000 and 10,000 iterations:
100 iterations:

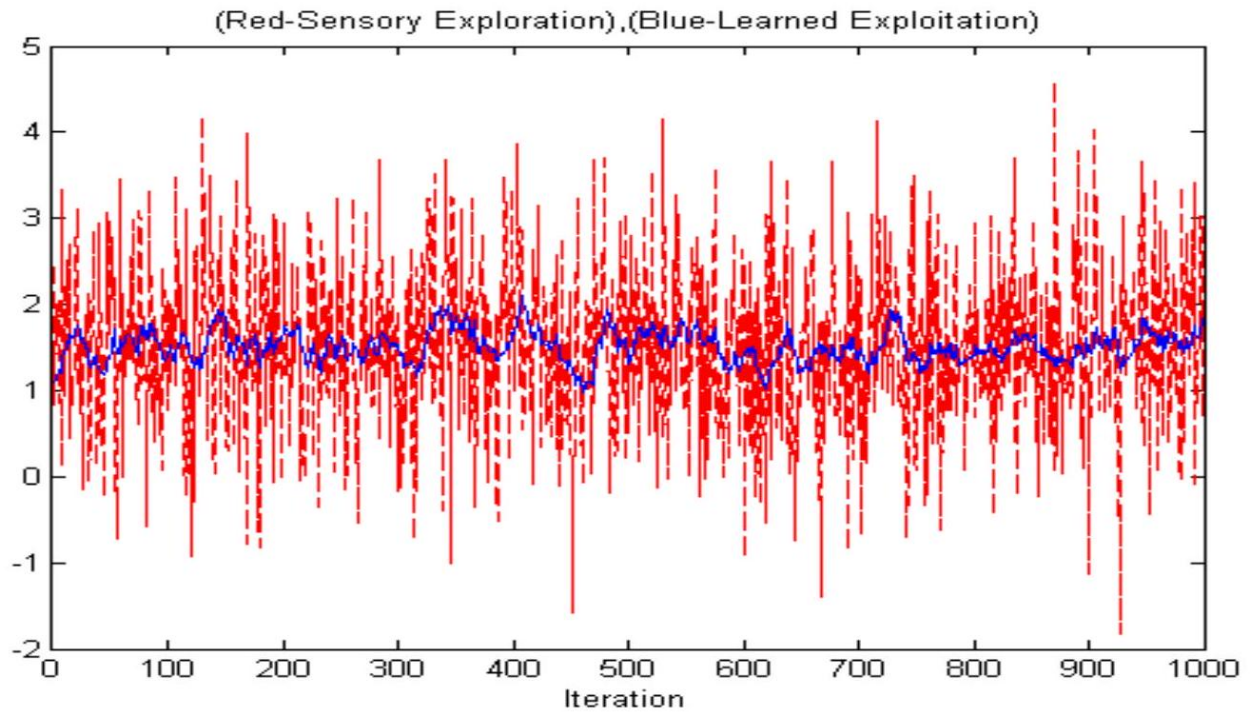
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for k = 1 : N % Generate sensed signals, given that the initial values are instincts to exploit
z(k,1) = sigmaz2(k,1) * (0.5 - randn); % Generate internal signalling
system based on activation function A(r)
y(k,1) = x(k,1) + z(k,1); ÷ Generate input - i.e. an external signal that
is sensed - exploration

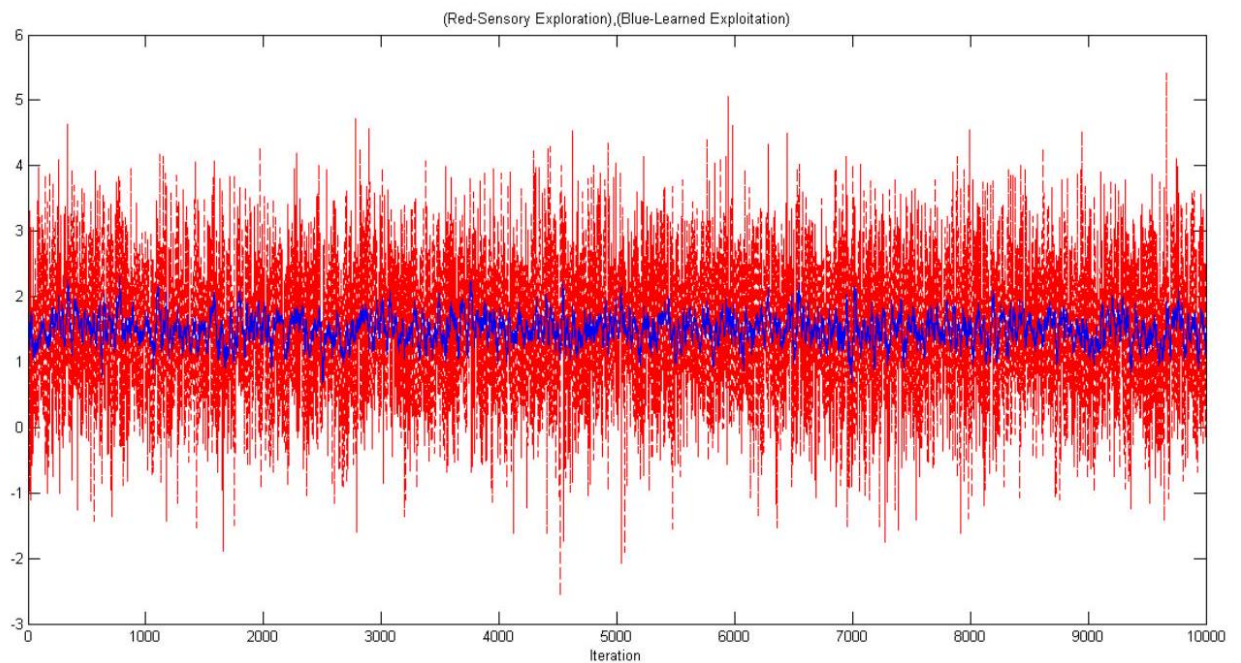
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1,000 iterations:



10,000 iterations:



As seen and discussed, there is a fine balance between the right amount of exploration and the right degree of exploitation. Despite its importance, there has never been any known practical guideline for this balance as regards to learning behaviour.

As we will find out, it is actually not practical at all to find a guideline based on the balances between exploration and exploitation by simply reading convergence plots, which are simply data, which is

at least partially recalcitrant as the overall self-organising behaviour of the system that follows the algorithm has not itself been pre-programmed but nor is the system based entirely on probability.

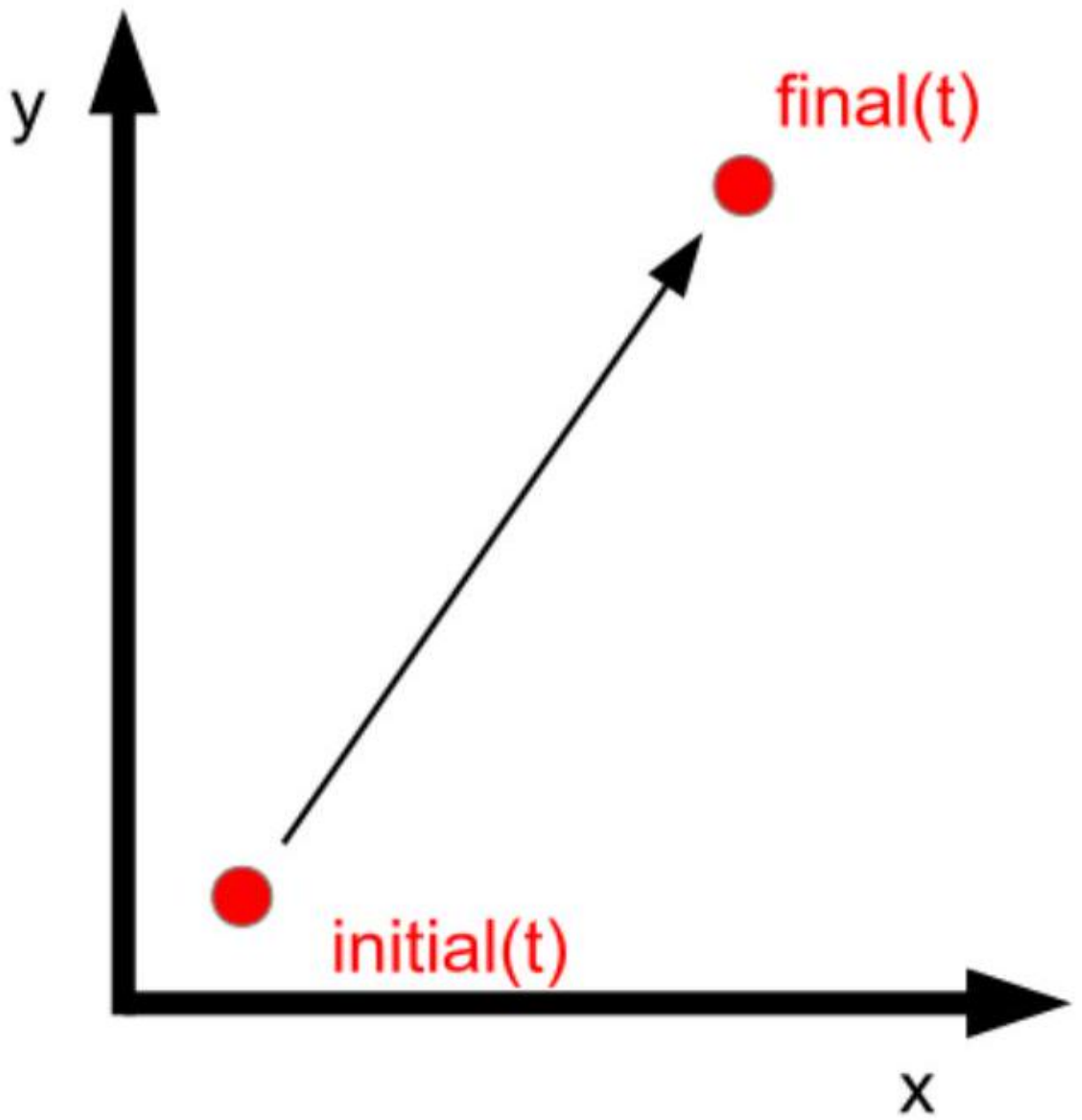
Therefore, it makes more sense to think about the behaviour and data we empirically observe using an abstract model of a physical system that reduces the

behaviour, otherwise taken for granted, to a simple signalling action principle based on the environmental parameters of the system. This then leads to very fascinating consequences and applications of metaheuristics in the realm of quantum information.

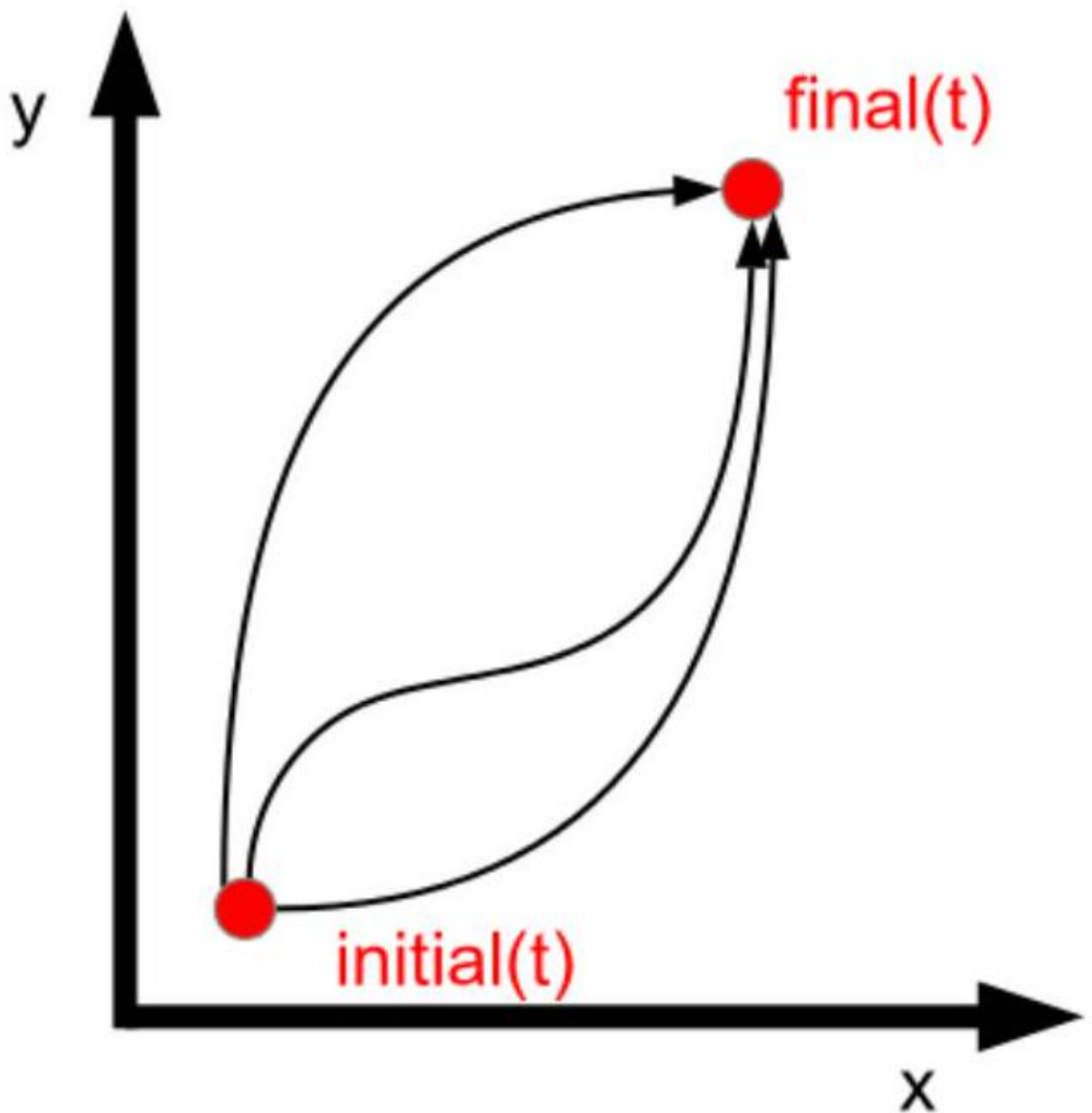
68 Abstraction - The Path Integral Interpretation of Quantum Mechanics:

Path-integral theory is a rigorous formulation of quantum theory that uses a particles path across space-time instead of wave functions. It has particular use in quantum many-body systems as explored in quantum field theory. [2][4]

Imagining a particle at some initial position in the (x,y)-plane, we want to know what path it will take to some final position. By the classical least action principle, the particle will take a path between the two positions that costs the least energy. But, if the particle is a quantum particle, it's not really localized at a point. Instead, the particle is a wave... and it doesn't take one path from the initial position to the final position, it takes all possible paths.



Classical particle



quantum particle

The basic idea then is to construct matrix elements of the time-evolution operator by summing all possible paths between two points and weighting the paths by the classical principle of least action. Remarkably, this gives identical results to solving the Schrödinger wave equation, but is much more concise a model.

The Hamiltonian generates the time evolution of quantum states. If $|\psi(t)\rangle$ is the state of the system at time t , then

$$H|\psi(t)\rangle = i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle$$

This equation is the Schrödinger wave equation

From the Schrödinger wave equation we of course get the time evolution operator to work with. Given the state at some initial time ($t = 0$), we can solve it to obtain the state at any subsequent time. In particular, if H is independent of time, then

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle.$$

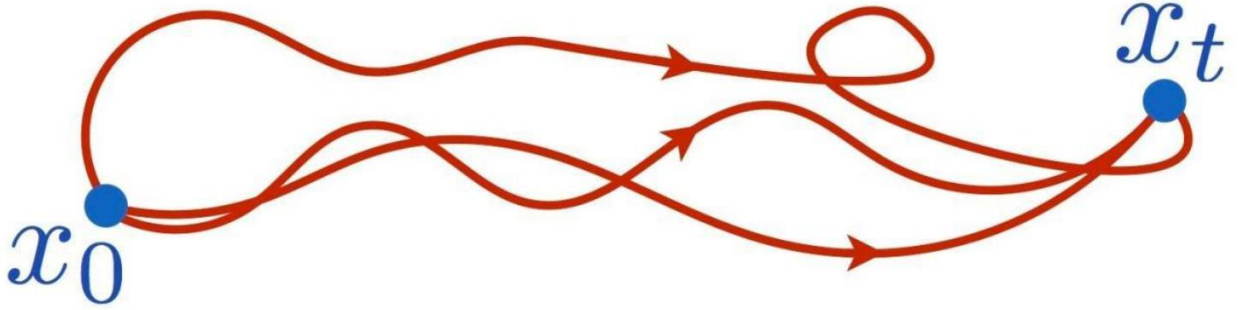
By the homomorphic property of the functional calculus, the operator

$$U = e^{-iHt/\hbar}$$

This is the time evolution operator, or propagator, of a closed quantum system.

Given the time evolution operator we can construct a matrix element which is a probability amplitude for the transition of a particle from state X_0 to X_t

$$\langle x_t | e^{-i\hat{H}t/\hbar} | x_0 \rangle =$$

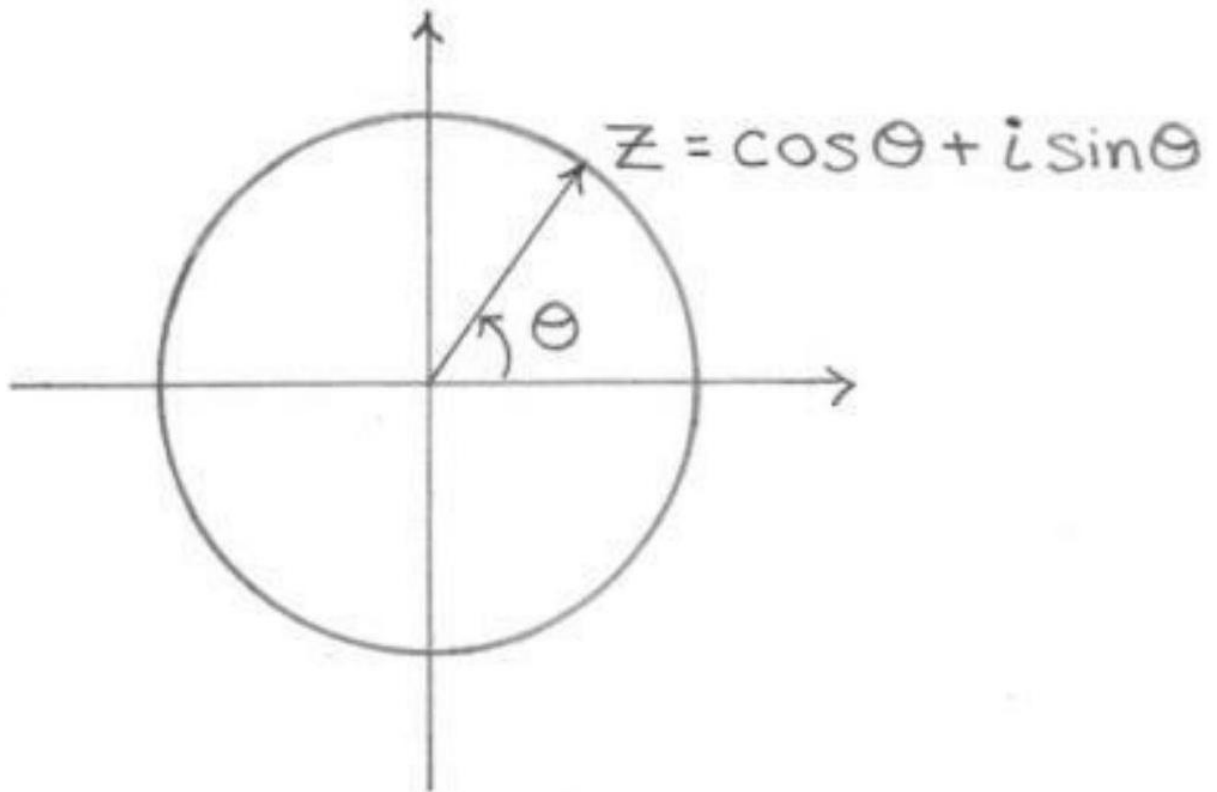


The quantum action, S , is represented here as being the phase of each path being determined by $\int L dt$, for that trajectory, where L is the Lagrangian. Lagrangian mechanics is used for discrete particles each with a finite number of degrees of freedom. Furthermore, the action, S , is a relativistically invariant operator. [2].

Quantum mechanics is a probabilistic theory, hence the principle of least action has to be written to make a probabilistic statement. Suppose we have a quantum particle with some wavelength or frequency (either will do-since they're inverses of each other) which we just measured at some initial point. We want to know the probability of finding it at some final point.

Each path the particle takes is equivalent to a probability wave as described by the Schrödinger wave equation. Waves are in fact represented here in the complex plane as being circles traced out by the wave-form.

This phenomenon is inherently related to the nature of the complex plane, where the complex number $z = \exp(i\theta) = \cos(\theta) + i \sin(\theta)$ acts like a unit vector with a phase angle θ in the complex plane:



Euler's Equation can then be used to establish the fundamental relationship between the trigonometric functions and the complex exponential function. Euler's formula states that, for any real number x :

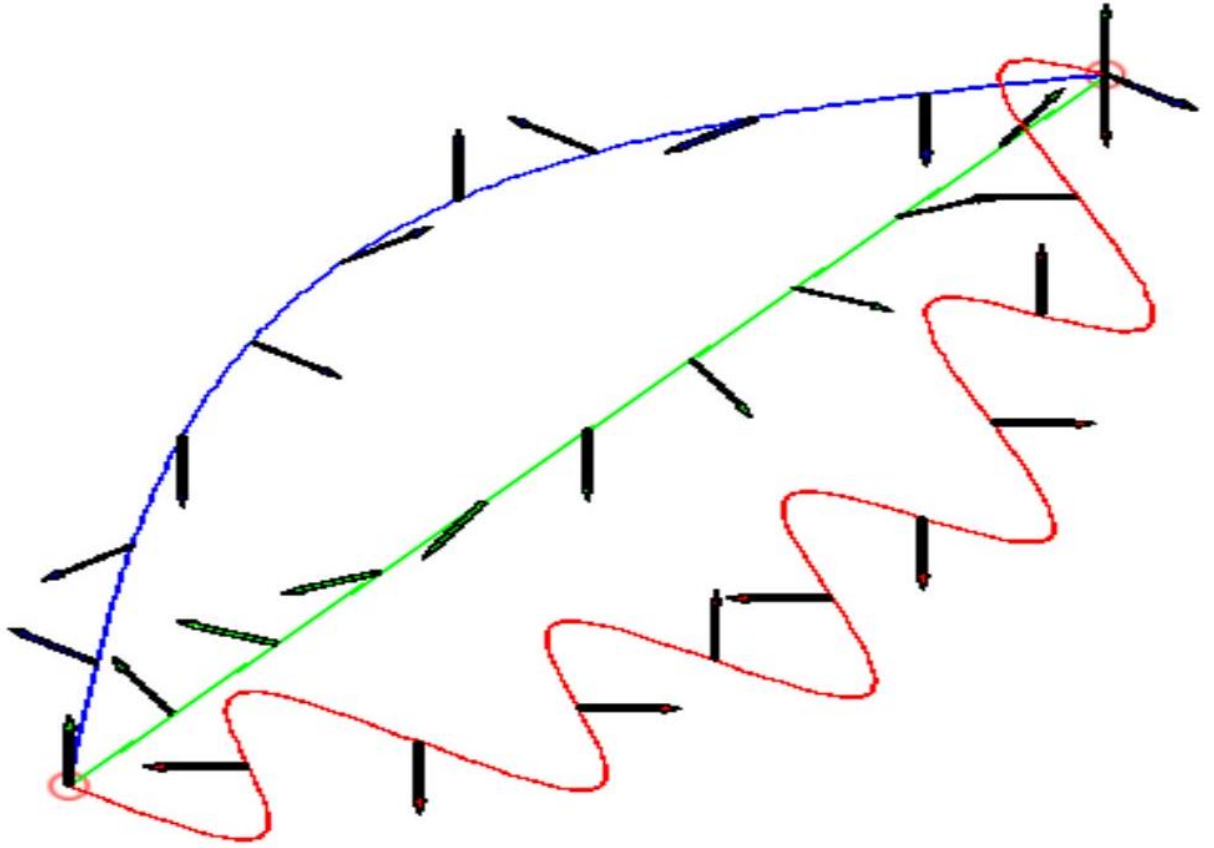
$$e^{ix} = \cos x + i \sin x$$

Where e is the base of the natural logarithm, i is the imaginary unit, and $\cos$ and $\sin$ are the trigonometric functions cosine and sine respectively, with the argument x given in radians.

The path of least action arises since the unit vector, z , tend to cancel out when the phase angles are out of phase with each other, and the vectors tend to add together when the phase angles are in phase with each other:

By taking each unit vector along probability wave represented by the path the particle takes, we form a series of arrows and make them oscillate around the complex unit circle at a frequency based on how difficult it is for a particle to travel through a given point-i.e. the harder the path to travel, the slower the arrow spins. (oscillating with frequency equal to the action.)

As discussed before that the particle takes all paths between the initial state and the final state. So we make our arrows follow each path and rotate them as we go along, as shown below.



Hence, in the path integral interpretation, the clockwork of unit vector arrows are along every possible path along the initial and final state (in this case 3 possible paths, red, green and blue). The paths are different only insofar as the arrows are rotated at different frequencies equal to the deviation from the path of least action, S . In other words the arrows are rotated with a frequency equal to that of the action imparted to deviate the particle from that of the least action possible. Therefore the path of least action itself should have the lowest frequency of unit vector rotation, where an increase in frequency corresponds to an increase in action.

Ref(3)

Using path-integrals has revolutionized quantum dynamics by by-passing the task to solve the Schrödinger equation itself (which is possible only for few-atom systems).

Avoiding going down the path of irreducible complexity, it is important to clarify that the least action principle is itself an extension of the principle of conservation of energy.

69 Viewing Meta-Heuristic Synchronization using the Path Integral :

The resemblance of the unitary time-evolution operator to the monotonically decreasing signalling function is obvious in their respective mathematical forms:

For one thing, the signalling algorithm is readily compatible with Lagrangian mechanics which are inherently used for discrete particles each with a finite number of degrees of freedom. Our Signalling Lagrangian, L_f , for each signalling node are in-effect integrated across the time-domain in the path it takes to achieve synchronization, hence leading to the total Signalling Action S_f , represented here

as being the phase of each path being determined by $\int \mathbf{L} \mathbf{f} d\mathbf{r}$, for that trajectory, where L_f is the Lagrangian.

$$S_f = \int L_f dr = \ln \left(\frac{I(r)}{I_0} \right) = -\gamma r^2$$

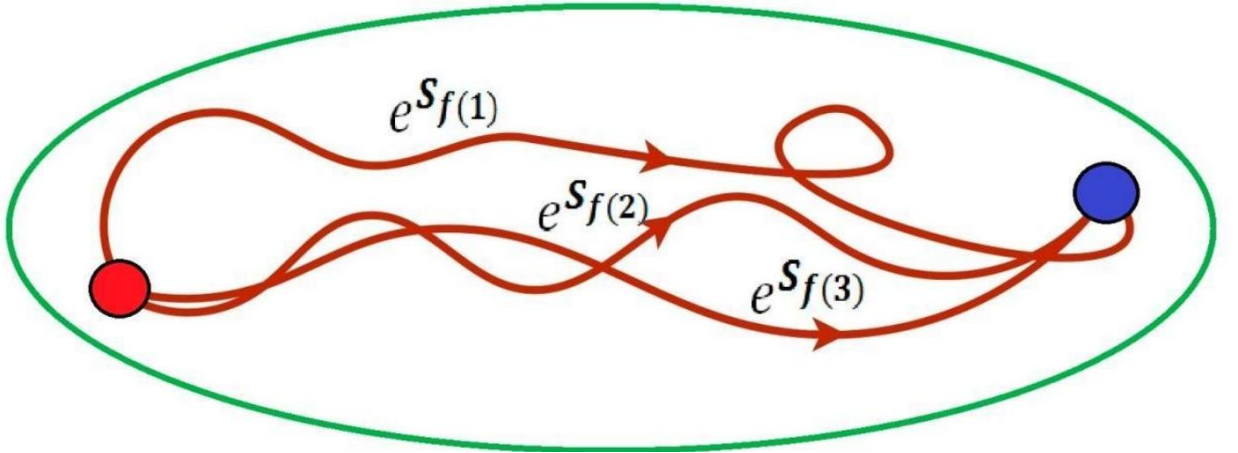
The frequency of signalling is then, as an oscillation in the complex plane:

$$e^{S_f}$$

However, we can investigate go further and compare the probabilistic representation of the path integral theory with the description of the algorithm as a system of discontinuous pas-coupling

Using this in the path-integral view we can represent the synchronization procedure as:

$$\langle \text{blue circle} | \beta_0 e^{-\gamma r^2} | \text{red circle} \rangle = \text{TOTAL Amplitude for Synchronization History}$$



CONFIGURATION SPACE}

In this convention, the signalling action, S_f , which is a real operator is defined as being essentially characteristic of the physical and environmental parameters of the system - namely the light absorption coefficient and the power law over the distance by which it is subject to

Since each deviation from the path of least action is, just like in the case with a quantum particle, proportional to the action imparted on the system, the signal activation function will activate with a frequency proportional to the action.

Therefore, the path of least possible action, which occurs during complete internal synchronization, should have the lowest frequency of signal function activation.

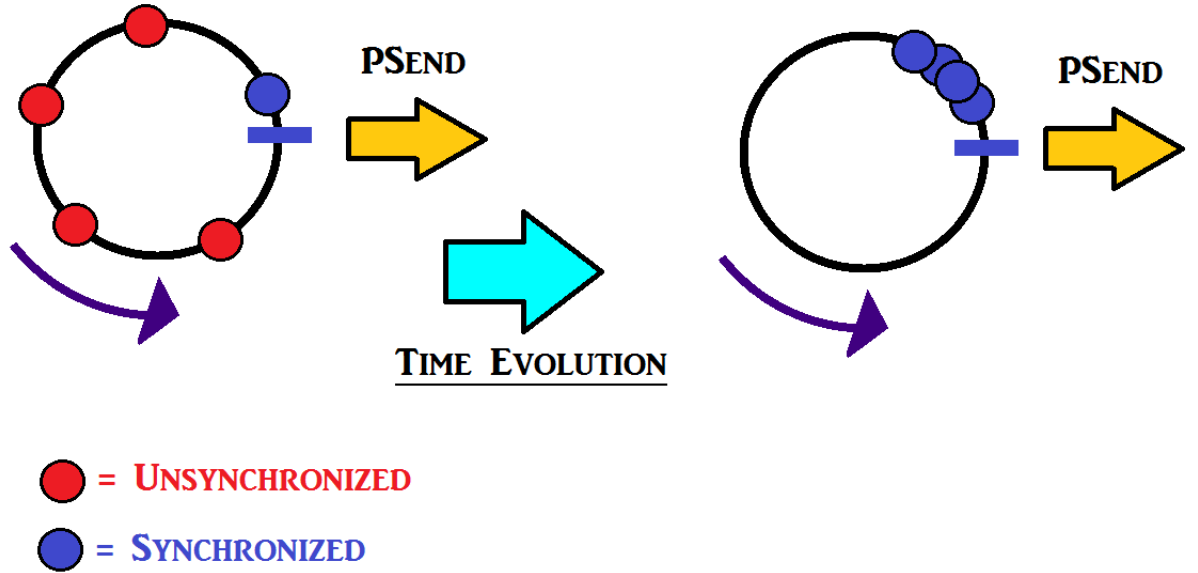
An increase in frequency of signal activation indicates a deviation from the path of least possible signalling action.

Therefore, when using any metaheuristic algorithm, when looking for any deviations, i.e. faults, between the coupled oscillators when internal synchronization is achieved we should just have to look for any changes in the frequency of the signalling between the different nodes.

This also explains how the self-organising system restores itself - as a fault is equivalent to a deviation from the principle of least action, which is energetically unfavourable.

69.1 What do I mean by this?

When an interaction occurs with our signalling model, the input deviation leads to a deviation from the path of least signalling action to achieve synchronization, as we have imparted our own equivalent signalling action, S , onto the system. This causes an initial synchronization collapse. However, the action, S , imparted towards the path will lead to an increase in the rate of cycles, i.e. an increase in the signalling action away from the least signalling action, and thus an increase the probability of achieving synchronization with neighbours again. In effect, internal processes are stimulated under any signalling action, increasing their state of energy, which is "unfavoured" - as the increasing energy state will itself drive the quantised nodes, which have a discrete threshold, that make up the system to spend more energy and thus increase cycling which overall increases the probability that the area under interaction will be synchronised with its neighbours again. Therefore the system is able to restore itself back to synchronization:



The fact that the system regulates itself to the most energy efficient configuration possible, which is to the state of uniform synchronisation is a key conclusion of studying the metaheuristic algorithm in the context of the sum over histories of the path integral representation.

69.2 Analogy of Fermi–Golden Rule for Optical Entanglement Transmission

In the context of quantum optical communication, the transfer of entanglement between two nodes—say, a source node and a receiver node—can be conceptually modeled akin to a nuclear transition governed by the Fermi–Golden Rule. Here, instead of a nuclear transition matrix element, we consider an effective transition amplitude $M_{s,d}$ that characterizes the fidelity or strength of entanglement transmission from the source (s) to the daughter (d) node.

69.2.1 Transition Rate and Entanglement Fidelity

Analogous to Fermi's Golden Rule, the "transition rate" $W_{s \rightarrow d}$ describing the effective entanglement transfer can be expressed as

$$W_{s \rightarrow d} = \frac{2\pi}{\hbar} |M_{s,d}|^2 \times (\text{Phase Space}), \quad (111)$$

where:

- $|M_{s,d}|^2$ is the squared transition amplitude representing the effectiveness of entanglement transfer between nodes.
- The phase space factor accounts for the available modes or channels for the optical signals, analogous to the density of states in nuclear transitions.

69.2.2 Transition Matrix Element and Signaling Analogy

The matrix element between the initial optical state (at the source) and the final state (at the destination), mediated by the optical channel, can be written as

$$|M_{s,d}|^2 = \left| \langle \psi_d \phi_{\text{pol}} \psi_{\text{env}} | \hat{H}_{\text{int}} | \psi_s \rangle \right|^2, \quad (112)$$

where:

- $|\psi_s\rangle$ and $|\psi_d\rangle$ are the quantum states of the source and destination nodes, respectively.
- ϕ_{pol} represents the polarization degree of freedom involved in entanglement.
- $\hat{H}_{\text{int}}$ is the effective interaction Hamiltonian governing the entanglement transfer, which can be modeled as

$$\hat{H}_{\text{int}} = G_V \hat{\mathbf{1}} \hat{\tau} + G_A \hat{\sigma} \hat{\tau}, \quad (113)$$

with analogous roles to the Fermi (vector) and Gamow–Teller (axial vector) components, representing different polarization or mode interactions.

69.2.3 Signaling Dynamics and Power-Law Attenuation

Mirroring the physical attenuation in nuclear transitions, the effectiveness of entanglement transmission diminishes with increasing optical path length r , following a power-law or exponential decay. The signaling strength $\beta(r)$ can be modeled as

$$\beta(r) = \beta_0 e^{-\gamma r^m}, \quad (114)$$

where:

- β_0 is the initial signaling strength at zero distance.
- γ is the absorption coefficient.
- m characterizes the power-law decay; for optical signals, typically $m = 2$ (inverse-square law).

69.2.4 Implications for Entanglement Protocols

This analogy suggests that the efficiency of entanglement distribution can be viewed as a transition process with a rate governed by an effective matrix element and attenuated by the physical medium, akin to nuclear decay processes. Optimizing $|M_{s,d}|^2$ and minimizing γ are thus crucial for robust quantum communication, much like controlling the interaction Hamiltonian is vital in nuclear physics.

This conceptual framework provides a valuable perspective linking nuclear transition theory to quantum optical signaling, emphasizing the fundamental role of interaction amplitudes and medium-induced attenuation in entanglement distribution fidelity.

70 Quantum Network Simulations using Kuramoto Oscillators

The path integral formalism hence offers a comprehensive framework for studying the dynamics of entangled networks undergoing synchronization. By representing the least signaling action in configuration space as a sum over functions, $\sum e^{P_{\text{send}}}$, we can extend this representation to phase space models akin to the Kuramoto model.

In the path integral formulation of quantum mechanics, a system can take any trajectory between two points of observation x_a and x_b , at times t_a and t_b , respectively. Denoting a path as $x(t)$ such that $x(t_a) = x_a$ and $x(t_b) = x_b$, the probability amplitude of the system taking this path is given by:

$$A[x(t)] \propto e^{-\frac{i}{\hbar} S[x(t)]}, \quad (115)$$

where the action $S[x(t)]$ is defined as:

$$S[x(t)] = \int_{t_a}^{t_b} dt L(\dot{x}(t), x(t)). \quad (116)$$

Therefore, the overall probability amplitude of the system starting at x_a at time t_a and ending up at x_b at time t_b is given by the sum over all possible paths:

$$\psi(x_b, t_b; x_a, t_a) \propto \sum_{x(t)} e^{-\frac{i}{\hbar} S[x(t)]}. \quad (117)$$

While it is challenging to rigorously define a sum over all paths, we can approximate paths $x(t)$ as continuous variables, treating the action $S[x(t)]$ as a function of this variable.

Although the propagator, $\psi(x_b, t_b; x_a, t_a)$ does not directly encapsulate effective coupling in oscillator networks, there exists an analogous model we can consider that can elucidate some of the emergent dynamics of synchronized states and collective oscillations represented in graph theory that can be bridged to represent dynamics in highly entangled networks [13, 16].

70.1 Kuramoto Model in Phase Space

The Kuramoto model provides a powerful framework for studying the synchronization of coupled oscillators in phase space. The dynamics of the Kuramoto model can be described by the following equation:

$$\dot{\theta}_i = \omega_i + \sum_{j=1}^N X_{ij} \sin(\theta_j - \theta_i), \quad (118)$$

where θ_i represents the phase of oscillator i , ω_i is the natural frequency of oscillator i , X_{ij} denotes the coupling strength between oscillators i and j , and N is the total number of oscillators in the network.

The Kuramoto model captures the collective behavior of coupled oscillators and how synchronization emerges through the interaction of phase differences between oscillators. By analyzing the phase space trajectories of the oscillators, one can observe the evolution of synchronization patterns and the emergence of coherent states in the network.

70.2 Effective Behaviors in a Network of Oscillators

The effective behaviors of oscillators within a network can be comprehensively analyzed through the lens of graph theory, with mathematical insights that draw from quantum mechanics, particularly regarding coherent states and entangled systems. The effective coupling strength between two oscillators in such a network can be articulated through their coordinated oscillation patterns represented on the graph.

70.2.1 Graph Representation

Consider an oscillator network represented as a graph $G = (V, E)$ where V denotes the set of vertices corresponding to oscillators (or modes), and E encapsulates the edges that define the coupling strengths X_{ij} connecting oscillators i and j . Each edge weight is indicative of the interaction strength between adjacent oscillators, thus affecting their collective dynamics [14].

70.2.2 Effective Coupling

The effective coupling strength $X_{\text{eff}}(i, j)$ between oscillators i and j can be determined using various methods, such as examining mode shapes or invoking Hamiltonian dynamics. A robust analytical method is to analyze the synchronization properties of the network. For instance, in a coupled oscillator model, one may express the effective coupling as:

$$X_{\text{eff}}(i, j) = \sum_{k \in N(i)} X_{ik} C_k, \quad (119)$$

where $N(i)$ is the set of neighbors of oscillator i , and C_k represents the contributions of these neighboring states to the effective coupling.

70.2.3 Connection to Synchronized States

In a synchronized oscillator system, the dynamics can be characterized by mutual synchronization, wherein the collective oscillations arise from the intricate coupling in the network. The propagator $\psi(x_b, t_b; x_a, t_a)$ encapsulates the amplitudes for transitions between coherent states of oscillators i and j . The synchronized behavior is effectively a manifestation of collective oscillation, which stems from the underlying graph structure.

70.2.4 Path Integral for Synchronization

Interpreting the effective coupling through the framework of path integrals lends further insight: we can view the effective coupling as arising from the cumulative effects of all possible paths linking oscillatory states. Each path's contribution is weighted by the respective coupling strengths along the paths:

$$X_{\text{eff}}(x_a, x_b) = \sum_{\text{paths}} W(\text{path}) \cdot \text{Amplitude}, \quad (120)$$

where $W(\text{path})$ represents the weight associated with the coupling strength along a given oscillatory path [15].

70.2.5 Mathematical Formalism

By employing graph theory in concert with the principles of oscillator dynamics, the effective coupling can be expressed in terms of the adjacency matrix A of the graph or through the eigenfunctions of the Hamiltonian associated with the system. Specifically, it can be represented as:

$$X_{\text{eff}}(x_a, x_b) = \frac{A_a}{B}, \quad (121)$$

where A_a represents the amplitude of the oscillators at positions a and b , and B is a measure of collective coupling strength across the involved oscillators.

The Kuramoto model provides a phase space description of coupled oscillators, where the dynamics of synchronization are governed by the phase differences between oscillators. In this model, the coupling strength between oscillators influences the emergence of synchronized states in the network.

71 Discontinuous Coupling in Phase Space

Let us take our oscillators running on a unit circle in phase-space. The oscillators represented on a circle in phase space. To achieve synchronization, the oscillators runs clockwise on the circle until it passes the threshold. Whenever it passes the threshold, it will emit a signal pulse, with some probability, P_{send} . When the oscillators are coupled, they will adjust their individual rates, or periods, of signaling to match with one another (encroaching closer together on the unit circle in phase-space) to achieve synchronization under a time evolution.

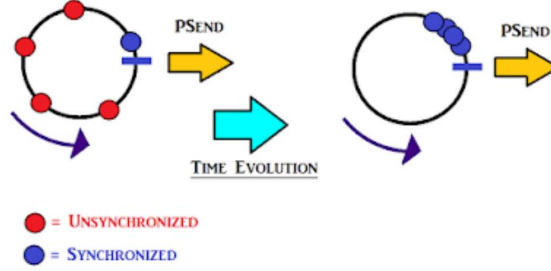


Figure 32: Oscillators running on a unit circle in phase space, emitting signal pulses when passing threshold.

Each oscillator will adjust its period of signaling according to our signaling activation function, which is, under the signaling field representation, a phase function:

$$\Phi(X_{i,j}) = A(X_{i,j}) \cdot \alpha$$

The function evolves linearly over time until it reaches a threshold value, defined by the vector weights, $\mathbf{X}_{i,j}$, which are translated into coupling strengths under a power law, i.e. the extinction ratio for light signals.

When the threshold is reached, a single oscillator fires the signal pulse and then resets its phase. If no coupling occurs, the oscillator will pulse with a period, $T(A)$. T is effectively the encoded time remaining until the next signal pulse.

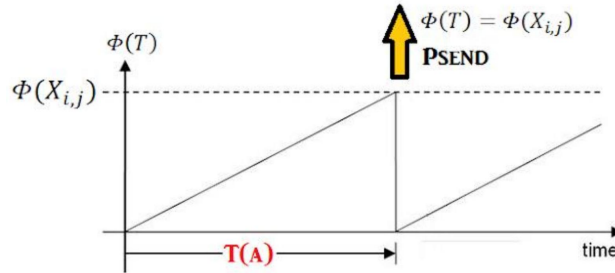


Figure 33: Signaling activation function at threshold.

When a phase-coupling occurs between oscillators, when an oscillator receives a pulse it will increment its phase function by an amount that itself depends on its current value and the change induced by the weights of the received signal. The oscillator will then pulse with a new period, $T(B)$:

This means that an oscillator that is in the lower half of the phase-space circle will jump forward, and an oscillator that is in the upper half will jump backwards in the circle in phase space. This is the main concept of a self-organizing update function, which brings the 2 oscillators in phase with one another and thus coupled.

The resemblance of the unitary time-evolution operator in quantum mechanics to the monotonically decreasing signaling function is obvious in their respective mathematical forms:

$$\beta = \beta_0 e^{-\gamma r^m} \text{ and } U = U_0 e^{\frac{-iHt}{\hbar}}$$

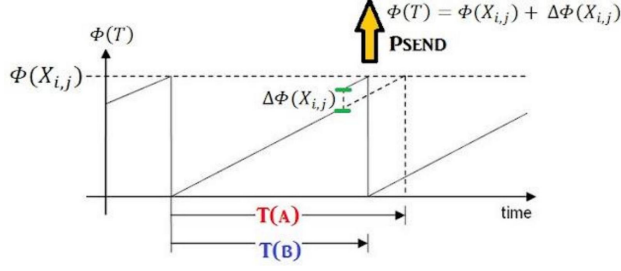


Figure 34: Phase updates in response to incoming signals.

Moreover, like the unitary time-evolution operator, the signaling algorithm is readily compatible with Lagrangian mechanics in the same way as the unitary time-evolution operator which are inherently used for discrete particles each with a finite number of degrees of freedom.

The classic Lagrangian is written as the difference of the kinetic and potential energy density of a particle.

$$L = K \cdot E - P \cdot E = \frac{mv^2}{2} - V(q)$$

The potential energy density of the particle is a field density, such as an electromagnetic field density in generalised coordinates $V(q)$.

Since $v = p/m$, the kinetic energy can, of course, be written in terms of momentum, p , which is more relevant for what we are describing.

$$L = \frac{p^2}{2m} - V(q), \quad p = \frac{dL}{dq}$$

The abbreviated action $\mathbf{S}_0$ is defined as the integral of the generalized momenta along a path in the generalized coordinates q :

$$S_0 = \int p \cdot dq = \int p_i dq_i$$

In our ad-hoc system of synchronization coupling, the momenta will be replaced by P_{send} and integrated over the time domain (dt) in the path it takes to achieve synchronization. This leads to the total Signaling Action, $\mathbf{S}_f$, being represented as the phase of each path being determined by $\int P_{\text{send}} dt$, for that trajectory.

$$S_f = \int P_{\text{send}} dt$$

The frequency of signaling is then, as an oscillation in phase-space:

$$e^{S_f}$$

The frequency of signaling is then an oscillation in the complex plane. In the nature of the path integral formalism, we consider the synchronization as:

$$\langle \bigcirc | \beta_0 e^{-\gamma t} | \bigcirc \rangle$$

which can be visualized as a sum of signaling histories as:

In this convention, the signaling action, $\mathbf{S}_f$, which is a real operator is defined as being essentially characteristic of the physical and environmental parameters of the system - namely the light absorption coefficient and the power law over the distance by which it is subject to.

from the Kuramoto model as described through the path-integral approach to synchronization we can develop several pseudo-entanglement networks to run simulations.

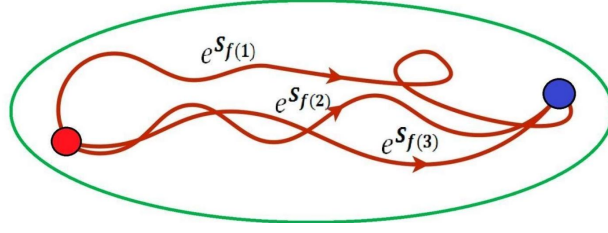


Figure 35: **Configuration space** containing the total amplitude for synchronization history; it encapsulates the cumulative effects of all allowed signals, each influenced by their respective signaling actions.

The phase space description enables us to analyze the collective behavior of coupled oscillators in our models and understand the mechanisms underlying synchronization and discordance in entangled networks.

To do this with a comprehensive scope a series of simulation codes are used in python building on the complex systems work of Hiroki Sayama et al [19].

The results of using Kuramoto oscillators in phase space is showcased here: <https://github.com/MuonRay/QuantumNetworkSimulations>

these series of codes use networkx in python to create network graphs with nodes and links subject to the description in the section III-A on Graph Representation.

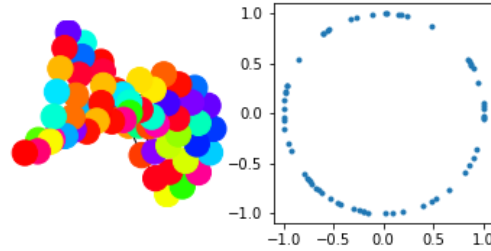


Figure 36: Left: graph $G = (V, E)$ where V denotes the set of vertices corresponding to oscillators (or modes), and E encapsulates the edges that define the coupling strengths X_{ij} connecting oscillators i and j . Right: Phase space distribution of oscillators

From this a series of experiments were undertaken to drive a system of synchronized oscillators and from using the network path-integral formalism discussed a process was developed to get to an action-based descriptor of the networks examined as they are driven towards synchronization from a initial condition of chaos towards an underdamped state, a critically damped threshold and ultimately a steady-state. The description of this network of oscillators as an underdamped system can be seen when the derivative of the action is plotted:

The evolution of this system of oscillators can be compared in the degree of chaotic phases of the oscillators, which relates to the frequency shifts per timestep.

It is possible combine these factors, which show relatable common trends, to give an overall configuration space trajectory of the system, where a colorbar can show exactly the critical point at which the maximum and minima actions drive the networked system of oscillators from underdamped to a steady state as they synchronize.

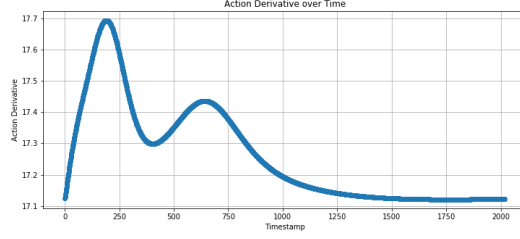


Figure 37: derivative of the action, S , plotted over the time of the evolution of synchronization, showing the underdamped nature of the system as well as the critical points of change

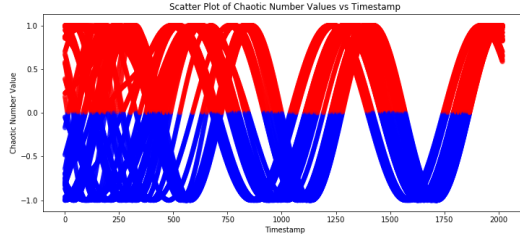


Figure 38: chaotic phases of the oscillators in an underdamped network approaching synchronization in a time evolution

The motivation for these simulations is that a coupled system of harmonic oscillators, such as an entanglement network, will need to be examined under the lens of control theory used in classical complex systems. For a salient description of how these entangled networks interact with the environment models must be considered that view them in terms familiar to analog systems of coupled oscillators that behave subject to perturbations, i.e. sudden changes or disturbances in the system, whether in the network parameters, external forces, or the configuration of the nodes/links in the network.

In networked structures, the coupling between different nodes means that a perturbation in one part of the system can influence others. Therefore, the response of the system (in terms of the action's derivative) can exhibit complex dynamics depending on the network topology and interactions.

The interconnectivity allows perturbations that do not just affect one node to propagate through the network, manifesting in collective phenomena such as synchronization, cascading failures, or robustness to disturbance.

72 Entangled Signal Interaction with Classical Synchronization Networks

One intriguing aspect of this interaction between entanglement and synchronization phenomena in networks of coupled oscillators is how entangled signals can serve as probes in classical synchronization scenarios. Entangled states, characterized by correlations that cannot be explained classically, carry unique phase properties that can influence the behavior of systems described as networks of harmonic oscillators.

As harmonic oscillators within a network approach thresholds of synchronization, the stability and coherence of their collective dynamics can be significantly affected by the presence of entangled signals. When entangled signals interact with these oscillators, they can facilitate synchronization under certain conditions, effectively acting as a catalyst for the emergence of synchronous behavior. This interaction is particularly relevant at what is termed the "edge of synchronization," where small

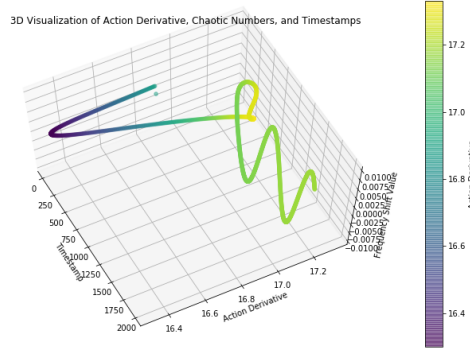


Figure 39: Trajectory of the Network of Oscillators in a high dimensional configuration space as synchronization emerges. It is possible for the trajectory to intersect itself, because the phase configuration of the system can go back to the same position after a given interval of time.

perturbations—such as those introduced by quantum entanglement—may shift the system into a regime of enhanced synchrony.

Moreover, the interplay between entangled signals and harmonic oscillators presents exciting possibilities for weak measurement probes, where the measurement process minimally disturbs the system being observed. In scenarios where phase dynamics are of primary interest, entangled signals can be deployed to extract phase information without fully collapsing the wave function of the oscillator network. This capability opens up avenues for leveraging entangled states to enhance our understanding of phase relationships and synchronization dynamics within quantum systems.

In the context of a network of oscillators, a weak measurement operator, denoted as A_w , acts on the eigenmodes of the system, offering insights into its phase dynamics. By utilizing weak values A_w , researchers can investigate how oscillators interact, particularly in entangled states, without collapsing the wavefunction to a definitive outcome. This capacity allows the exploration of the intricate nature of their synchronizing relationships, reminiscent of the synchrony observed in classical oscillators, albeit with the added complexity of quantum behavior.

When weak measurement operators are reformulated within the framework of the path integral approach, they illuminate the probable trajectories that oscillators might take during synchronization. Each trajectory can be interpreted as a quantum path, weighted by its action S :

$$\mathcal{Z} = \int \mathcal{D}[r(t)] e^{\frac{i}{\hbar} S[r(t)]}$$

By measuring the dynamics of each state using weak measurement operators, one can establish a connection between quantum measurements and classical paths in configuration space.

The resemblance of the time-dependent weak measurement operator to the monotonically decreasing signaling function is obvious in their respective mathematical forms:

$$\hat{M}(t) = \int_{-\infty}^t K(t, s) \hat{O}(s) ds, \quad (122)$$

where $\hat{M}(t)$ denotes the weak measurement operator at time t , $\hat{O}(s)$ represents the observable being measured at time s , and $K(t, s)$ is the kernel function that defines the influence of the past measurements on the present measurement outcome.

The eigenmodes can be quantified as variables measured in a weak measurement operator, $\hat{M}(t)$.

The operator $\hat{M}(t)$ then acts on the eigenmodes, with the initial and final states of the propagator $\psi(x_f, t_f; x_i, t_i)$:

$$\hat{M}(t) = \frac{\langle \psi_f | \hat{O}(s) | \psi_i \rangle}{\langle \psi_f | \psi_i \rangle} \quad (123)$$

with the observable operation $\hat{O}(s)$.

In comparison, the monotonically decreasing signaling function, $f(t)$, can be expressed as:

$$f(t) = \beta_0 e^{-\gamma t}, \quad (124)$$

where β_0 is the initial signal strength and γ is a positive constant that dictates the rate of decay.

Both expressions illustrate a dependency on past values and exhibit a form of time evolution characterized by an integral or exponential decay, highlighting their analogous behavior in temporal dynamics.

This is readily compatible with Lagrangian mechanics which are inherently used for discrete particles each with a finite number of degrees of freedom. Our Signalling Lagrangian, L_f , for each signaling node is in effect integrated across the time-domain in the path it takes to achieve synchronization, hence leading to the total Signalling Action S_f , represented here as being the phase of each path being determined by $\int L_f d\mathbf{r}$, for that trajectory, where L_f is the Lagrangian.

$$S_f = \int L_f dt = \ln \left(\frac{I(t)}{I_0} \right) = -\gamma t$$

In this convention, the signaling action, S_f , which is a real operator is defined as being essentially characteristic of the physical and environmental parameters of the system - namely the light absorption coefficient and the power law over the distance by which it is subject to.

Since each deviation from the path of least action is, just like in the case with a quantum particle, proportional to the action imparted on the system,

the signal activation function will activate with a frequency proportional to the action. Therefore, the path of least possible action, which occurs during complete internal synchronization, should have the lowest frequency of the signaling function.

72.1 Linking Deviation in Frequency to Action

In weak measurement theory, we consider a system (such as a photon with spin states) that is slightly perturbed by an interaction with a measurement device. The initial state of the system can be written as:

$$|\psi\rangle = \alpha|s_1\rangle + \beta|s_2\rangle$$

where $|s_1\rangle$ and $|s_2\rangle$ are the eigenstates corresponding to spin up and spin down, respectively, and α and β are complex amplitudes with the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

The interaction Hamiltonian H' can be represented as:

$$H' = -\hbar\Delta\omega\hat{P}$$

where $\hat{P}$ is the operator corresponding to the observable being measured (here, it may relate to the spin projection), and $\Delta\omega$ represents the perturbation in frequency due to the measurement process.

The energy associated with this perturbation is:

$$\Delta E \approx \hbar\Delta\omega$$

The integration of weak measurement techniques within this framework allows for the observation of these subtle variations in frequency without significantly disturbing the system. Under this paradigm, we can effectively monitor synchronization integrity through measurable frequency shifts related to the action's time derivative [18].

When a synchronized system experiences a fault, it deviates from the path of least action. This deviation can be quantified in terms of a change in frequency $\Delta\omega$:

$$\Delta\omega \propto \left. \frac{\partial S}{\partial t} \right|_{\text{faults}}.$$

This implies that as the action changes due to faults in the synchronization, the frequency of the system's oscillations will also change.

The proportionality indicates that greater deviations from the optimal path correspond to more significant changes in frequency [15].

$$\Delta\omega \propto \frac{\partial S}{\partial t}$$

This notion allows for the monitoring of synchronization integrity through observable frequency shifts with respect to a drive factor .

Thus, the weak measurement framework, when aligned with the path integral formalism, affords novel insights into the delicate synchronization dynamics of both classical and quantum systems. This approach serves as a mechanism for testing theories of synchronization across both domains, leveraging weak values to elucidate the underlying structures and dynamics governing synchronized states.

The potential applications of this work are broad, ranging from weak measurement approaches to quantum information to studies of biological systems exhibiting synchronized behavior. As such, understanding the dynamics of entangled signals within ordered systems of harmonic oscillators paves the way for innovative approaches to control and measure both classical and quantum complex systems.

73 A Metaheuristic View Of Quantum Network Dynamics : Boltzmann Machines

An interesting consequence of this new point of view we are having is the fact that the representation works both ways and it makes sense to think of the quantum transitions themselves, i.e. an evolution from state x_0 to x_t , can be thought as being in-effect, characteristic of a meta-heuristic function. This has interesting ramifications in the field of quantum computing, in which it has already been proven experimentally that meta-heuristic algorithms can be in fact solved using certain transitions on certain machinery, with the problem represented as matrix elements, An example in quantum computing is representing a computational procedure as a series of quantum tunnelling on a Josephson Junction network.

Josephson Junction circuits which in fact perform meta-heuristic calculations, such as simulated annealing, have already been developed by D-Wave Systems in Vancouver, Canada. Our model at the very least provides a way, using the path-integral interpretation, an abstract model of what the physical mechanism is in the ability of these systems to solve these meta-heuristic algorithms, such as simulated annealing, using quantum transitions rather than simply take it for granted that the devices just happen to perform these tasks.

Simulated annealing is probably the best example of modern metaheuristic algorithms, and it was developed by Kirkpatrick, Gelatt and Vecchi in 1983 [8], inspired by the annealing process of metals during heat treatment and Metropolis algorithms for Monte Carlo simulations.

The basic idea of the simulated annealing algorithm is similar to dropping some bouncing balls over a landscape, and as the balls bounce and loose energy, they will settle down at some local minima, sometimes referred to as a spurious attractor. If the balls are allowed to bounce enough times and loose energy slowly enough, some of the balls will eventually fall into the globally lowest locations, and hence the minimum will be reached. Of course, we can use many balls (parallel simulated annealing), or use a single ball to trace its trajectory (standard simulated annealing). The optimization process typically starts from an initial guess with higher energy. It then moves to other locations randomly with slightly reduced energy. The move is accepted if the new state has lower energy and the solution improves with a better objective or lower value of the objective function for minimization.

With non-stochastic networks, the appearance of local energy minima as a result of the network internally signalling along the path of the least signalling action possible can cause the signalling between the different nodes to get stuck in a region which do not satisfy the real constraints of the network very well.

This can be seen in the following network, simulated in Matlab.

In the simulation, we have a grid network of 10*10 nodes with input signal in the form of an input vector which is weighted by the size of the width 2D grid:

74 Weights = 2Dweights(GridWidth);

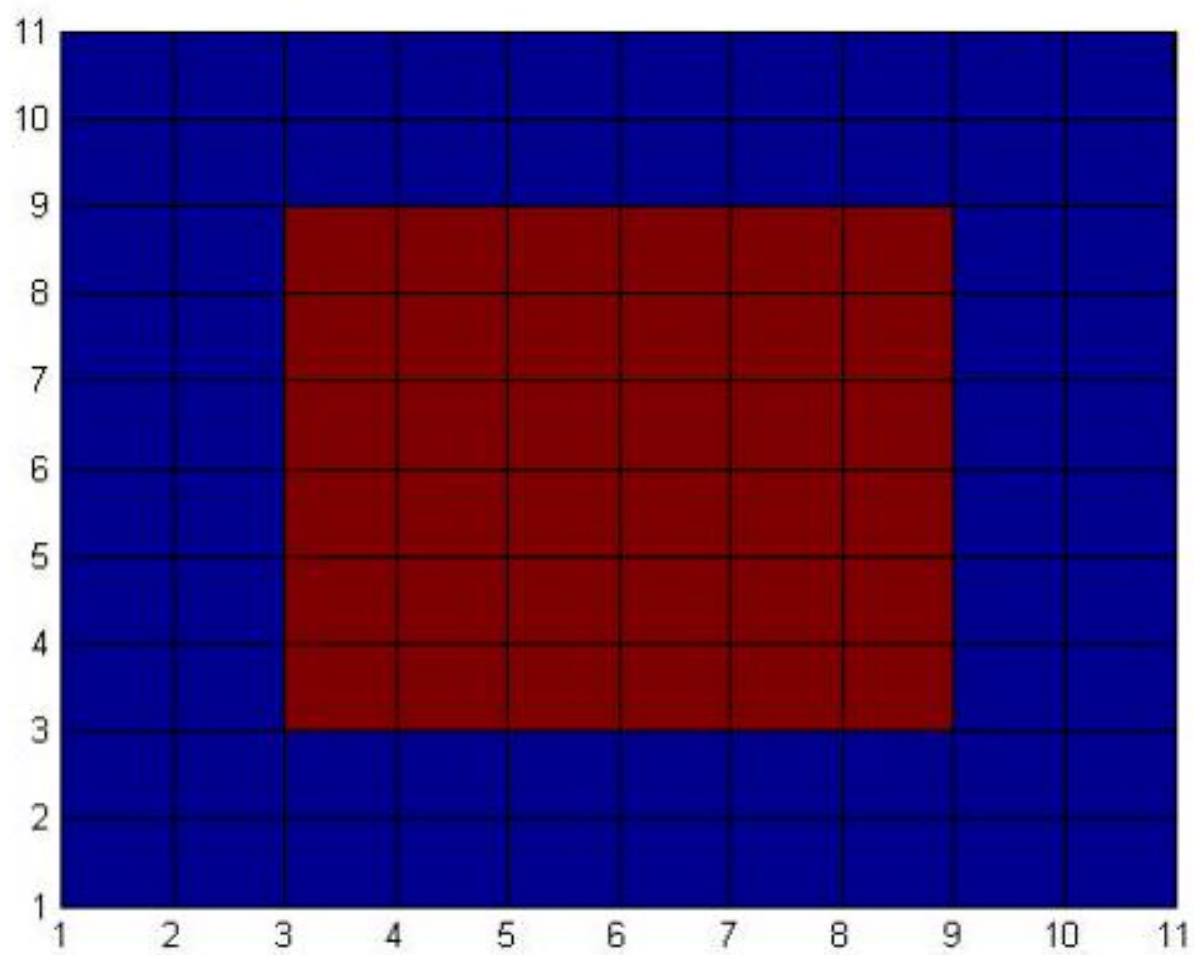
This allows the plotting of the input vector as an inputWidth * inputWidth grid

75 ActivationGrid = zeros(GridWidth);

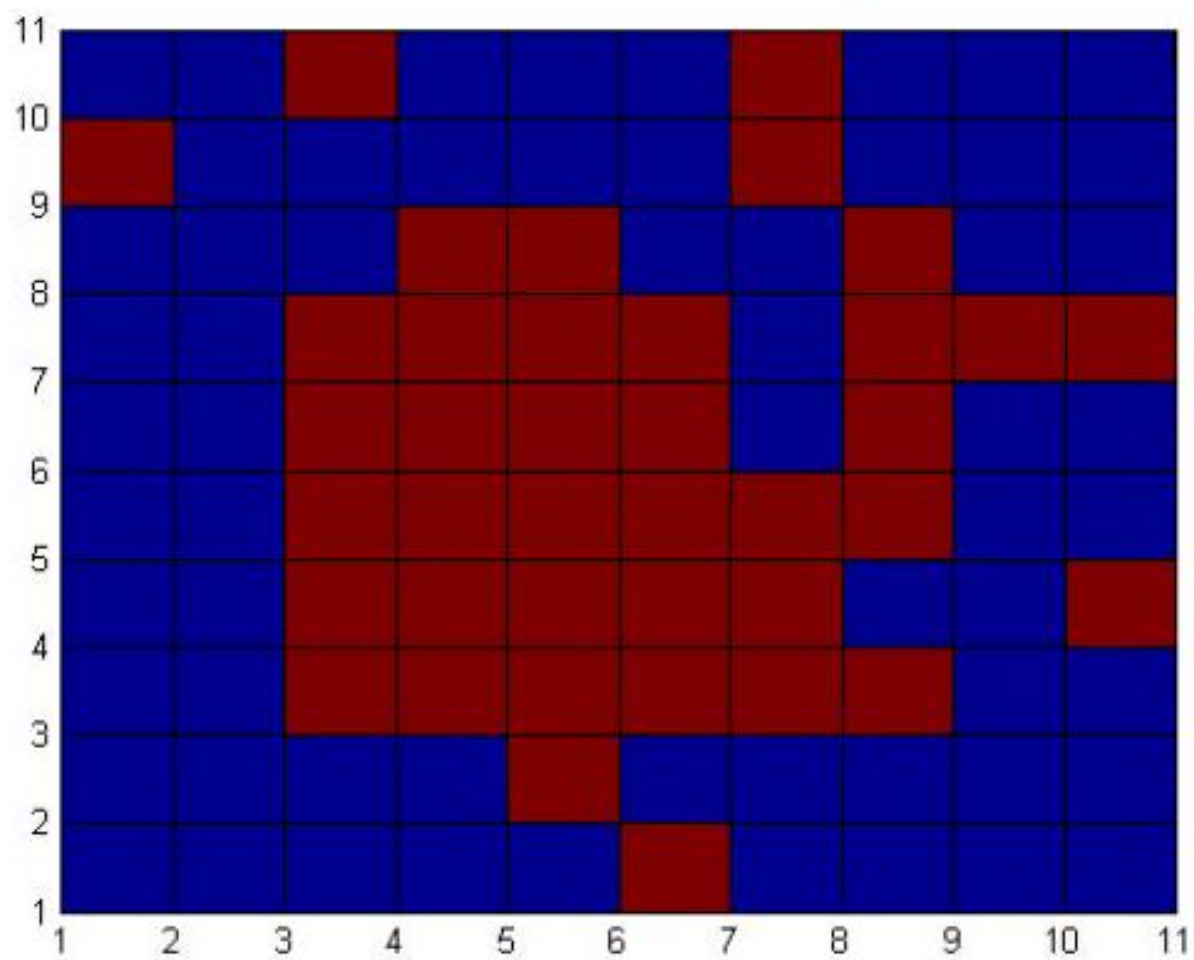
we use our signalling activation function as

```
%% as a 2D image, and think of the network as de-noising the image.
activations = ones(nInputs,1) * (-1);
for col = 3:(GridWidth-2),
    for row = 3:(GridWidth-2),
        activations((col - 1)*GridWidth + row) = 1;
    end;
end
for i = 1:nInputs,
    if rand(1,1) < SignalActionLevel
        activations(i,1) = activations(i,1) * (-1);
    end
end
```

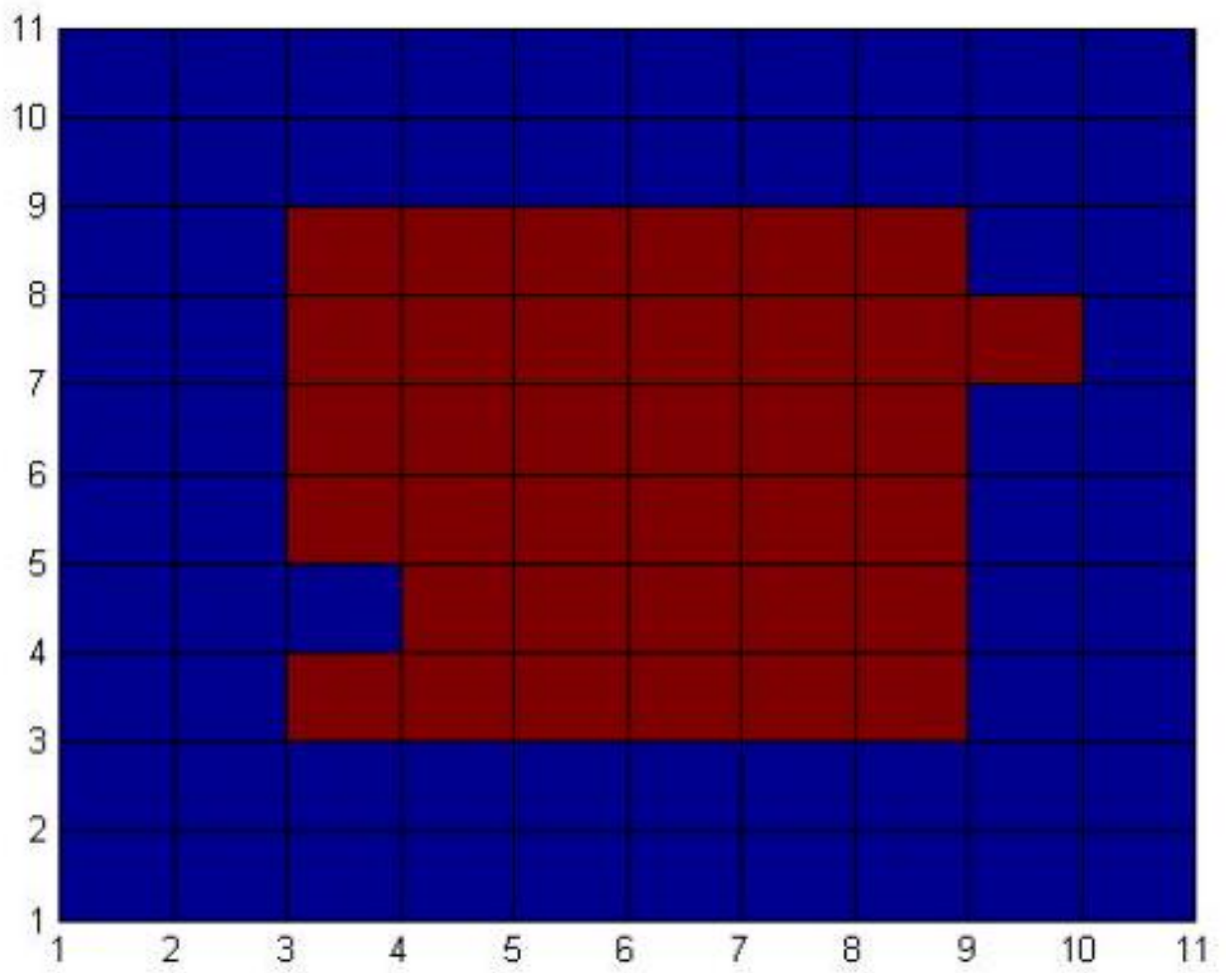
Signal Action Level (0->0.001):



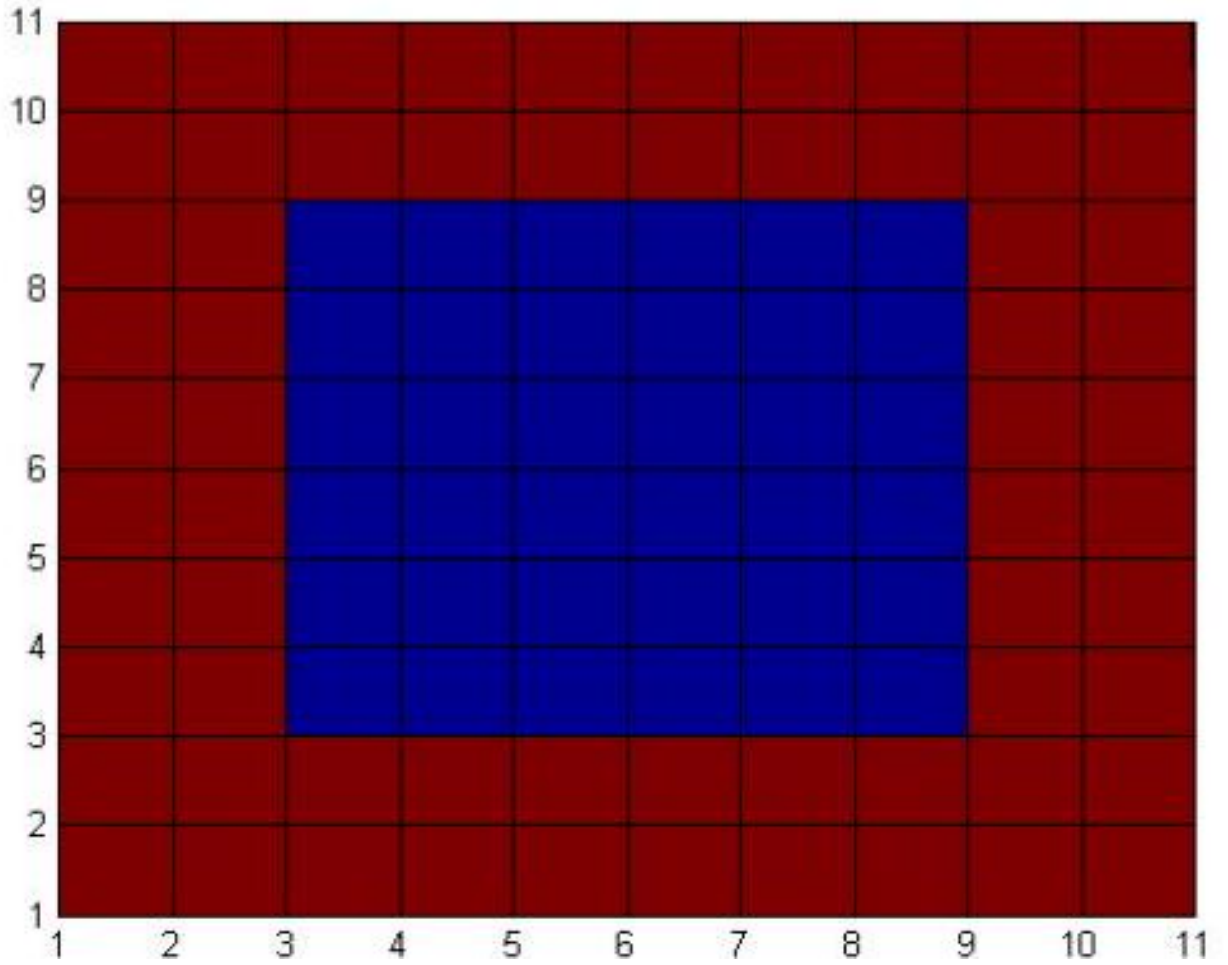
Signal Action Level (0.1):



Signal Action Level (0.01):



Signal Action Level ($1 \rightarrow \infty$) :



The signalling activations are attracted to the center of the network, which is where the local energy minima - i.e. the path of least signalling, is located. This can only disappear in the network by introducing more of an external signalling action.

Another way to think about it is if we think of the network as achieving

synchronization in order to remove noise from an externally imposed signal, passed as a input vector through the network, then when the network gets itself trapped in a local energy minimum by the principle of least signalling action the only way to get it out of this minimum is to impose action, i.e. noise, into the network.

The main failing of this is the fact that we have not properly defined what the network constraints are in terms of network structures. The constraints themselves are really imposed by the weights of the inputs, $W_{i,j}$.

For a given input vector:

$$X_{i=1,2,...N}^0$$

In a neural network structure we weight the inputs and sum them through k-number of hidden layers in the following summation:

$$X_j^{(k)} = A \left(W_{0j}^{(k)} + \sum_{i=1}^{M_{k-1}} W_{ij}^{(k)} \cdot X_i^{(k-1)} \right)$$

Our activation function which passes the weighted values through the k-number of layers, $\mathbf{A}(\mathbf{x})$, is then in the same form as the signalling basis function for a metaheuristic algorithm, which is based on the Gaussian form of the signalling intensity which is monotonically decreasing.

The activation then takes the form:.

$$A(x) = (1 + e^{-x})^{-1}$$

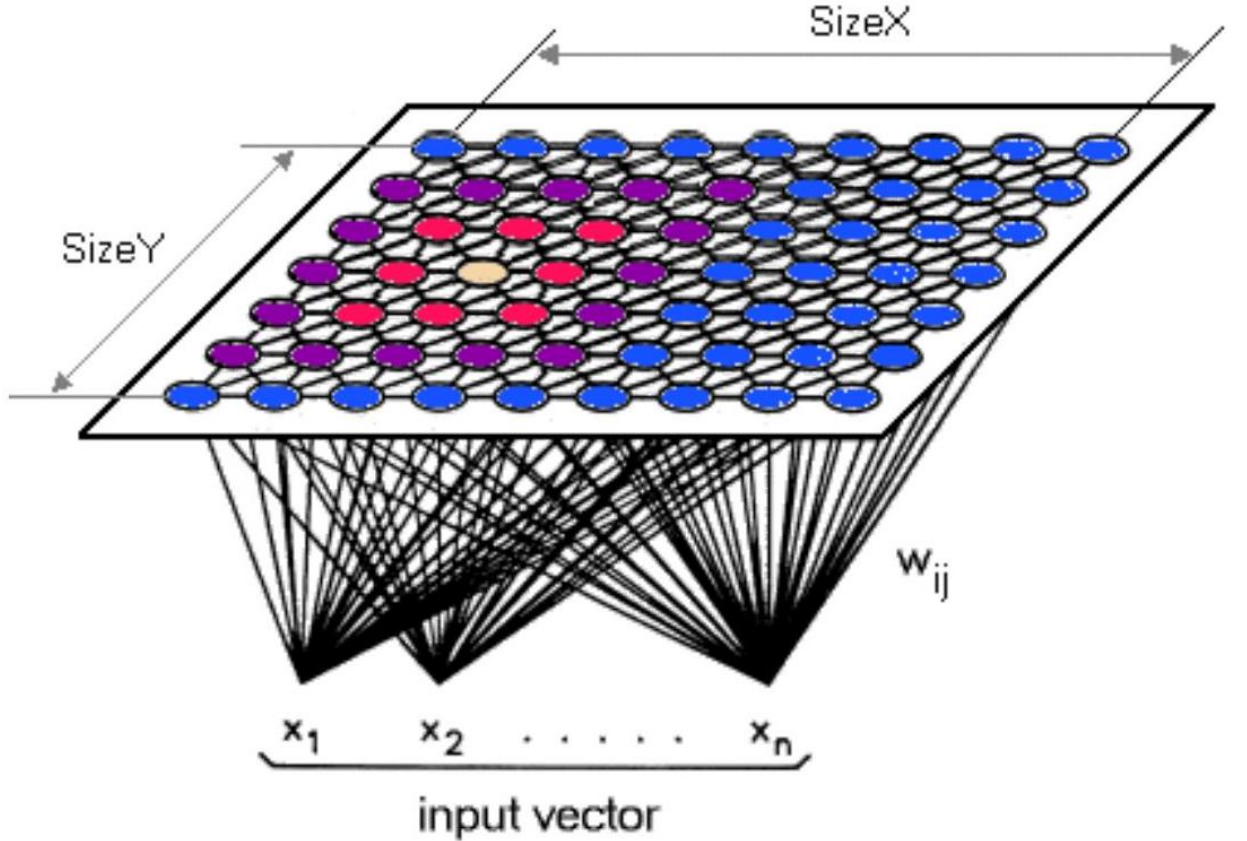
Which is a sigmoidal activation function. This is exactly the same form of equation as our metaheuristic activation function.

Hence the "hidden" layers of the neural network will be a sum over histories of all possible weights the inputs take across the sigmoidal activation function in the neural network structure.

The output is then a finite discrete integer response, which could be represented on a digital number line, characteristic of the initial input vector, that has been effectively broadened by the k-layered neural network:

$$X_{1,2,...N}^{(k+1)}$$

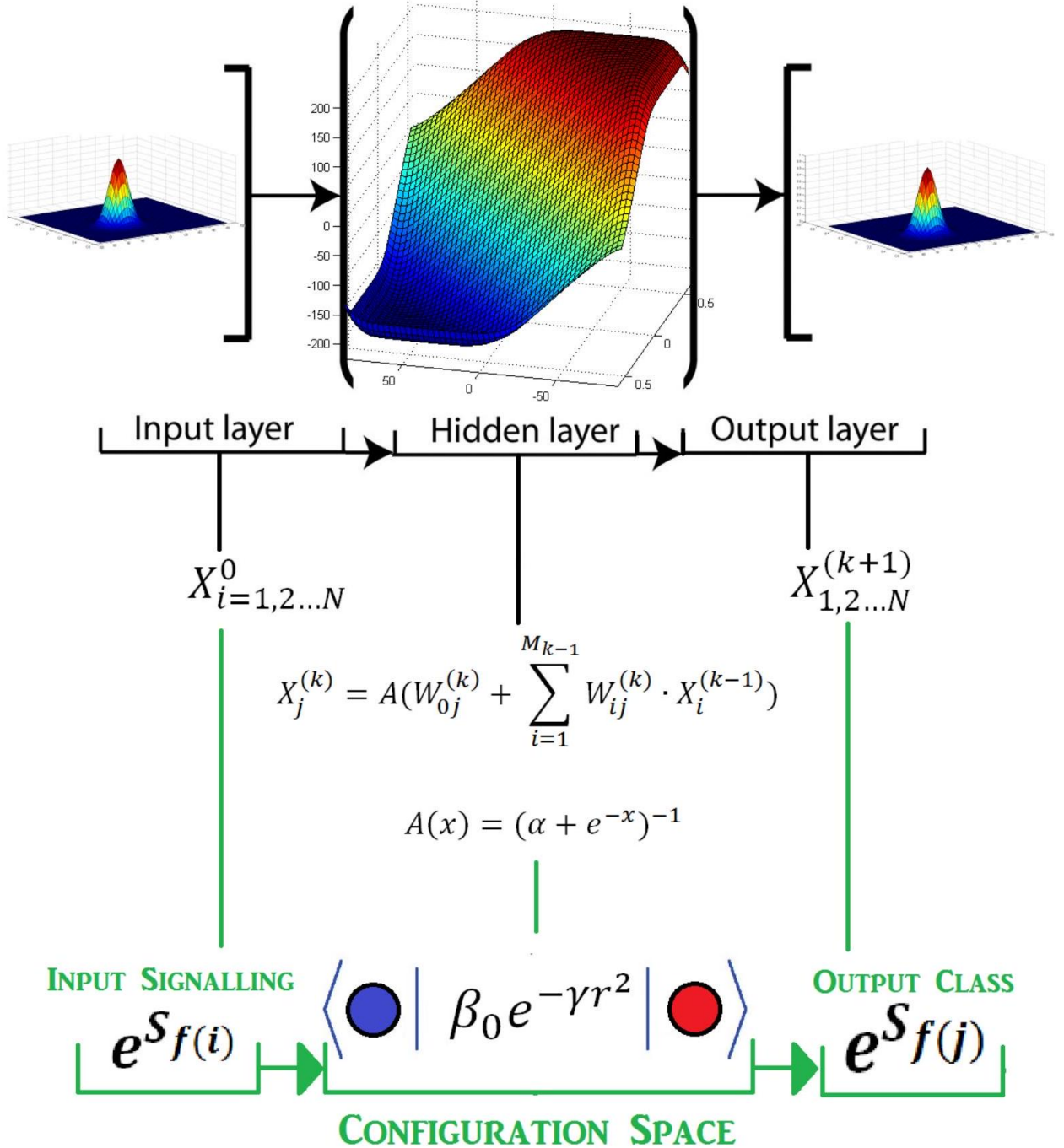
This framework forms the classic and familiar neural network structure.



Using this information in our view of meta-heuristics under the path integral interpretation, we can then reduce the input vectors to an input signalling action, $S_{f(i)}$, oscillating at a certain input frequency in the complex plane, $e^{S_{f(i)}}$, which passes through the network, for example under sigmoidal or other Gaussian analog activation functions, in our path integral interpretation. This then gives

an output signalling action, $S_{f(j)}$, of certain nodes in the network which will now carry a frequency, $e^{S_{f(j)}}$, characteristic of the input signal as interpreted by the neural network. The weights the input vector acts on in the neural network picture are then replaced by the paths in the sum over all possible taken in the time-evolution to achieve synchronization within the network.

I represent this construct in the diagram below, in which I have created a Radial Basis Function synchronised neural network in Matlab which receives a signal and creates an output response characteristic of this input, broadened by the k-number of paths of sigmoidal activation function signalling within the network:



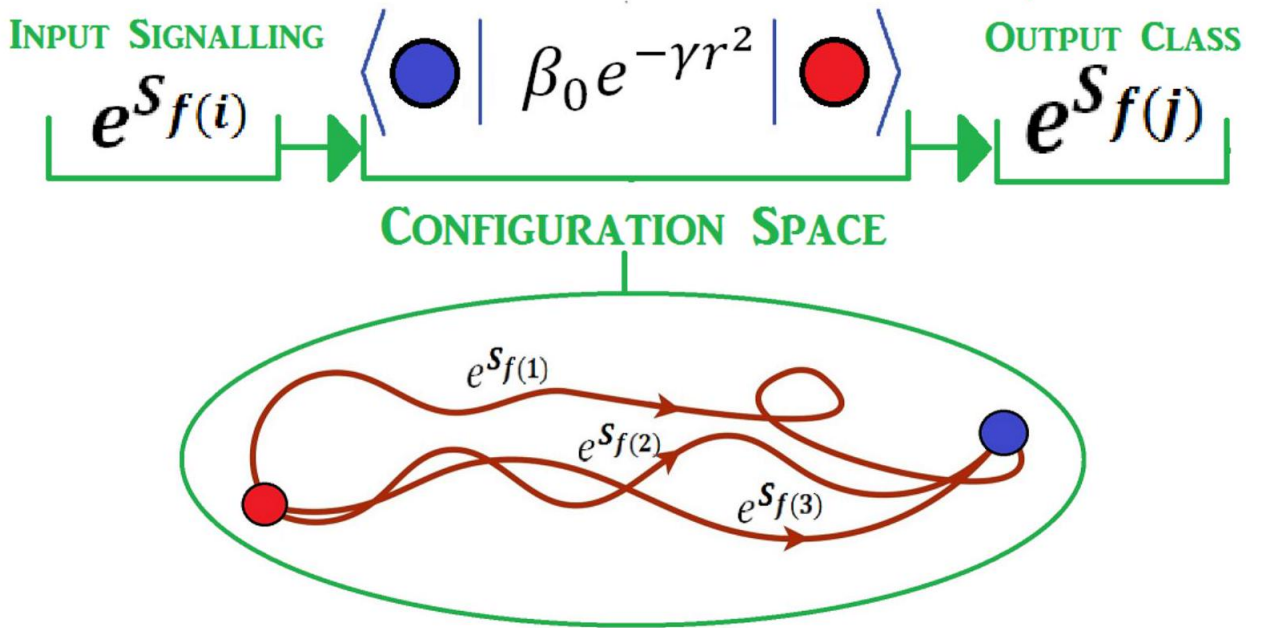
We then say there is an output "class" of internal node signalling action in the network from being stimulated by an external signalling action, in which sense our input signalling action is interpreted

as a "class" by the nodes after passing through the neural network.

It is perhaps important to note that the signalling action towards the generation of nodal output classes is clearly not the core function of the neural network and is merely ancillary to it. By this I mean that the vector inputs and their associated weights cause action on the network but in this case are not in fact forming the network structure itself. We are reminded that the network structure is formed, independently, by the time-evolution of the discontinuous pas-coupling generated by the meta-heuristic nature of the internal signalling along the path of least signalling, which is an extension of the principle of conservation of energy is itself as discussed earlier.

Hence we can now think of the established meta-heuristically synchronized system as an independent computational architecture, based on the path integral interpretation:

PATH INTEGRAL ARCHITECTURE



In the classical picture of such a network, which have local energy minima, we have a network made up of binary nodes (activations either 0 or 1), where the probability of achieving a coupling between the nodes would be set to 1 if and only if the connected nodes have the same activation, i.e. $A_1=A_2$, i.e. the 2 activation levels have an allowed crossing. This has the effect of greatly limiting the nodal capacity of such systems.

To solve this problem we must relate our quantum mechanical system to equilibrium dynamics which will allow us to create a meta-heuristic representation of quantum diabatic behaviour which will induce the necessary condition of avoided crossing between nodes in the network.

Diabatic transitions can in fact be represented as a meta-heuristic algorithm therefore one need not go through the trouble of constructing a quantum computer to simulate annealing in the optimum way. For example, one thing that a metaheuristic, under the lens of the path integral interpretation of quantum mechanics, can do is a form of self-generating quantum noise correction, just in the same way as true adiabatic quantum computer would.

Using quantum dynamics, we can say that the probability of a Diabatic Transition between 2 discrete energy levels E_1 and E_2 is given by the solution to the Landau-Zener Formula, solved by contour integration with the contour chosen to be in accord with the perturbative limit [9]

The Landau-Zener equation is:

$$P_{(E2)} = P_{(E1)} e^{-\Gamma 2\pi}$$

or

$$P_{D(E2-E1)} = e^{-\Gamma 2\pi}$$

Where:

$$\Gamma = \frac{g^2}{\hbar \left| \frac{dq}{dt} \frac{\partial}{\partial q} (E_2 - E_1) \right|} = \frac{g^2}{\hbar |\alpha|}$$

Where $\frac{dq}{dt}$ is the Landau-Zener Velocity for the perturbation variable q (where q is either a quantisation, i.e. a charge, of an electric or magnetic field, the length scale of a molecular bond, or any other perturbation to the system), and E1 and E2 are the energies of the two diabatic (crossing) states.

The quantity g is the off-diagonal coupling element of the two-level system's Hamiltonian that couples the crossing states, and as such it is half the distance between the two unperturbed eigenenergies at the avoided crossing, when $E1 = E2$.

We can write the transition probability as:

$$P_D = e^{-\frac{g^2 2\pi}{\hbar |\alpha|}}$$

The adiabatic theorem in quantum mechanics therefore tells us, that as the difference between the energy states, α , Goes to 0, then the transition probability P_D goes to 1 (unity).

This means that the quantised transitions between the energy states themselves are suppressed at the point of avoided crossing where 2 or more eigenvalues of the Hamiltonian cannot become equal in value.

This implies reversibility - i.e. no net work done, in a cyclic process.

In terms of thermal dynamics then we examine the transition probability in terms of the Energy Hamiltonian

Thermal Equilibrium Dynamics:

$$z = \text{Tr } e^{-\beta H} = \text{Tr } e^{-\beta e(n)}$$

Since the occupation space can only be $|0\rangle$ or $|1\rangle$ for fermions,

$$\begin{aligned} z &= \langle 0 | e^{-\beta e(0)} | 0 \rangle + \langle 1 | e^{-\beta e(1)} | 1 \rangle \\ &= 1 + e^{-\beta e} \end{aligned}$$

An imaginary time path integral approach to thermal equilibrium dynamics can be performed by the fact that our canonical density operator, $e^{-\beta H}$, Is related to the time evolution operator in the Schrödinger scheme, $e^{-\frac{iHt}{\hbar}}$ By the analytic continuation of time, we define:

$$t = -i\hbar\beta$$

This is known in field theory as a Wick rotation from a Minkowskian to a Euclidean field theory [9].

However, as we have shown, we can write the coordinate matrix elements of either $e^{-\beta H}$, $e^{-\frac{iHt}{\hbar}}$, $e^{-\frac{g^2 2\pi}{\hbar |\alpha|}}$ as a sum over histories in the path integral interpretation and furthermore can represent the path integral as a meta-heuristic signalling scheme in a neural network.

We can then unite the approach to achieve meta-heuristic synchronization with the adiabatic theorem of equilibrium dynamics which is where the longer the thermal system is allowed to transition to reach its final state in the time evolution from x_0 to x_t , the more likely it will have stayed in the ground state (or, said another way, the longer the system is allowed to run, the less likely there is to be random excitations, which basically translate to noise).

Therefore, in order to reduce noise or error in such a system, all you need to do with a metaheuristic algorithm is to simply run the algorithm in the adiabatic fashion for a longer time. i.e. if you ran it for $T = \text{infinity}$, you'd be 100% accurate.

This is exactly the criterion for the standard metaheuristic signalling algorithm, as the signalling actions move towards the least signalling, and thus most energetically conservative, action possible.

This leads us to thinking of the neural network as a Boltzmann Machine Framework which can overcome the problem of systems of binary nodes getting stuck in local energy minima by the addition of three key features:

- Stochastic activation function: the state a unit is in is probabilistically related to its Energy gap. The bigger the energy gap between its current state and the opposite state, the more likely the unit will flip states.
- Temperature and simulated annealing: the probability that a node is on is computed according to a activation function of its total weighted summed input divided by T . If T is large, the network behaves very randomly. T is gradually reduced and at each value of T , all the units' states are updated. Eventually, at the lowest T , units are behaving less randomly and more like binary threshold units.
- Contrastive Learning: A Boltzmann machine is trained in two phases, "clamped" and "unclamped". It can be trained either in supervised or unsupervised mode; this type of training proceeds as follows, for each training pattern:
 1. Clamped Phase: The input units' states are clamped to (set and not permitted to change from) the training pattern, and the output units' states are clamped to the target vector. All other units' states are initialized randomly, and are then permitted to update until they reach "equilibrium" (simulated annealing). Then learning is applied.
 2. Unclamped Phase: The input units' states are clamped to the training pattern. All other units' states (both hidden and output) are initialized randomly, and are then permitted to update until they reach "equilibrium". Then anti- learning (learning with a negative sign) is applied.

The above two-phase learning rule must be applied for each training pattern we feed into the network and for a great many iterations through the whole training set. Eventually, the output units' states should become identical in the clamped and unclamped phases, and so the two learning rules exactly cancel one another. Thus, at the point when the network is always producing the correct responses, the learning procedure naturally converges and all weight updates approach zero.

The stochasticity enables the Boltzmann machine to overcome the problem of getting stuck in local energy minima, while the contrastive learning rule allows the network to be trained with hidden features and thus overcomes the capacity limitations.

Boltzmann learning is very powerful, but the complexity of the algorithm increases exponentially as more neurons are added to the network. To reduce this effect, a Reduced Boltzmann machine (RBM) can be used. The hidden nodes in an RBM are not interconnected as they are in regular Boltzmann networks. Once trained on a particular feature set, these RBM can be combined together into larger, more diverse machines, such as polymer rings of different nodes.

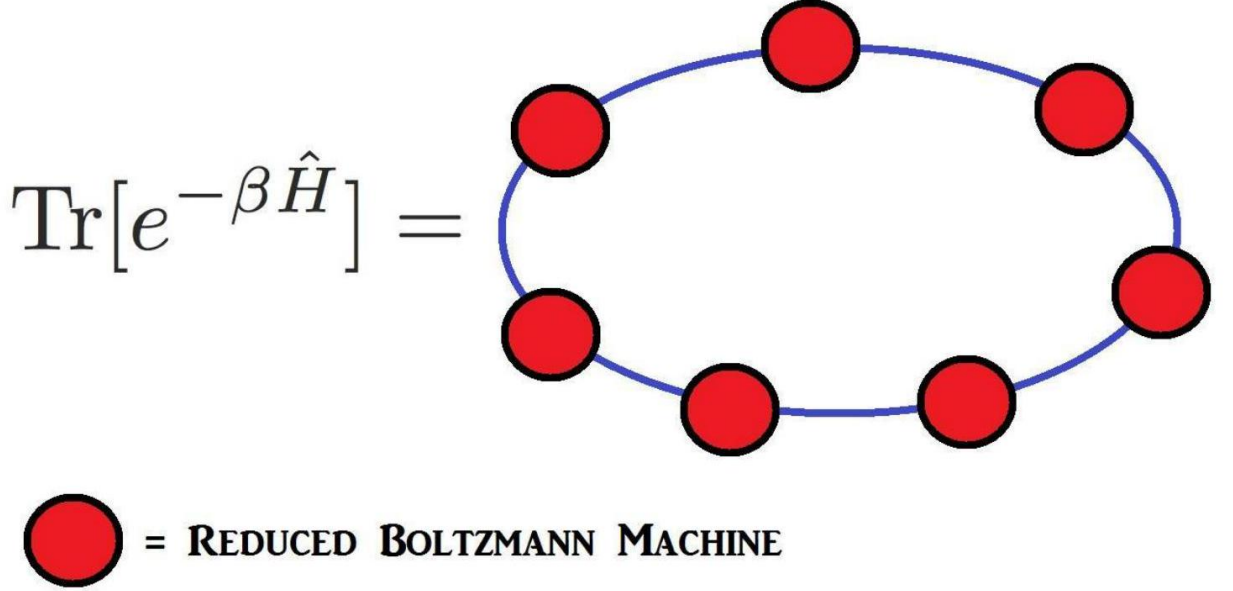
Because Boltzmann machine weight updates only require looking at the expected distributions of surrounding neurons, it is a plausible model for how actual biological neural networks learn.

Boltzmann learning is statistical in nature, and is derived from the field of thermodynamics. It is similar to error-correction learning and is used during supervised training. In this algorithm, the

state of each individual neuron, in addition to the system output, are taken into account. In this respect, the Boltzmann learning rule is significantly slower than the error-correction learning rule.

Reduced Boltzmann Machine learning is particular interest when considering so-called polymer rings and the means used to simulate them.

Polymer rings are essentially compact Gaussian chains of individual reduced Boltzmann machines subject to strong interactions with neighbouring chains due to the presence of topological constraints.



Therefore, one might assume that ring polymer simulations that use a Reduced Boltzmann Machine metaheuristic are potentially much more accurate and consistent with nature than using the current artificial dynamical Monte Carlo simulations. We also need not even think about in the sense of performing quantum evolution in terms of path integrals on polymer chain simulations, as doing a metaheuristic simulation is functionally equivalent as a quantum path integral. In effect, every time we do a metaheuristic simulation, we are in fact performing a quantum path integral.

Entanglement is the core feature of quantum phenomena, as it displays how quantum behaviour is irreducibly random with non-locality being a core feature of the probabilistic wavefunction.

The Von-Neuman entropy gives us a comparable way to measure the degree of entanglement of a multipartite system, such as the Boltzmann machines.

The entanglement for a pure bipartite state $|\psi_{AB}\rangle$ is given by the von Neumann entropy of the subsystem state ρ_A or ρ_B :

$$S(|\psi_{AB}\rangle) = -\text{Tr} \rho_A \log \rho_A = -\text{Tr} \rho_B \log \rho_B$$

It is possible to use metaheuristics to develop Quantum computing of a kind using complex networks, although interesting in and of itself, has very broad applications in any field involving a networked topology, which covers topics such as polymer interactions under certain input actions and possibly much more in the field of complex systems with respect to quantum chemistry.

This is too broad a field to cover in this thesis however it cannot be fully ignored as a "network as computing resource" may be useful in quantum networking between purpose built quantum computing resources themselves especially in data centers wherein the network really would be treated as a distributed extension of the computer, sharing entanglement via gate based qubit operations locally, but with global entanglement and hence networked memory, synchronization, etc with "flying qubits".